Distributionally Robust Group Testing with Correlation Information

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Motivated by the need for more efficient and reliable methods of group testing during widespread infectious outbreaks, this paper introduces a novel operational improvement to the widely-used Dorfman's group testing procedure, where a single test is conducted on the pooled sample, followed by individual testing of positive pools. Our method minimizes a weighted sum of testing volume and misclassifications, predicated on prevalence rates and *inter-individual Pearson correlation coefficients*. Recognizing the inherent ambiguity in the population-level joint probability distribution of infection statuses that rises from the correlations between individuals, our approach leverages a distributionally robust optimization (DRO) framework to counteract the worst-case probability distribution. In the single-cluster case, where each pair of subjects are equally correlated, we establish uniform group sizes and connect our analysis to Nash equilibrium principles. Larger testing groups are generally favored under high correlation, whereas individual testing becomes optimal under high prevalence. In the multi-cluster case, where the population is formed by several intra-correlated but inter-independent clusters, we highlight the effectiveness of mixed-cluster testing strategies, particularly at lower levels of prevalence and correlation. This is a notable addition to the prevailing view that advocates pooling correlated individuals. For both single- and multi-cluster cases, we develop polynomial-time solutions and offer practical insights and policy implications for optimal pooling strategies. We demonstrate the trade-offs and benefits of our DRO approach through a thorough comparison with stochastic alternatives, and demonstrate the impact of incorporating correlated infections in a case study using a COVID-19 dataset from Hong Kong.

1. Introduction

We have recently emerged from a strenuous battle against COVID-19. The pandemic has left indelible marks on society and the economy—it triggered an economic downturn and a stock market crash in 2020, dampened global business activity during lockdown periods, and precipitated a supply chain upheaval alongside an inflation spike. Throughout the pandemic, large-scale screening has been a notably common and effective approach (Mercer and Salit 2021, Barak et al. 2021) that has been instrumental in identifying infections and facilitating timely medical intervention, hence

containing the virus. However, administering tests to an extensive population can be fraught with significant logistical complexities due to its extensive demand for medical supplies and healthcare professionals. For example, Todd (2020) highlights the challenging situation in the US during the onset of the pandemic due to scarce testing capacity and difficulties in managing limited resources. Given the fast-spreading nature of contagious viruses, it is vital to adopt more effective mass testing methods to manage the spread of infectious diseases, especially when resources are limited.

One such method is the group testing strategy, which has been widely implemented (see, for example, NBC News (2020) for HIV and Global News (2022), USC Schaeffer Center News (2020) for COVID-19[1]). In this testing scheme, all subjects are assigned to groups, and all subjects in the same group are initially pooled for testing, i.e., the group sample (consisting of the mixed samples for all subjects in this group) is tested once. If the test result for the group sample is positive, the subjects in the corresponding group are invited to the second-stage individual tests. Otherwise, all subjects in the group are declared negative. Figure 1 depicts a straightforward scenario of the process, where samples are partitioned into non-overlapping testing groups. Rooted in the seminal work by Dorfman (1943), we refer this as Dorfman's procedure throughout our discussion.

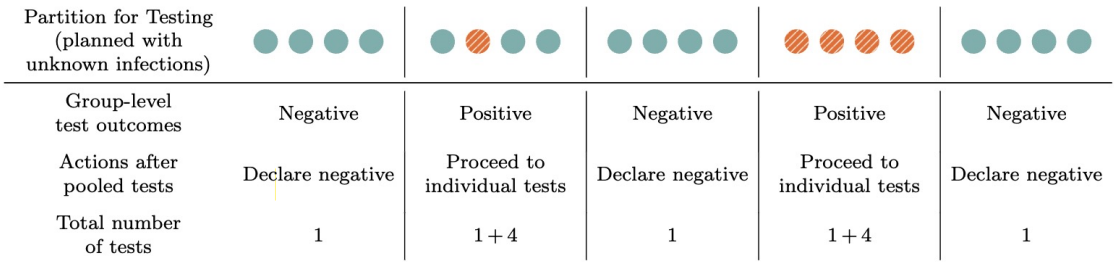


Figure 1 **An illustration of Dorfman's group testing procedure. A population of 20 people is divided into 5 groups of size 4 for the pooled tests, with the infected individuals represented in orange. In this scenario, the total number of tests amounts to 1+5+1+5+1=13, which is notably lower than the 20 tests required if each individual were tested separately.**

Despite its widespread application and considerable societal impact, the classic Dorfman testing scheme exhibits two significant limitations. Firstly, tests are usually assumed to have perfect accuracy, and the only objective in designing a testing scheme is to minimize the total number of tests, overlooking the false negatives and false positives. However, in practice, false negatives are common due to the dilution effect and low viral load at the beginning of infection (Hwang 1976), and improper operations and sample contamination may lead to false positives. Therefore, it is crucial to balance accuracy (misclassifications) with efficiency (total number of tests). Secondly, infection statuses are assumed to be statistically independent with each other. This assumption becomes

[1] https://en.wikipedia.org/wiki/List_of_countries_implementing_pool_testing_strategy_against_COVID-19

questionable in many contexts, including contagious diseases where infection likelihood is influenced by close physical interactions (see news.gov.hk (2022) for an example). Furthermore, such correlation can significantly impact the choice of the testing strategy. As illustrated in Example 1, a perfect positive correlation between subjects shifts the optimal testing scheme from individual testing to group testing.

Example 1 *(The effect of correlation) Consider a population of two individuals, where each individual has an infection rate of* $p = 0.4$*. As shown in Table 1, in the independent case, the population has four possible scenarios for its infection status; if group testing is adopted, only the first scenario will be exempted for further individual tests, and the total number of tests will be* $1 + 2 = 3$ *for all other scenarios. As a result, the expected number of tests over all possible scenarios exceeds 2, indicating that individual testing is more efficient in saving testing resources. However, if the two individuals are perfectly correlated, only two scenarios remain possible—both uninfected, or both infected. Consequently, the likelihood of saving the second-stage individual tests increases, and the expected number of tests for group testing drops below 2. This implies that group testing in this case is more efficient than individual testing.*

Table 1 **The effect of correlation illustrated using a 2-individual case. In all possible scenarios, infected individuals are represented in orange, while uninfected individuals are shown in light blue.**

	Possible Scenarios	Number of Tests (if Group Testing)	Testing Choice
Independent ($\rho = 0$)	with Prob. 0.36	1	$\mathbb{E}[\text{Number of Tests}] = 2.28 > 2$, *Individual Testing* is more efficient
	with Prob. 0.24	3	
	with Prob. 0.24	3	
	with Prob. 0.16	3	
Perfectly Correlated ($\rho = 1$)	with Prob. 0.6	1	$\mathbb{E}[\text{Number of Tests}] = 1.8 < 2$, *Group Testing* is more efficient
	with Prob. 0.4	3	

Nevertheless, incorporating the impact of correlation can be challenging, as it introduces inherent ambiguity in the distribution of infection status. While this may not seem problematic in Example 1, where the population size is 2 and the correlation coefficient is either 0 or 1, the joint distribution becomes ambiguous when these conditions are relaxed. Generally, multiple joint distributions can match the same rate and correlation parameters. This contradicts the common assumption in stochastic optimization that the joint probability distribution is precisely known, making it imperative to develop a reliable decision to account for potentially pessimistic scenarios.

In this paper, we introduce a novel approach to improve the classic Dorfman's procedure by *simultaneously* tackling these limitations and challenges. In particular, for a partition of the population, we account for the misclassifications by minimizing a weighted sum that includes the expected counts of tests, false negatives and false positives. To address the ambiguity incurred by correlated infections, we adopt a minimax strategy that aims to identify the most adverse joint distribution scenario and formulate an optimal pooling strategy to safeguard against this worst-case distribution. Our main contributions are summarized as follows.

- We bring a fresh perspective to the current literature by developing an effective and reliable design of two-stage group testing with the consideration of correlation and false testing results. Specifically, we address the inherent ambiguity in the joint distribution of the population's infection status by developing a distributionally robust optimization (DRO) problem, with the only known information being the prevalence rate and pairwise correlation coefficients.
- In the single-cluster case where all pairs of subjects exhibit uniform correlation coefficient, we first show the optimality of equal-sized pooling and establish a connection between our model and a non-cooperative game between policy makers and a hypothetical "nature" adversary. This offers theoretical backing for the empirical adoption of equal-sized pooling. By further deriving an explicit form of the worst-case joint distribution, we develop an efficient $O(1)$-time solution for determining optimal group size, i.e., the computation time remains constant regardless of changes in population size. Our analysis reveals that larger groups are preferred at high correlation levels, or when minimizing the number of tests is the main objective; by contrast, individual testing is advantageous at high prevalence rates, or when reducing false negatives is the priority.
- We extend our results to the multiple-cluster case, where the population consists of M intra-correlated but inter-independent clusters. We show that the optimality of homogeneous testing groups remains in a broader context—first, testing groups within an optimal partition share not only a uniform size, but also a consistent mix of subjects from various clusters; second, while it is not necessary for an optimal testing pool to include subjects from every cluster, each involved cluster should be represented equally in each pool, with the count differing by at most one. This is a notable addition to the prevailing view in existing literature that advocates pooling correlated individuals. In particular, we demonstrate the effectiveness of mixed-cluster testing at low prevalence rates and low intra-cluster correlation levels. Computationally, we show that these structural characteristics reduce the problem to $O(M)$ time complexity.
- Our study highlights the importance of considering correlated infections and the benefits of using a robust optimization framework. We illustrate the minor losses and considerable gains in tractability and reliability by comparing our model's outcomes with those of several stochastic alternatives. To demonstrate the benefits of incorporating correlations, we apply our model to

a 2020 COVID-19 dataset from Hong Kong and showcase the efficiency of our group testing strategy in a real-life setting with correlated infections.

The remainder of this paper is organized as follows. We review the relevant literature in Section 2 and introduce the single-cluster model in Section 3. In Section 4, we derive the equal-sized property of an optimal partition and provide an efficient algorithm for finding the optimal group size. We extend our model and analysis to the multi-cluster case in Section 5. We conduct numerical experiments and a case study using COVID-19 data in Section 6 and conclude in Section 7.

2. Literature Review

Our paper is related to the literature on diagnostic testing in pandemics (see, for example, Kelly-Cirino et al. (2019) and Rosenthal (2020)). In policy design and the economic effects of diagnostic testing, Dai and Singh (2020) and Dai and Singh (2024) discuss how mandating the disclosure of viral load and manipulating diagnosis criteria can affect test demand, testing capacity decisions, and the spread of infections; Serra-Garcia and Szech (2023) explore how behavioral interventions (such as monetary incentives and default testing options) influence vaccine uptake and testing participation. Additionally, a broader stream of literature examines the supply chain aspects of testing and resource distribution (Mak et al. 2022, Wang et al. 2023, Hu et al. 2023). We distinguish our work from these streams of studies by focusing on a more localized scale and addressing universal screening through the group testing approach.

Building on the seminal work of Dorfman (1943), group testing has garnered extensive research as an efficient tool for large-scale screening in various contagious diseases, including malaria (Hsiang et al. 2010), HIV (Pilcher et al. 2002, Aprahamian et al. 2016), hepatitis viruses (Weusten et al. 2011, Aprahamian et al. 2016), genetic screening(El Hajj et al. 2022b), and the recent COVID-19 (Abdalhamid et al. 2020, Mutesa et al. 2021, Augenblick et al. 2022). Notably, among studies on the operations of Dorfman's testing procedure, most assume perfect tests (e.g., Saraniti 2006, Feng et al. 2010, Li et al. 2014). Our research is more closely aligned with a number of studies that relax this assumption. For example, Aprahamian et al. (2019) consider two variant models, separately being cost-efficient (i.e., minimizing a weighted sum of misclassifications and number of tests) and budget-efficient (i.e., minimizing misclassifications with the constraint of test capacity). Lin et al. (2020) study the minimization of false negatives under the constraints of both test capacity and false positives. While these studies approach the problem from a stochastic programming perspective (i.e., the distributions of infections are assumed to be perfectly known), our study adopts a DRO framework and acknowledges that the underlying probability distributions can be ambiguous.

Among the various group testing methods studied, research on the effects of correlation in group testing remains limited. Lendle et al. (2012) show the expected number of tests per subject decreases

in correlation via simulating from known distributions. Augenblick et al. (2020) illustrates the general intuition that correlation increases the group size and reduces the total number of tests required. As we formalize this result in our analysis, we will show that it does not always hold. In a different vein, Comess et al. (2022) model the non-independence by network effect and quantify the level of correlation by the probability of disease transmission. Lin et al. (2021) apply the correlation coefficients to a social graph and capture the benefit of pooling correlated samples in a dynamic setting. However, our investigation into the multi-cluster correlation structure reveals that the optimal grouping strategy often includes a combination of both correlated and independent subjects. This insight adds a refined understanding to the existing literature. From a policy-design perspective for the *optimal* group testing strategy, Zhang and Heath (2023) propose an adaptive scheme that groups the most closely connected individuals in a social contact network and maximize the total number of detected infections within a limited testing budget. Basso et al. (2022) search for the optimal group size by simulation when pairwise correlation coefficients are given, but the joint probability distribution of the infection is limited to binomial or beta-binomial distributions.

Our paper also belongs to the stream of literature that applies robust optimization technique to large-scale testing problems. For example, Jain et al. (2024) propose a robust optimization methodology for combining various POC tests (with different costs and accuracy levels) to enhance diagnostic precision in the absence of PCR testing. The robustness of their model counters the uncertainty in test sensitivity and specificity. El Hajj et al. (2022a) study the selection of a panel of mutations among hundreds of variants to screen newborns for cystic fibrosis. With the goal of minimizing false negatives under a testing budget, they consider robustness against misspecification of the prevalence rate of each variant. In the group testing literature, Aprahamian et al. (2020a) design a Dorfman's procedure with robustness against prevalence rate misspecifications, and Aprahamian et al. (2020b) additionally consider robustness against the variation of estimation error of the prevalence rate. Bish et al. (2023) propose robust cost- and capacity-efficient models for multi-disease testing, considering both sample pooling and disease bundling. All these studies focus their uncertainty analysis only on prevalence rates/accuracy levels under the independence assumption, thereby neglecting a vital aspect of the problem—the potential interdependence among subject risks. In contrast, our work explores the impact of correlation structure by characterizing the ambiguity of the joint distribution across the entire population.

Beyond public health surveillance, this paper also connects to the broader literature on the applications of group testing in other operational domains. For instance, Cheng (2011) investigates randomized group testing methods for efficiently identifying defective products in quality control settings. Broeckling et al. (2023) review the use of pooled testing in an analytical quality control approach with applications spanning agriculture, ecology, environmental sciences, microbiology,

and more. Similarly, group testing has been widely employed for monitoring the environment for trace organic compounds present at low levels (Schaeffer et al. 1982, Patil 1995, Patil et al. 2010). In addition, Ibarrondo et al. (2023) apply group testing to accelerate identification processes in biometric databases. In cybersecurity, Hahn-Klimroth et al. (2024) leverage group testing to locate suspicious behaviors and malicious nodes in computer networks.

3. Model Description

We consider a population of N subjects, denoted by $[N] = \{1, \ldots, N\}$. Each subject in the population is either infected or not (i.e., positive or negative, respectively), which can be considered as a random event until a certain test unveils its realization. Given any $i \in [N]$, the infection of subject i can be represented by a Bernoulli random variable $\tilde{\xi}_i$, with the tilde sign representing its random nature. A subject i is positive if $\tilde{\xi}_i$ is realized to be $\xi_i = 1$, and negative if $\xi_i = 0$. We further define $\tilde{\boldsymbol{\xi}} = \left(\tilde{\xi}_1, \cdots, \tilde{\xi}_N\right)$, with the boldfaced character representing a vector and nonboldfaced ones representing scalars. Let $\mathbb{P}$ be the probability distribution of $\tilde{\boldsymbol{\xi}}$.

As introduced in Section 1, our goal is to identify an optimal partition of the population, denoted by $\mathcal{X}$, such that i) a weighted sum of the expected counts of tests $\mathbb{E}_{\mathbb{P}}[T(\mathcal{X}, \tilde{\boldsymbol{\xi}})]$, false negatives $\mathbb{E}_{\mathbb{P}}[FN(\mathcal{X}, \tilde{\boldsymbol{\xi}})]$ and false positives $\mathbb{E}_{\mathbb{P}}[FP(\mathcal{X}, \tilde{\boldsymbol{\xi}})]$ is minimized; and ii) robustness is ensured over all probability distributions $\mathbb{P}$ that share the same information on prevalence rate and correlation coefficient. Formally, we consider the following optimization problem,

$$\min_{\mathcal{X} \in \mathcal{B}} \sup_{\mathbb{P} \in \mathcal{P}} \lambda_1 \mathbb{E}_{\mathbb{P}}\left[FN(\mathcal{X}, \tilde{\boldsymbol{\xi}})\right] + \lambda_2 \mathbb{E}_{\mathbb{P}}\left[FP(\mathcal{X}, \tilde{\boldsymbol{\xi}})\right] + (1 - \lambda_1 - \lambda_2) \mathbb{E}_{\mathbb{P}}\left[T(\mathcal{X}, \tilde{\boldsymbol{\xi}})\right], \tag{1}$$

where λ_1, λ_2 are weights assigned to false testing results and satisfy $\lambda_1, \lambda_2 \geq 0$, $\lambda_1 + \lambda_2 \leq 1$. Sets $\mathcal{B}$ and $\mathcal{P}$ are feasible sets, respectively, for partition $\mathcal{X}$ and the population-level joint distribution $\mathbb{P}$.

In this section, we will first elaborate on the partitioning decision $\mathcal{X}$ and its feasible set $\mathcal{B}$, followed by explicit expressions of $FN(\mathcal{X}, \tilde{\boldsymbol{\xi}}), FP(\mathcal{X}, \tilde{\boldsymbol{\xi}}), T(\mathcal{X}, \tilde{\boldsymbol{\xi}})$. Then, we will discuss a pivotal assumption and define the feasible set $\mathcal{P}$ for the joint probability distribution $\mathbb{P}$.

3.1. Partitioning Decision

In terms of the partitioning decision, we divide the population into a number of disjoint groups, denoted as $\Omega_1, \ldots, \Omega_W \subseteq [N]$ for some positive integer W. Note that group testing bears a recognized constraint — the dilution effect, where an increase in group size results in a diminished testing accuracy, as discussed in Wein and Zenios (1996) and Lin et al. (2020). To mitigate this dilution effect, we impose a constant maximum size Γ for all testing groups[2], where the value of Γ is

[2] Setting the upper limit Γ does not completely eliminate the dilution effects in group testing. Additionally, various other factors can lead to false testing results, such as improper cycle threshold values, operational errors, and sample contamination. Consequently, it is essential to include the consideration of misclassifications in our analysis.

typically small compared to N. For example, at the onset of COVID-19, Perchetti et al. (2020) implemented 4-way pooled testing to more than 100,000 samples, and Hogan et al. (2020) reported massive testing by pooling 9 or 10 individual samples in the San Francisco Bay Area, CA. From a biochemical perspective, Verdun et al. (2021) suggest that the pool size for COVID-19 tests can be up to 10. Consequently, we formulate the feasible set $\mathcal{B}$ as follows,

$$\mathcal{B} = \left\{ \mathcal{X} = \{\Omega_1, \ldots, \Omega_W\} \;\middle|\; \begin{array}{ll} \Omega_1 \cup \cdots \cup \Omega_W = [N] & \\ \Omega_{w_1} \cap \Omega_{w_2} = \varnothing, & \forall w_1, w_2 \in [W],\ w_1 \neq w_2 \\ |\Omega_w| \leq \Gamma, & \forall w \in [W] \\ W \in [N] & \end{array} \right\},$$

where $|\Omega_w|$ stands for the cardinality of Ω_w, and $[W]$, similar to $[N]$, is defined to be the set of positive running indices $\{1, \ldots, W\}$.

3.2. Objectives

Our objective contains three components, of which the total number of tests can be calculated straightforward. To model the mis-specified test outcomes, we denote the probability of a false negative as β_n and a false positive as β_p for any single test involving either an individual sample or a mixed sample of size less than Γ. We impose $\beta_n + \beta_p < 1$ so that the tests have a discriminative ability better than random chance.

Lemma 1 *Given a partition $\mathcal{X} = \{\Omega_1, \ldots, \Omega_W\}$ and a joint distribution $\mathbb{P}$, we have $\mathbb{E}_{\mathbb{P}}[FN(\mathcal{X}, \tilde{\boldsymbol{\xi}})] = \sum_{i=1}^{N} \mathbb{E}_{\mathbb{P}}[FN_i(\mathcal{X}, \tilde{\boldsymbol{\xi}})]$, $\mathbb{E}_{\mathbb{P}}[FP(\mathcal{X}, \tilde{\boldsymbol{\xi}})] = \sum_{i=1}^{N} \mathbb{E}_{\mathbb{P}}[FP_i(\mathcal{X}, \tilde{\boldsymbol{\xi}})]$, $\mathbb{E}_{\mathbb{P}}[T(\mathcal{X}, \tilde{\boldsymbol{\xi}})] = \sum_{w=1}^{W} \mathbb{E}_{\mathbb{P}}[T_w(\mathcal{X}, \tilde{\boldsymbol{\xi}})]$, where $FN_i(\mathcal{X}, \tilde{\boldsymbol{\xi}})$, $FP_i(\mathcal{X}, \tilde{\boldsymbol{\xi}})$ and $T_w(\mathcal{X}, \tilde{\boldsymbol{\xi}})$ represent the number of false negative classification for any subject $i \in [N]$, the number of false positive classification for any subject $i \in [N]$, and the number of test for group $w \in [W]$ under partition $\mathcal{X}$ and random classification vector $\tilde{\boldsymbol{\xi}}$, respectively, and can be formulated as follows,*

$$FN_i(\mathcal{X}, \tilde{\boldsymbol{\xi}}) = \begin{cases} \beta_n & \text{if } \tilde{\xi}_i = 1 \text{ and } i \text{ is tested individually in } \mathcal{X}, \\ \beta_n(2-\beta_n) & \text{if } \tilde{\xi}_i = 1 \text{ and } i \text{ is tested in a group in } \mathcal{X}, \\ 0 & \text{if } \tilde{\xi}_i = 0. \end{cases}$$

$$FP_i(\mathcal{X}, \tilde{\boldsymbol{\xi}}) = \begin{cases} \beta_p & \text{if } \tilde{\xi}_i = 0 \text{ and } i \text{ is tested individually in } \mathcal{X}, \\ \beta_p^2 & \text{if } \tilde{\xi}_i = 0 \text{ and } i \text{ is tested in a non-infected group in } \mathcal{X}, \\ (1-\beta_n)\beta_p & \text{if } \tilde{\xi}_i = 0 \text{ and } i \text{ is tested in an infected group in } \mathcal{X}, \\ 0 & \text{if } \tilde{\xi}_i = 1. \end{cases}$$

$$T_w(\mathcal{X}, \tilde{\boldsymbol{\xi}}) = \begin{cases} 1 & \text{if } |\Omega_w| = 1, \\ 1 + \beta_p|\Omega_w| & \text{if } |\Omega_w| \geq 2 \text{ and all subjects in } \Omega_w \text{ are non-infected}, \\ 1 + (1-\beta_n)|\Omega_w| & \text{if } |\Omega_w| \geq 2 \text{ and } \Omega_w \text{ includes infected subjects.} \end{cases}$$

Lemma 1 delivers a formal definition of the objective in our model. Our evaluation of FN, FP, T follows a similar idea to Aprahamian et al. (2019), with the key difference being our relaxation

of the independence assumption between individuals. Despite being straightforward, this form is visibly impractical for further investigation in analysis. We present a more tractable equivalent definition in Section 4.1.

Remark 1 *Lemma 1 formulates the total number of test, i.e.,* $\mathbb{E}_{\mathbb{P}}[T(\mathcal{X},\tilde{\boldsymbol{\xi}})]$*, by setting the same weights of the test numbers of the first and second stages. However, in practice, additional operational costs can occur to organize the second-stage testing for positive groups. We consider an extended case where the second-stage cost differs from the first stage in Appendix EC.A.1.*

3.3. Population-level Joint Distribution

We consider an identical infection rate p across the population, which is a common assumption in real-world scenarios where testing agencies may lack detailed information to differentiate various subjects. Consider any distinct subjects $i,j\in[N]$ with the Pearson correlation coefficient between $\tilde{\xi}_i,\tilde{\xi}_j$ being ρ, then we have $\mathbb{P}\left(\tilde{\xi}_i=1,\tilde{\xi}_j=1\right)=p^2+\rho p(1-p)$. We assume that the correlations between all pairs of subjects are identical. This assumption will be relaxed in Section 5, where we consider multiple intra-correlated but inter-independent clusters of population. To avoid trivial cases, we let $p\in(0,1)$ and $\rho\in[0,1)$.

Besides the information on the prevalence rate and correlation coefficient, the following assumption about homogeneity between subjects is also vital to our analysis.

Assumption 1 *All subjects are exchangeable, i.e., for any* $\boldsymbol{\xi}\in\{0,1\}^N$ *and permutation matrix* $\boldsymbol{P}_\pi$*, we have* $\mathbb{P}\left(\tilde{\boldsymbol{\xi}}=\boldsymbol{\xi}\right)=\mathbb{P}\left(\tilde{\boldsymbol{\xi}}=\boldsymbol{P}_\pi\boldsymbol{\xi}\right)$.

Assumption 1 indicates that any two population-level infection statuses (i.e., $\boldsymbol{\xi}',\boldsymbol{\xi}''\in\{0,1\}^N$) share the same probability, provided that the number of infected subjects is the same (i.e., $\mathbf{1}^\top\boldsymbol{\xi}'=\mathbf{1}^\top\boldsymbol{\xi}''$). Essentially, it operates on the principle that the specific identities of subjects do not affect their likelihood of infection. For instance, the probability of only the first two subjects being infected is equal to that of only the last two subjects being infected. We base this assumption on the premise that subjects are indistinguishable and hence interchangeable in terms of probability, and the same assumption has also been utilized in works such as Lendle et al. (2012).

The prevalence rate p, correlation coefficient ρ and Assumption 1 define the feasible set $\mathcal{P}$ (or termed the *ambiguity set*), which contains every admissible probability distribution. Formally, $\mathcal{P}$ can be formulated as follows,

$$\mathcal{P}=\left\{\mathbb{P}\;\middle|\;\begin{array}{ll}\mathbb{P}\left(\tilde{\xi}_i=1\right)=p, & \forall i\in[N]\\ \mathbb{P}\left(\tilde{\xi}_i=1,\tilde{\xi}_j=1\right)=p^2+\rho p(1-p), & \forall i,j\in[N],i\neq j\\ \mathbb{P}\left(\tilde{\boldsymbol{\xi}}=\boldsymbol{\xi}\right)=\mathbb{P}\left(\tilde{\boldsymbol{\xi}}=\boldsymbol{P}_\pi\boldsymbol{\xi}\right), & \forall\boldsymbol{\xi}\in\{0,1\}^N \text{ and permutation matrix } \boldsymbol{P}_\pi\end{array}\right\}. \quad (2)$$

It is important to note that the conditions in (2) does *not* fully determine a population-level probability distribution. In fact, $\mathcal{P}$ contains infinitely many distributions, including a family of binomial mixtures, e.g., the beta-binomial distribution. Among the numerous admissible distributions, our robust approach seemingly adopts the most pessimistic one. We will assess the trade-offs of this robust framework in Section 6.1, particularly in the presence of distributional misspecifications.

4. Optimal Grouping Strategy

Solving Problem (1) is considerably challenging, due to the large-scale combinatorial nature of partitioning decisions $\mathcal{X}$, the intricate computation of $FN(\mathcal{X},\tilde{\boldsymbol{\xi}}), FP(\mathcal{X},\tilde{\boldsymbol{\xi}}), T(\mathcal{X},\tilde{\boldsymbol{\xi}})$, and the vast array of feasible distributions $\mathbb{P}\in\mathcal{P}$. In this section, we use the unique properties of Problem (1) to reformulate the problem in Section 4.1, offer an efficient solution method for the optimal grouping strategy in Sections 4.2 and 4.3 and discuss its structural properties in Section 4.4.

4.1. Model Reformulation

Observing the intractability that stems from the high dimensions of $\mathcal{X}$ and $\mathbb{P}$, in this section, we provide lower-dimension alternatives for both variables and reformulate Problem (1) into a more manageable scale.

We start by addressing the population-level joint distribution $\mathbb{P}$. Although it could theoretically be represented by a 2^N-dimensional vector (given that the sample space of $\tilde{\boldsymbol{\xi}}$ is $\{0,1\}^N$), the exponentially high dimension can pose significant practical challenges. To address this, we define an $(N+1)$-dimensional vector

$$\boldsymbol{d}_{\mathbb{P}} = (d_{\mathbb{P},0}, d_{\mathbb{P},1}, \ldots, d_{\mathbb{P},N}) = \left(\mathbb{P}\big(\tilde{\boldsymbol{\xi}} = \textstyle\sum_{i=1}^{0} \boldsymbol{e}_i\big), \mathbb{P}\big(\tilde{\boldsymbol{\xi}} = \textstyle\sum_{i=1}^{1} \boldsymbol{e}_i\big), \ldots, \mathbb{P}\big(\tilde{\boldsymbol{\xi}} = \textstyle\sum_{i=1}^{N} \boldsymbol{e}_i\big)\right),$$

where we denote by $\boldsymbol{e}_i$ the vector of all zeros except for 1 at position i, and let $\sum_{i=1}^{0} \boldsymbol{e}_i = \boldsymbol{0}$ for notational consistency. In essence, $d_{\mathbb{P},k}$ represents the probability that the first k subjects are positive while the remaining subjects are all negative under $\mathbb{P}$. More precisely, consider $N+1$ distinct sets, $H_0,\ldots,H_N$, each defined as $H_k = \{\boldsymbol{\xi}\in\{0,1\}^N : \boldsymbol{1}^\top\boldsymbol{\xi} = k\}$. Each H_k contains $\binom{N}{k}$ possible scenarios, and any potential realization $\tilde{\boldsymbol{\xi}}$ must belong to one of the sets. Under the exchangeability assumption, the probability mass assigned to each possible realization within H_k is uniform and is represented by the k-th element of $\boldsymbol{d}_{\mathbb{P}}$. Letting

$$\mathcal{D} = \left\{ \boldsymbol{d} = (d_0, d_1, \ldots, d_N) \in \mathbb{R}^{N+1} \;\middle|\; \begin{array}{l} \sum_{k=0}^{N} \binom{N}{k} d_k = 1 \\ \sum_{k=1}^{N} \binom{N-1}{k-1} d_k = p \\ \sum_{k=2}^{N} \binom{N-2}{k-2} d_k = p^2 + \rho p(1-p) \\ \boldsymbol{d} \geq \boldsymbol{0} \end{array} \right\}, \tag{3}$$

We will formally establish an equivalence between $\mathbb{P}\in\mathcal{P}$ and $\boldsymbol{d}\in\mathcal{D}$ in the following Proposition 1.

Next, we simplify the partition decision $\mathcal{X}$ and reformulate its feasible region $\mathcal{B}$. With Assumption 1 in place, the identities of individual subjects become irrelevant, allowing us to focus only on the size of each group for an optimal partition. Consequently, we introduce a Γ-dimensional non-negative integer vector, $\boldsymbol{z} \in \mathbb{Z}^{\Gamma}$ to represent an arbitrary partition, where each z_i represents the count of groups containing exactly i subjects. (Recall that the maximum group size limit is Γ.) As a result, the original partition decision $\mathcal{X} \in \mathcal{B}$ can be reformulated as $\boldsymbol{z} \in \mathcal{Z}$, where

$$\mathcal{Z} = \left\{ \boldsymbol{z} \in \mathbb{Z}_{+}^{\Gamma} \;\middle|\; \sum_{n=1}^{\Gamma} n \cdot z_n = N \right\}.$$

With the adoption of $\boldsymbol{z} \in \mathcal{Z}$ and $\boldsymbol{d} \in \mathcal{D}$, we next present a streamlined reformulation of our model.

Proposition 1 *Problem (1) can be simplified to the following problem,*

$$\min_{\boldsymbol{z} \in \mathcal{Z}} \max_{\boldsymbol{d} \in \mathcal{D}} \; \lambda_1 FN_{\boldsymbol{d}}(\boldsymbol{z}) + \lambda_2 FP_{\boldsymbol{d}}(\boldsymbol{z}) + (1 - \lambda_1 - \lambda_2) T_{\boldsymbol{d}}(\boldsymbol{z}), \tag{4}$$

where

$$\begin{aligned}
&FN_{\boldsymbol{d}}(\boldsymbol{z}) = N\beta_n(2-\beta_n)p - \beta_n(1-\beta_n)pz_1, \\
&FP_{\boldsymbol{d}}(\boldsymbol{z}) = N(1-\beta_n)\beta_p(1-p) + \beta_n\beta_p(1-p)z_1 - \beta_p(1-\beta_n-\beta_p)\textstyle\sum_{n=2}^{\Gamma}\left(z_n n\left(\sum_{k=0}^{N-n}\binom{N-n}{k}d_k\right)\right), \\
&T_{\boldsymbol{d}}(\boldsymbol{z}) = N(1-\beta_n) + \beta_n z_1 + \textstyle\sum_{n=2}^{\Gamma}\left(z_n - (1-\beta_n-\beta_p)z_n n\left(\sum_{k=0}^{N-n}\binom{N-n}{k}d_k\right)\right).
\end{aligned}$$

Moreover, we can further rearrange the terms and simplify Problem (4) as the following one.

Corollary 1 *Problem (4) can be equivalently reformulated as*

$$NC_0 + \min_{\boldsymbol{z} \in \mathcal{Z}} \max_{\boldsymbol{d} \in \mathcal{D}} \sum_{n=1}^{\Gamma} n z_n h(n, \boldsymbol{d}), \tag{5}$$

where the function $h : [\Gamma] \times \mathcal{D} \to \mathbb{R}$ maps the group size n and distribution $\boldsymbol{d}$ to a real number, and is defined by

$$h(n, \boldsymbol{d}) = \begin{cases} C_1 & \text{if } n = 1, \\ \frac{C_2}{n} - C_3 \sum_{k=0}^{N-n}\binom{N-n}{k} d_k & \text{if } n \geq 2. \end{cases} \tag{6}$$

Here, parameters C_0, C_1, C_2, C_3 are independent of both the probability distribution $\boldsymbol{d}$ and the decision $\boldsymbol{z}$, and are defined as

$$\begin{aligned}
C_0 &= \lambda_1\beta_n(2-\beta_n)p + \lambda_2(1-\beta_n)\beta_p(1-p) + (1-\lambda_1-\lambda_2)(1-\beta_n), \\
C_1 &= -\lambda_1\beta_n(1-\beta_n)p + \lambda_2\beta_n\beta_p(1-p) + (1-\lambda_1-\lambda_2)\beta_n, \\
C_2 &= 1 - \lambda_1 - \lambda_2, \\
C_3 &= (1-\lambda_1-\lambda_2(1-\beta_p))(1-\beta_n-\beta_p).
\end{aligned} \tag{7}$$

In the rest of Section 4, we will focus on Problem (5) to determine the optimal $\boldsymbol{z}$ (hence the optimal partition) for the group testing problem. Specifically, we show in Section 4.2 that an optimal partition of the population follows the structure of equal group size across all testing groups; then, in Section 4.3, we propose an optimization procedure to find the optimal group size.

4.2. Optimality of Equal Sized Partitions

In this section, we investigate the optimality structure of our model. Our first result captures the characteristics of worst-case distribution.

Lemma 2 *For any $\boldsymbol{z} \in \mathcal{Z}$, we have $\max_{\boldsymbol{d}\in\mathcal{D}} \sum_{n=1}^{\Gamma} n z_n h(n, \boldsymbol{d}) = \sum_{n=1}^{\Gamma} n z_n h(n, \boldsymbol{d}^*)$. Here $\boldsymbol{d}^* \in \mathcal{D}$ is independent from $\boldsymbol{z}$ and defined as*

$$\begin{aligned}
d_{\bar{k}}^* &= \frac{\bar{k}!(N-\bar{k}-1)!}{(N-1)!}(1-p)\left(\bar{k}+1-(N-1)p(1-\rho)\right) \\
d_{\bar{k}+1}^* &= \frac{(\bar{k}+1)!(N-\bar{k}-2)!}{(N-1)!}(1-p)\left(-\bar{k}+(N-1)p(1-\rho)\right) \\
d_N^* &= \frac{\bar{k}^2+(1-2Np)\bar{k}+N(N-1)(p^2+\rho p(1-p))}{(N-\bar{k})(N-\bar{k}-1)} \\
d_k^* &= 0 \qquad \forall k \in \{0,1,...,N\}\backslash\{\bar{k},\bar{k}+1,N\}
\end{aligned} \tag{8}$$

where $\bar{k} = \lfloor (N-1)p(1-\rho) \rfloor$, and we define $0! = 1$ by convention.

Lemma 2 provides a closed-form expression for the worst-case distribution, which assigns non-zero probabilities only to three specific scenarios: a) a group of $\bar{k}$ subjects being infected, b) a group of $\bar{k}+1$ subjects being infected, and c) all subjects being infected. Note that both $\bar{k}$ and the worst-case distribution $d_{\bar{k}}^*, d_{\bar{k}+1}^*, d_N^*$ are exclusively determined by the parameters N, p, ρ, but remain independent of the specific partition that we ultimately select for the pooled tests. As will be demonstrated in the following result, this property substantially simplifies the computational effort needed to identify an optimal solution.

Theorem 1 *Problem (5) has an optimal solution $\boldsymbol{z}^*$ which has only one non-zero component.*

Theorem 1 indicates that there exists an optimal partition such that all groups in it are the same size. That is, to achieve the lowest possible overall testing cost in the worst-case scenario, the ideal partition strategy involves groups of a uniform size n^*, indicating that the inherent homogeneity of subjects translates into the formation of testing groups. As this might seem intuitive, interestingly, we will show that this uniformity in optimal testing groups persists in a non-homogeneous population with multiple intra-correlated but inter-independent clusters as well. The detailed analysis will be presented in Section 5.

Remark 2 *The results established above can also be interpreted through a game-theoretic lens. We could show that the game between a test designer and an adversary choosing an infection distribution has a Nash equilibrium, and the following equality holds.*

$$\min_{\boldsymbol{z}\in\mathcal{Z}} \max_{\boldsymbol{d}\in\mathcal{D}} \sum_{n=1}^{\Gamma} n z_n h(n, \boldsymbol{d}) = \max_{\boldsymbol{d}\in\mathcal{D}} \min_{\boldsymbol{z}\in\mathcal{Z}} \sum_{n=1}^{\Gamma} n z_n h(n, \boldsymbol{d}). \tag{9}$$

For completeness, a rigorous proof of Equation (9) *is shown in Appendix EC.C.5. Dividing both sides of Equation* (9) *by* N *reveals that the game is equivalent to*

$$\min_{\boldsymbol{z}\in\mathcal{Z}}\max_{\boldsymbol{d}\in\mathcal{D}}\sum_{n=1}^{\Gamma}\frac{nz_n}{N}h(n,\boldsymbol{d})=\max_{\boldsymbol{d}\in\mathcal{D}}\min_{\boldsymbol{z}\in\mathcal{Z}}\sum_{n=1}^{\Gamma}\frac{nz_n}{N}h(n,\boldsymbol{d})=\max_{\boldsymbol{d}\in\mathcal{D}}\min_{\boldsymbol{w}\in\Delta}\sum_{n=1}^{\Gamma}w_n h(n,\boldsymbol{d})=\max_{\boldsymbol{d}\in\mathcal{D}}\ h(n^*(\boldsymbol{d}),\boldsymbol{d}), \tag{10}$$

where we define $\Delta=\{\boldsymbol{w}\in[0,1]^{\Gamma}:\sum_{n=1}^{\Gamma}w_n=1\}$ *and let* $n^*(\boldsymbol{d})\in\arg\min_{n\in[\Gamma]}h(n,\boldsymbol{d})$ *for any given* $\boldsymbol{d}$*. Equation* (10) *suggests that there is an optimal* $\boldsymbol{w}^*=\boldsymbol{e}_{n^*}$ *and a dominant strategy* $\boldsymbol{d}^*\in\mathcal{D}$ *so that a Nash equilibrium (i.e., a saddle point) can be found in* $(\boldsymbol{w}^*,\boldsymbol{d}^*)$*. In other words, for an optimal* $\boldsymbol{z}$*, we have* $z_{n^*}=N/n^*$ *and* $z_i=0$ *for all* $i\neq n^*$ *for certain* $n^*\in[\Gamma]$*.*

4.3. Optimal Group Size

With the optimality of the equal sized grouping structure, we next investigate the optimal group size for the pooled tests. Recall that the worst-case $\boldsymbol{d}^*$ is independent with any grouping strategy, and there are at most three non-zero entries in a worst-case $\boldsymbol{d}^*$. Using these properties, we first provide a further simplified reformulation of Problem (5), followed by demonstrating that this reformulation finds us an optimal group size with a time complexity of $O(1)$.

Proposition 2 *The optimal group size* n^* *can be derived by solving the following problem*

$$\min_{n\in[\Gamma]}h(n,\boldsymbol{d}^*), \tag{11}$$

with the overall testing cost $NC_0+Nh(n^*,\boldsymbol{d}^*)$*, where function* $h(n,\boldsymbol{d}^*)$ *can be reformulated as follows,*

$$h(n,\boldsymbol{d}^*)=\begin{cases} C_1 & \text{if } n=1,\\ \frac{C_2}{n}-C_3(1-p)\left(\bar{k}+1-(N-1)p(1-\rho)\right)\prod_{\kappa=1}^{n-1}\frac{N-\bar{k}-\kappa}{N-\kappa} & \\ \qquad -C_3(1-p)\left(-\bar{k}+(N-1)p(1-\rho)\right)\prod_{\kappa=1}^{n-1}\frac{N-1-\bar{k}-\kappa}{N-\kappa} & \text{if } 2\le n\le N-\bar{k}-1,\\ \frac{C_2}{n}-C_3(1-p)\left(-\bar{k}+(N-1)p(1-\rho)\right)\prod_{\kappa=1}^{n-1}\frac{N-1-\bar{k}-\kappa}{N-\kappa} & \text{if } n=N-\bar{k},\\ \frac{C_2}{n} & \text{if } n\ge N-\bar{k}+1. \end{cases} \tag{12}$$

Given that Γ is a relatively small fixed number, Equation (11) suggests that the optimal group size can be determined simply by enumerating $h(n,\boldsymbol{d}^*)$ for all $n\in[\Gamma]$. In particular, although the evaluation of $h(n,\boldsymbol{d})$, as defined in (6), involves factorials of large numbers and hence is computationally challenging, the neat structure of $\boldsymbol{d}^*$ renders the calculation of $h(n,\boldsymbol{d}^*)$ quite straightforward. In the new formulation of $h(n,\boldsymbol{d}^*)$ in Proposition 2, the number of terms in the factorial product is now independent of N. As a result, the reformulation reduces the time complexity of evaluating $h(n,\boldsymbol{d}^*)$ from $O(N)$ to $O(1)$, allowing efficient computation when N becomes large.

Remark 3 *We have efficiently identified the optimal grouping strategy through our derivation in Sections 4.2 and 4.3. It is worth mentioning that, while our results are established in a DRO model setup, they can be applied to stochastic setups as well, where the population-level joint distribution is assumed to be fully known. Firstly, the equal sized property of an optimal partition transfers from DRO to stochastic setups. To see this, notice that Theorem 1 is essentially derived from* $\boldsymbol{w}^* = \boldsymbol{e}_{n^*}$ *of Equation* (10)*. The fact that this equality holds independent of the choice of* $\boldsymbol{d}$ *enables us to consider any arbitrary* $\boldsymbol{d}^{stoc} \in \mathcal{D}$*. Here* $\boldsymbol{d}^{stoc}$ *represents a feasible joint distribution with exchangeability, rate* p *and correlation* ρ*. Consequently, our result validates the practice of uniform group sizing. As a result, we lend a theoretical support to the substantial body of existing literature and practices that have largely based on empirical observations to adopt the equal sized grouping strategies.*

Secondly, our approach to finding the best group size applies to a stochastic counterpart model by replacing $h(n, \boldsymbol{d}^*)$ *with* $h(n, \boldsymbol{d}^{stoc})$ *in formulation* (11)*. Nevertheless, without the specific structure of* $\boldsymbol{d}^*$ *from Lemma 2,* $h(n, \boldsymbol{d}^{stoc})$ *must be calculated using Equation* (6) *rather than Equation* (12)*, which necessitates* $O(N)$ *time. While* $O(N)$ *time may appear manageable, the practicality of this approach can be significantly hindered by the computational challenges associated with large combinatorial numbers for a large* N*. Specifically, the calculation based on Equation (12) enables our robust model to outperform the stochastic one in the aspect of scalability: solving stochastic models involves calculations with complexity* $N!$*, while our robust model only requires* $\Gamma!$*. In other words, the stochastic model needs to solve a large-scale non-linear optimization problem, and remains tractable only for small* N *in practice. We will detail the computational discussion in Section 6.1.*

Theorem 1 holds strictly valid when N/n^* is an integer. This is a commonly seen assumption in the group testing literature (e.g. Aprahamian et al. 2020a, Basso et al. 2022). Considering that the population size N is typically much larger than Γ (as explained in Section 3.1), this condition can be assumed to hold with negligible loss of generality. Specifically, when $\Gamma \ll N$, even if it is not a multiple of n^*, the majority of the groups could still be of size n^* with potentially only one smaller group. Therefore, overall performance is predominantly influenced by the groups of size n^*, making it reasonable to overlook the smaller group. Indeed, we can theoretically bound the relative optimality gap of relaxing this assumption by $O(1/N)$, within which the constant coefficient is relatively small for a wide range of reasonable values of p, ρ. The result is formally presented as follows.

Suppose that N is not a multiple of the optimal group size n^*. We hereby define a partition $\boldsymbol{z}^0$, where $\boldsymbol{z}^0$ only have two non-zero elements: $z^0_{n^*} = \lfloor \frac{N}{n^*} \rfloor$ and $z^0_{N - \lfloor \frac{N}{n^*} \rfloor n^*} = 1$. Under partition $\boldsymbol{z}^0$, we assign subjects into $\lfloor \frac{N}{n^*} \rfloor$ groups with size n^* and one group with size $N - \lfloor \frac{N}{n^*} \rfloor n^*$. Then, we introduce the following proposition to compare the performance of $\boldsymbol{z}^0$ and the true optimal

partition $\boldsymbol{z}^*$, which is defined as the optimal solution to Problem (5). For convenience, we denote the objective values of Problem (5) under solution $\boldsymbol{z}^0$ and $\boldsymbol{z}^*$ as f^0 and f^*, respectively.

Proposition 3 *The objective relative ratio, $\frac{f^0-f^*}{f^*}$, is bounded by $C\frac{\Gamma}{N}$, where C is a constant as*

$$C = \frac{\max\{C_1, C_2/2\}\Gamma - C_2 + C_3\Gamma(1-p-p(1-p)(1-\rho))}{C_0\Gamma + C_2 - C_3\Gamma(1-p-p(1-p)(1-\rho))},$$

where C_0, C_1, C_2, C_3 are defined in Equation (7).

Proposition 3 shows that implementing $\boldsymbol{z}^0$ achieves a satisfactory performance level, where the gap to the optimal value is bounded by a scalar multiplication of Γ/N. Typically, the scalar C is a small value independent with N. We launch numerical tests with randomly generated $\lambda_1, \lambda_2, \beta_n, \beta_p, p, \rho$ to check the value of constant C. After $10,000$ numerical tests under the condition that $\beta_n \leq 0.2, \beta_p \leq 0.1$ and $p \leq 0.2$, the mean value of the constant C is 1.2540. Moreover, since population size N is a large number, while group size limit Γ is a smaller number to avoid dilution effect, e.g., $\Gamma = 10$ in Verdun et al. (2021) under the context of COVID-19, the bound derived in Proposition 3 is acceptable.

4.4. Sensitivity Analysis

Having presented an efficient approach to finding the optimal grouping strategy, in this section, we investigate how the optimal group size changes as a function of other parameters. We will first focus on β_n, β_p and λ_1, λ_2, and then discuss the sensitivity with respect to p, ρ.

Theorem 2 *The optimal group size n^* has the following properties:*

(i) When $\lambda_1 = 1$ and $\lambda_2 = 0$, $n^* = 1$; when $\lambda_1 = 0$ and $\lambda_2 = 1$, $n^* = 2$;

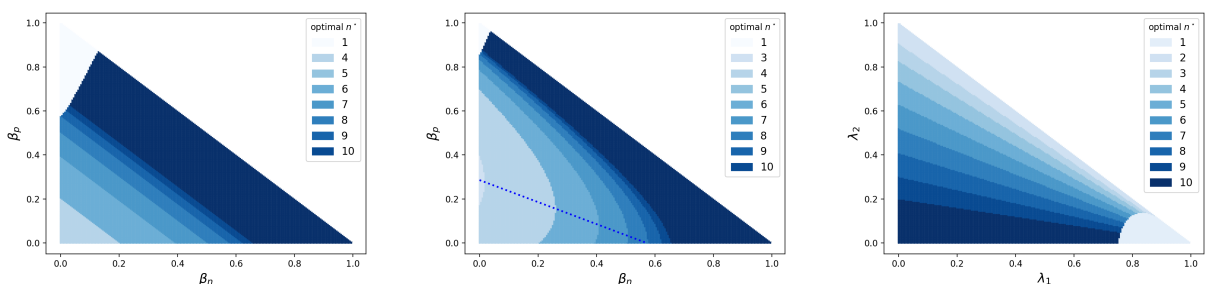
(ii) When $\lambda_2 = 0$, there exists $\underline{\lambda}_1 \in [0,1]$ such that $n^* = 1$ for any $\lambda_1 \geq \underline{\lambda}_1$ and n^* increases in $\beta_n + \beta_p$ for $\lambda_1 < \underline{\lambda}_1$;

(iii) When $n^* \geq 2$, we have a) n^* increases in β_p if and only if $\lambda_2 = 0$ or $\beta_p \geq \frac{1-\beta_n}{2} - \frac{1-\lambda_1-\lambda_2}{2\lambda_2}$, and b) n^* increases in β_n;

(iv) When $n^* \geq 2$, for any fixed (β_n, β_p), n^* is dependent on the ratio $\frac{\lambda_2}{1-\lambda_1}$ rather than on the individual values of (λ_1, λ_2) jointly.

Theorem 2 sheds light on how optimal testing strategies shift based on the prioritization of false negatives, false positives and testing costs. When policymakers in universal screening are risk-averse and prioritize diagnostic accuracy (this is commonly seen when managing highly contagious diseases in vulnerable populations), Statement (i) dictates that the optimal testing strategy is either individual or paired testing. On the other hand, Statement (ii) shows that when minimizing false positives is less of a priority, larger group sizes are more favorable for optimizing testing

resources. In real-world testing scenarios, this often applies during resource-constrained phases of pandemics, where the test practitioners tend to emphasize cost-efficiency over absolute accuracy. More precisely, Statement (ii) shows a monotonic increase in the optimal pool size when test accuracy declines (within acceptable limits). This relationship is visually represented in Figure 2(a)—apart from the northwest corner where individual testing is optimal, the optimal group size increases with $\beta_n + \beta_p$. Of the two types of misclassification, false negatives and false positives arise at different stages in group testing—false positives occur only during second-stage individual tests, while false negatives can occur at both the pooled and individual stages. This asymmetry influences testing strategies differently: while the cost of false negatives driven by increased β_n is consistently offset by the benefits of reduced test volume through larger pools, Statement (iii) reveals that false positives (via β_p) can erode these gains. The two opposing effects converge at $\beta_p = \frac{1-\beta_n}{2} - \frac{1-\lambda_1-\lambda_2}{\lambda_2}$, as illustrated by the dotted line in Figure 2(b). Statement (iv) further justifies this trade-off by demonstrating that the influence of (λ_1, λ_2) on the optimal testing strategy is mediated only by $\lambda_2/(1-\lambda_1)$. This ratio effectively represents the balance between false positive costs (weighed by λ_2) and the test volume (weighed by $1-\lambda_1-\lambda_2$), which implies that false positives play a more decisive role in determining the optimal testing strategy.

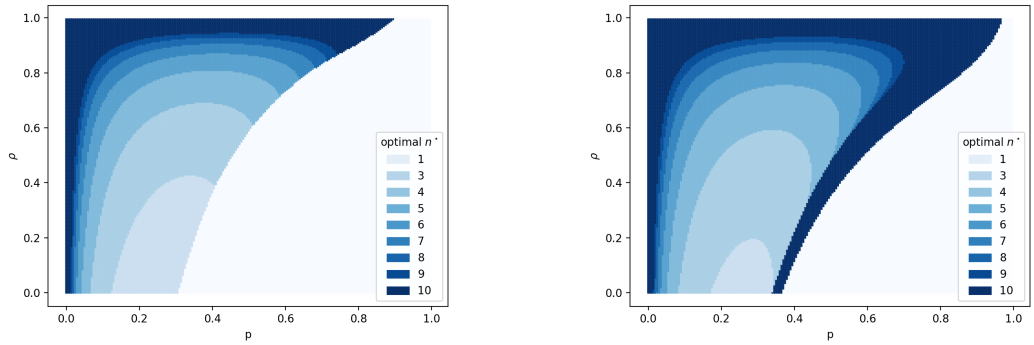


(a) $\lambda_1 = 0.5, \lambda_2 = 0$ (b) $\lambda_1 = 0, \lambda_2 = 0.7$ (c) $\beta_n = 0.1, \beta_p = 0.6$

Figure 2 **The change of optimal group size with respect to β_n, β_p (subfigures (a), (b)) and λ_1, λ_2 (subfigure (c)) when $p = 0.25, \rho = 0.5$ and $N = 15,000$, $\Gamma = 10$.**

Next, we examine the optimal group size as a function of p and ρ in Figure 3. As demonstrated by the figure, no monotonicity remains true, and there could be sudden shifts in the optimal value as p, ρ change, mainly due to (1) a discontinuity in the cost function when switching between individual and pooled testing, and (2) the non-quasi-convexity of the cost function of pooled testing. Our first observation is that, in most cases, the optimal group size increases with correlation ρ. Arguably, the main mechanism behind group testing's efficiency is the likelihood of testing an all-negative pool, and this probability persists more effectively in larger groups when correlation is high. This increased efficiency typically outweighs the potential rise in false positives and leads

to larger optimal pool sizes. However, there are exceptions where the trade-off reverses (e.g. when $p > 0.4$ in Figure 3(b)). Secondly, as long as pooled testing ($n > 2$) is favored over individual testing ($n = 1$), the optimal group size exhibits a decrease-then-increase pattern in p. This is driven by two competing forces—while increasing pool size reduces the chance of all-negative pools, which leads to more second-stage tests, it also lowers the total number of groups, which reduces overall test volume. The increased second-stage testing dominates at low p, and the reduction in group count takes precedence at high p. Lastly, pooled testing is ineffective when p is high and ρ is weak. Under high infection rates and low correlation, it is unlikely that a negative result is declared even for the smallest pool.



(a) $\beta_n = 0, \lambda_1 = 0, \beta_p = 0, \lambda_2 = 0$ (b) $\beta_n = 0.1, \lambda_1 = 0.2, \beta_p = 0.05, \lambda_2 = 0.1$

Figure 3 **The change of optimal group size with respect to p, ρ.**

Remark 4 *We acknowledge that imperfect information on p and ρ may occur in practice. To address this concern, we introduce a robustness analysis to discuss the influence of misspecifying the parameters p and ρ in Appendix EC.B.1. Our experiments demonstrate that although the optimal group size is sensitive on p and ρ as Figure 3 shows, a slight deviation from the true value of p or ρ does not result in a variation in the overall testing cost. However, if the misspecification exceeds a predetermined safe interval, the corresponding total testing cost will experience an increase. This observation serves as another rationale for considering p and ρ: it would be meaningless to consider p and ρ if their values do not affect the objective.*

5. Optimal Grouping Strategy for Multiple Clusters

In the previous sections, we have focused on cases where all subjects in the population are correlated with each other. This can be unrealistic however, especially when the population is large. In this

section, we relax this assumption and consider populations that include multiple intra-correlated but inter-independent clusters, i.e., correlation only exists between subjects in the same cluster. We will first formulate the model and provide an efficient solution method building on the results from Section 4. Subsequently, we will discuss the dependence of the optimal grouping strategy on infection rates, correlations and inaccurate tests.

5.1. Optimality of Consistent and Balanced Group Compositions

We consider a population that can be separated into M clusters, each of size N. Similar to the single-cluster case, subjects in each cluster are interchangeable, with prevalence rate p and Pearson correlation coefficient ρ. Here we assume different clusters share the same values of p, ρ and N for notational brevity. As the analysis unfolds, we will demonstrate the applicability of our results when these parameters differ across clusters. Subjects from different clusters are assumed to be independent. We denote by $\tilde{\xi}_{m,n}$ the Bernoulli random binary variable for any $m \in [M]$ and $n \in [N]$ to indicate the random infection status of the n-th subject from cluster m, and define the realization of $\tilde{\xi}_{m,n}$ by $\xi_{m,n}$. The joint distribution of the random matrix $(\tilde{\xi}_{m,n})_{m\in[M],n\in[N]}$ is denoted by $\mathbb{P}^{\mathcal{M}}$, where we use the superscript $\mathcal{M}$ to differentiate this joint distribution from the one defined in the single-cluster model, as introduced in Section 3. Within each correlated cluster, we denote by $\tilde{\boldsymbol{\xi}}^m = (\tilde{\xi}_{m,1}, \ldots, \tilde{\xi}_{m,N})$ the Bernoulli random binary identification vector of cluster m with realization $\boldsymbol{\xi}^m$ and joint distribution $\mathbb{P}^m$, for any $m \in [M]$. Mathematically, according to the independence assumption, we have $\mathbb{P}^{\mathcal{M}} = \mathbb{P}^1 \times \cdots \times \mathbb{P}^M$.

Similar to Equation (2), the joint distribution is assumed to be in the following ambiguity set,

$$\mathcal{P}^{\mathcal{M}} = \left\{ \mathbb{P}^{\mathcal{M}} \;\middle|\; \begin{array}{ll} \mathbb{P}^m\left(\tilde{\xi}_{m,n} = 1\right) = p & \forall n \in [N], m \in [M], \\ \mathbb{P}^m\left(\tilde{\xi}_{m,n_1} = 1, \tilde{\xi}_{m,n_2} = 1\right) = p^2 + \rho p(1-p) & \forall n_1, n_2 \in [N], n_1 \neq n_2, m \in [M], \\ \mathbb{P}^m\left(\tilde{\boldsymbol{\xi}}^m = P_\pi \boldsymbol{\xi}^m\right) = \mathbb{P}^m\left(\tilde{\boldsymbol{\xi}}^m = \boldsymbol{\xi}^m\right) & \forall P_\pi, \boldsymbol{\xi}^m \in \{0,1\}^N, m \in [M], \\ \mathbb{P}^{\mathcal{M}}\left(\tilde{\boldsymbol{\xi}}^1 = \boldsymbol{\xi}^1, \ldots, \tilde{\boldsymbol{\xi}}^M = \boldsymbol{\xi}^M\right) = \prod_{m=1}^M \mathbb{P}^m\left(\tilde{\boldsymbol{\xi}}^m = \boldsymbol{\xi}^m\right) & \forall \boldsymbol{\xi}^1, \ldots, \boldsymbol{\xi}^M \in \{0,1\}^N. \end{array} \right\},$$

where the first three constraints follow from the definition of $\mathcal{P}$ in Equation (2), describing the distributional information for each cluster. The last constraint is to characterize the independence between subjects from different clusters.

Next, we define the decision variables, i.e., the partition of the whole population. For this population including MN subjects, there are at most MN groups in a feasible partition, which occurs when assigning each subject with an individual group. We use binary variables $\boldsymbol{w} = (w_{m,n,l})_{m\in[M],n\in[N],l\in[MN]}$ to indicate whether to assign the subject (m, n) to the l-th testing group,

for any $m \in [M], n \in [N], l \in [MN]$. Thus, to describe a feasible partition, the decision vector $\boldsymbol{w}$ belongs to the following feasible region,

$$\mathcal{W} = \left\{ \boldsymbol{w} \;\middle|\; \begin{array}{ll} \sum_{l=1}^{MN} w_{m,n,l} = 1 & \forall m \in [M], n \in [N] \\ \sum_{m=1}^{M} \sum_{n=1}^{N} w_{m,n,l} \leq \Gamma & \forall l \in [MN] \\ w_{m,n,l} \in \{0,1\} & \forall m \in [M], n \in [N], l \in [MN] \end{array} \right\},$$

where the first constraint implies every subject must be assigned into exactly one group, and the second constraint imposes the upper bound Γ of group size. Consequently, the DRO partition problem with partial correlation can be written as follows,

$$\min_{\boldsymbol{w} \in \mathcal{W}} \sup_{\mathbb{P}^{\mathcal{M}} \in \mathcal{P}^{\mathcal{M}}} \lambda_1 \mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[FN(\boldsymbol{w}, \tilde{\boldsymbol{\xi}})] + \lambda_2 \mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[FP(\boldsymbol{w}, \tilde{\boldsymbol{\xi}})] + (1 - \lambda_1 - \lambda_2) \mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[T(\boldsymbol{w}, \tilde{\boldsymbol{\xi}})], \tag{13}$$

where $\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[FN(\boldsymbol{w}, \tilde{\boldsymbol{\xi}})]$, $\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[FP(\boldsymbol{w}, \tilde{\boldsymbol{\xi}})]$ and $\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[T(\boldsymbol{w}, \tilde{\boldsymbol{\xi}})]$ respectively denote the expected number of false positives, false negatives, and total tests required under partition $\boldsymbol{w}$. Similar to the single-cluster cases, for Problem (13), we will derive the population-level joint worst-case distribution and reformulate the original partition decision $\boldsymbol{w}$ to the decision $\boldsymbol{v}$, where $\boldsymbol{v}$ represents the number of subjects assigned to each group. We construct the worst-case distribution and reformulation of Problem (13) by the following Proposition 4.

Proposition 4 *The worst-case distribution to Problem (13), denoted by $\mathbb{P}^{\mathcal{M}*}$, satisfies that $\mathbb{P}^{\mathcal{M}*}\left(\tilde{\boldsymbol{\xi}}^m = \boldsymbol{\xi}\right) = d^*_{\mathbf{1}^\top \boldsymbol{\xi}}$ for any cluster $m \in [M]$, where $\boldsymbol{d}^* = (d^*_0, d^*_1, \ldots, d^*_N)$ is defined in Lemma 2. Moreover, Problem (13) is equivalent to*

$$\min_{\boldsymbol{v} \in \mathcal{V}} \sum_{l=1}^{MN} g_l(\boldsymbol{v}), \tag{14}$$

where

$$\mathcal{V} = \left\{ \boldsymbol{v} \in \mathbb{Z}_+^{M \times MN} \;\middle|\; \begin{array}{ll} \sum_{l=1}^{MN} v_{m,l} = N & \forall m \in [M], \\ \sum_{m=1}^{M} v_{m,l} \leq \Gamma & \forall l \in [MN], \\ v_{m,l} \in \{0, 1, \ldots, \Gamma\} & \forall m \in [M], l \in [MN] \end{array} \right\},$$

$$g_l(\boldsymbol{v}) = \begin{cases} 0, & \text{if } \sum_{m=1}^{M} v_{m,l} = 0, \\ C_0 + C_1, & \text{if } \sum_{m=1}^{M} v_{m,l} = 1, \\ C_0 \sum_{m=1}^{M} v_{m,l} + C_2 - C_3 \left(\sum_{m=1}^{M} v_{m,l} \right) \prod_{m=1}^{M} \left(\sum_{k=0}^{N - v_{m,l}} \binom{N - v_{m,l}}{k} d^*_k \right), & \text{otherwise.} \end{cases}$$

and C_0, C_1, C_2, C_3 are constants defined in Equation (7).

Proposition 4 establishes the worst-case joint distribution of Problem (13) and simplifies the original partitioning decision $\boldsymbol{w}$ into the number of subjects assigned to each group, $\boldsymbol{v}$. The simplified decision $\boldsymbol{v} \in \mathcal{V}$, defined by $v_{m,l} = \sum_{n \in [N]} w_{m,n,l}$, stands for the number of subjects from cluster

m that are represented in the l-th testing group. $\boldsymbol{v}$ is sufficient to fully capture any information from $\boldsymbol{w}$, following a similar logic as the pairwise-correlated case. The worst-case joint distribution for the entire population aligns with the product of the worst-case distributions for each individual cluster, i.e., statistical independence between different clusters is maintained over the inner maximization process. Notably, Proposition 4 can be extended to cases where different clusters have varying prevalence p, correlation ρ, and size N—the cluster-wise worst-case joint distributions defined by Lemma 2 remain applicable because of the statistical independence. We formalize this in Appendix EC.A.3.

Proposition 4 significantly reduces the complexity of Problem (13). Nevertheless, the highly complicated landscape of $\mathcal{V}$ still makes it impractical to find the optimal partition through direct enumeration of all feasible $\boldsymbol{v}$. To navigate through this, our next result captures a structural property of the optimal partition $\boldsymbol{v}^*$, showing that all testing groups share a consistent structure of composition defined by

$$\boldsymbol{y}^* \in \arg\min_{\boldsymbol{y}\in\mathcal{Y}} \; f(\boldsymbol{y}), \tag{15}$$

where $\mathcal{Y} = \left\{\boldsymbol{y} \in \mathbb{Z}_+^M : 1 \le \sum_{m=1}^M y_m \le \Gamma\right\}$,

$$f(\boldsymbol{y}) = \begin{cases} C_0 + C_1, & \text{if } \sum_{m=1}^M y_m = 1, \\ C_0 + \frac{C_2}{\sum_{m=1}^M y_m} - C_3 \prod_{m=1}^M \sum_{k=0}^{N-y_m} \binom{N-y_m}{k} d_k^*, & \text{if } 2 \le \sum_{m=1}^M y_m \le \Gamma. \end{cases}$$

The following theorem formally establishes this property and constructs an optimal solution $\boldsymbol{v}^*$ based on $\boldsymbol{y}^*$.

Theorem 3 *A partition decision $\boldsymbol{v}^* \in \mathcal{V}$ is optimal to Problem* (14) *if for any $l \in [MN]$, the vector $\boldsymbol{v}_l^* = (v_{1,l}^*, v_{2,l}^*, \ldots, v_{M,l}^*)$ is either $\mathbf{0}$ or a permutation of $\boldsymbol{y}^*$. Specifically, to find an optimal partition $\boldsymbol{v}^* \in \mathcal{V}$, we can obtain $\boldsymbol{k} = (k_1, \ldots, k_{M!})$ as any solution of the equation*

$$k_1 P_\pi^1 \boldsymbol{y}^* + \cdots + k_{M!} P_\pi^{M!} \boldsymbol{y}^* = N\mathbf{1} \tag{16}$$

where $P_\pi^1, \ldots, P_\pi^{M!}$ are all the $M!$ distinct $M \times M$ permutation matrices in an arbitrary order. Then,

$$(v_{1,l}^*, v_{2,l}^*, \ldots, v_{M,l}^*) = \begin{cases} P_\pi^i \boldsymbol{y}^* & \text{if } \exists i \in [M!] \text{ such that } l \in \{\sum_{s=1}^{i-1} k_s + 1, \ldots, \sum_{s=1}^{i} k_s\} \\ \mathbf{0} & \text{if } l > \mathbf{1}^\top \boldsymbol{k} \end{cases} \qquad \forall l \in [MN]$$

is an optimal solution to Problem (14)*. Moreover, the optimal total testing cost is $MNf(\boldsymbol{y}^*)$.*

With Equation (16) specifies the partitioning constraint that a valid grouping strategy must satisfy (i.e., allocating each subject to exactly one testing group), Theorem 3 states that the cluster

compositions of an optimal partition are permutations of a template structure $\boldsymbol{y}^*$. For instance, consider a population with 3 clusters, each consisting of 5 subjects (i.e., $M = 3, N = 5$), and suppose that the optimal group structure is $\boldsymbol{y}^* = (3, 2, 0)$. Representing subjects from cluster 1, 2 and 3 respectively by circles, Figure 4 depicts that the optimal partition includes 3 groups whose compositions form a rotation of $(3, 2, 0)$. Computationally, Theorem 3 simplifies the decision by reducing it from an entire partition to a unified pattern of group composition. Conceptually, it reinforces the idea of broader homogeneity across testing groups as the population extends beyond a single cluster. Building on this, our next result extends the notion of homogeneity from between testing groups to within them.

Figure 4 **Illustration of an optimal partition when $\boldsymbol{y}^* = (3, 2, 0)$. While any testing group with composition being a permutation of $\boldsymbol{y}^*$ is admissible in an optimal partition, here, we illustrate the partition formed by $(3, 2, 0)$, $(2, 0, 3)$ and $(0, 3, 2)$. Note that a rotation of any permutation of $\boldsymbol{y}^*$, such as $(0, 2, 3)$, $(2, 3, 0)$ and $(3, 0, 2)$, is equivalently effective.**

Theorem 4 *For any two non-zero elements $y^*_{m_1}, y^*_{m_2}$ in an optimal structure $\boldsymbol{y}^*$, we have $|y^*_{m_1} - y^*_{m_2}| \leq 1$.*

Theorem 4 implies that in an optimal partition with mixed pooling, each pool should evenly represent the included clusters, with subject numbers varying by at most one. In other words, while not every cluster needs to be present in an optimal testing pool, those included should be represented in a balanced manner. As such, the principle of homogeneity is not only preserved across different testing groups in an optimal partition, but also distinctly manifests within each individual testing group. Remarkably, as the feasible region for an optimal $\boldsymbol{y}^*$ becomes more constrained, Theorem 4 further reduces the computational complexity of finding $\boldsymbol{y}^*$—for a given group size of n, there are at most $\min\{n, M\}$ candidate $\boldsymbol{y}$'s that conform to the equal-split structure. For instance, when creating a group of size $n = 10$ for a population of $M = 4$ clusters, permissible compositions might include $(0, 0, 0, 10)$, $(0, 0, 5, 5)$, $(0, 3, 3, 4)$, $(2, 2, 3, 3)$ and their permutations; because each $\boldsymbol{y}$ effectively represents all its permutations in forming a testing group, it suffices to consider only 4 choices of $\boldsymbol{y}$ for each group size. Coupled with the predetermined maximum group size Γ, it suffices to enumerate over $\sum_{n=1}^{\Gamma} \min\{n, M\}$ candidates in order to derive the optimal $\boldsymbol{y}^*$, and the overall time complexity for determining the optimal partition is reduced to $O(M)$.

Theorems 3 and 4 together provide theoretical support for mixing samples from different clusters in group testing, which marks a notable addition to the current literature on correlated group

testing, which has primarily focused on pooling within the same cluster (see, e.g., Lendle et al. 2012, Augenblick et al. 2020, Zhang and Heath 2023).

Remark 5 *When N is not a common multiple of $\{1, \ldots, \Gamma\}$, a the similar deviation as Proposition 3 yields that the approximation error in this case is still bounded by a scalar multiplication of $\frac{\Gamma}{N}$. The detailed proposition and proof can be found in Appendix EC.A.2.*

5.2. Sensitivity of the Optimal Group Composition

In this section, we discuss how the optimal group composition $\boldsymbol{y}^*$ can be shaped by variations in test functionalities (β_n, β_p), weights placed on false testing results (λ_1, λ_2), and infection-related parameters (p, ρ).

We first investigate the dependence of an optimal group composition on weight parameters (λ_1, λ_2) and accuracy parameters (β_n, β_p). For a given infection characterization, i.e., for fixed p and ρ, we establish this sensitivity result by the following proposition.

Proposition 5 *For any given N and (p, ρ), $\boldsymbol{y}^*$ has the following properties with respect to $\beta_n, \beta_p, \lambda_1, \lambda_2$:*

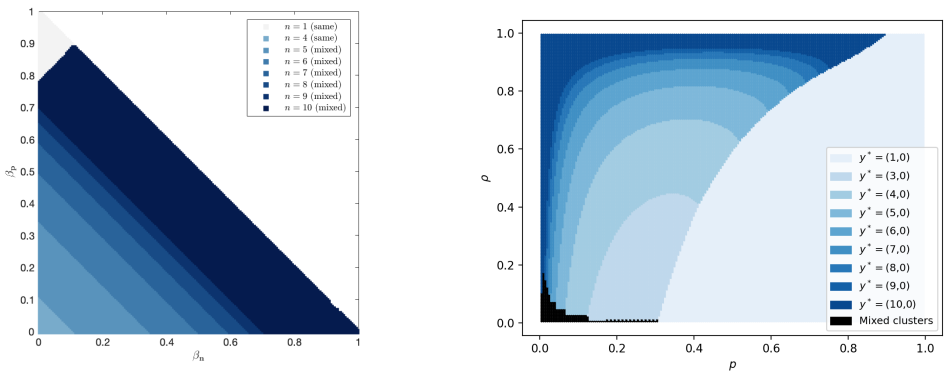
(i) In an optimal partition, given a pool size n, a composition structure $\boldsymbol{y}^*$ can be uniquely determined and independent of the specific values of $\beta_n, \beta_p, \lambda_1, \lambda_2$;

(ii) If $\boldsymbol{y}^*$ indicates the optimality of same-cluster pooling, i.e., it features only one non-zero component, then the optimal pool size n^* shares all properties as described in Theorem 2.

Statement (i) illustrates that, given an optimal size of pooling, test accuracy parameters do not change the optimal pooling structure, e.g. same- or mixed-cluster strategies. Put another way, misclassifications affect the optimal grouping strategy only through the optimal group size. In Figure 5(a), for each fixed group size, instead of exhibiting a variety of compositions under different $\lambda_1, \lambda_2, \beta_n, \beta_p$, the optimal grouping strategy is either same-cluster pooling or a fixed way of mixing up samples from different pools. We also observe that mixed-pooling strategy tends to be advantageous when test accuracy drops. Statement (ii) discusses the sensitivity of the optimal pool size when same-cluster pooling is preferable. Similar to the intuition of Theorem 2, in the trade-off between test efficiency and accuracy, the optimal same-cluster group size increases in the impact of former and decreases in the impact of latter.

Next, we discuss optimal composition as a function of prevalence rate p and partial correlation ρ. Figure 5(b) considers a two-cluster population, where different values of (p, ρ) are indicated by different shades of blue (same-cluster testing is optimal) and black (mixed-cluster testing is optimal). When p is low and the same-cluster correlation is strong, the pronounced effect of correlation

suggests that the multi-cluster population should be treated as M single-cluster sub-populations, i.e., same-cluster group testing is optimal. Conversely, when p is high and correlation is weak, the dominant influence of the high infection rate overshadows the impact of correlations, making individual testing more favorable. Mixed-cluster testing is optimal primarily when both prevalence and correlation are low. Weak correlation lowers the tendency of pooling from the same group, and low prevalence promotes pooling regardless of their cluster affiliation. As a result, testing groups that "partially bundle" samples from various clusters become advantageous. Demonstrating the effectiveness of mixed-cluster testing (particularly at low prevalence rates and low correlation levels, which are common in real-world outbreaks), we note that this insight adds an important dimension to the existing literature that predominantly advocates pooling correlated individuals (see, for example, Lin et al. (2021) and Zhang and Heath (2023)).



(a) Optimal pool size in β_n, β_p (b) Optimal group composition in p, ρ

Figure 5 **(a) The change of optimal grouping strategy, when $p = 0.08, \rho = 0.04$, $\lambda_1 = 0.6, \lambda_2 = 0$, and $M = 2$, $N = 100$, $\Gamma = 10$. Here $n = i$ (same) and $n = j$ (mixed) means the optimal grouping policy is to perform the same-cluster grouping with size i and a mixed-cluster grouping with pool size j. (b) The change of optimal grouping structure with respect to p, ρ. Lower left corners in both figures represent the notable ranges of (p, ρ) such that mixing up samples from different intra-correlated clusters in each testing group is optimal.**

6. Numerical Studies

In this section, we conduct a two-fold numerical investigation. In Section 6.1, we investigate the trade-offs of adopting our robust optimization framework through several numerical experiments. Our results show that implementing robust optimization presents minor costs in average performances, but offers substantial benefits in stability and tractability. In Section 6.2, we conduct a comprehensive study based on massive testing cases conducted during a COVID-19 ourbreak in Hong Kong. All numerical studies are conducted on a laptop with an M1 chip and 8GB random-access memory.

6.1. Adopting Robust Optimization: Losses and Gains

We examine the robustness/conservativeness of our pooling strategy by comparing our DRO grouping strategy with that solved from our model's stochastic programming counterparts. These stochastic models assume exactly known joint probability distributions that have rate p, correlation ρ, and the exchangeability property (as specified in Section 3.3). As such, any joint distribution adopted by a stochastic counterpart model must also be feasible to our ambiguity set $\mathcal{P}$. Similar to the analysis in Section 4.1, the stochastic model can be reformulated as

$$\min_{\boldsymbol{z}\in\mathcal{Z}} \ \lambda_1 FN_{\boldsymbol{d}^{stoc}}(\boldsymbol{z}) + \lambda_2 FP_{\boldsymbol{d}^{stoc}}(\boldsymbol{z}) + (1-\lambda_1-\lambda_2)T_{\boldsymbol{d}^{stoc}}(\boldsymbol{z}), \tag{17}$$

where $\boldsymbol{d}^{stoc} = (d_k^{stoc})_{k\in\{0,\ldots,N\}}$, and d_k^{stoc} is the probability that an arbitrary selection of k individuals is exactly all the infected ones in the population. $(d_k^{stoc})_{k\in\{0,\ldots,N\}}$ should satisfy the constraints in Equation (3). We follow Lendle et al. (2012) and consider a family of qualified $(d_k^{stoc})_{k\in\{0,\ldots,N\}}$ as

$$d_k^{stoc} = \mathbb{E}[\pi^k(1-\pi)^{N-k}], \quad k\in\{0,\ldots,N\},$$

where π is a random variable with support $[0,1]$, and the expectation is taken over π. In other words, the choice of $\boldsymbol{d}^{stoc}$ is determined by the distribution of π, for which we consider the following three options that have been discussed in the literature.

- $\pi = p$ with probability $1-\rho$, $\pi = 0$ with probability $\rho(1-p)$ and $\pi = 1$ with probability ρp, as discussed in Madsen (1993). In this case, $d_k^{stoc} = (1-p)p^k(1-p)^{N-k} + \rho(1-p)0^k + \rho p 0^{N-k}$.
- $\pi = p + (1-p)\sqrt{\rho}$ with probability p and $\pi = p - p\sqrt{\rho}$ with probability $1-p$, as described in Morel and Neerchal (1997). In this case, $d_k^{stoc} = p(p+(1-p)\sqrt{\rho})^k(1-p-(1-p)\sqrt{\rho})^{N-k} + (1-p)(p-p\sqrt{\rho})^k(1-p+p\sqrt{\rho})^{N-k}$.
- π follows a Beta distribution (α,β) with mean p and variance $\rho p(1-p)$, i.e., parameters $\alpha = (1-\rho)p/\rho$ and $\beta = (1-\rho)(1-p)/\rho$. Such π leads to a Beta-binomial joint distribution. Formally, we have $d_k^{stoc} = \int_0^1 \pi^k(1-\pi)^{N-k}\frac{\pi^{\alpha-1}(1-\pi)^{\beta-1}}{\mathrm{B}(\alpha,\beta)}\mathrm{d}\pi = \frac{\mathrm{B}(k+\alpha,N-k+\beta)}{\mathrm{B}(\alpha,\beta)}$, where $\mathrm{B}(\cdot,\cdot)$ stands for the Beta function.

As introduced in Lendle et al. (2012), all the three joint distributions have mean values p and correlation coefficients ρ, thereby leading to three stochastic models in the form of Problem (17). In the following, we will denote these models separately as <u>Madsen</u>, <u>Morel & Neerchal</u> and <u>Beta-binomial</u>, and use them as benchmarks to evaluate our <u>DRO</u> grouping strategy.

Robustness and stability. We set up our experiments with $p = 0.1, 0.2$, $\rho = 0.25, 0.5, 0.75$ and $\beta_n = \beta_p = 0.05$ or 0.1. We further set the weight parameters to $\lambda_1 = \beta_n, \lambda_2 = \beta_p$. For each combination of parameter values, we solve the optimal grouping strategies from the four models, and test their performances under distribution misspecifications. Specifically, we randomly sample 50

distributions from the ambiguity set and calculate the expected testing costs for each pooling method over these distributions. Given the same information of p and ρ, the evaluation setup mimics real-life scenarios where the true underlying joint distribution can be any arbitrary choice from the ambiguity set. Based on this setup, we report the mean and standard deviation of these 50 expected values for each method.

While the DRO model theoretically guarantees optimal performance only under worst-case distributions, our results indicate that DRO consistently achieves the best performance across 50 evaluated feasible distributions. As shown in Table 2, the DRO pooling strategy yields the lowest per-individual testing cost in most cases, with only three parameter instances showing a performance gap of around 3%. Furthermore, we observe that the performance of our DRO pooling strategy remains stable against distributional changes. Specifically, for all tested parameter combinations, the standard deviation of the DRO model's performance is consistently the lowest, with the stochastic strategies exhibiting variability that is significantly higher. This is consistent with the discussion in the seminal robust optimization paper by Ben-Tal and Nemirovski (1998). In summary, while potentially incurring a minor increase in average testing costs, our DRO pooling strategy offers solid performance, enhanced reliability and strong stability.

Computational tractability. As presented in Sections 4.3 and 5.1, our approach provides an $O(1)$-time solution under the single-cluster correlation structure. In contrast, as discussed in Remark 3, a stochastic counterpart model (that assumes an exactly known joint distribution) has a time complexity of $O(N)$. While this polynomial-time complexity may appear efficient, these models are computationally limited by large combinatorial factors. In our experiments, they are tractable only when $N \leq 1,030$. By comparison, our DRO model can handle values of N that reach at least one million, solving in 0.0001 seconds. In a separate experiment where we test the impact of changing M, the average computation time is around 0.0001 second when $M = 2, N = 1,000$ and 0.0006 second when $M = 10, N = 1,000$. Notably, as M further increases, the computation time remains consistent due to the limit of the maximum group size Γ. In comparison, stochastic models struggle with high dimensionality in the variable $\boldsymbol{v}$ for Problem (14) and become intractable even for a two-cluster case ($M = 2$).

6.2. Capturing Correlation: A Case Study on the Benefits

In this section, we further evaluate the effectiveness of factoring correlated infections for group testing strategy design in a complex real-world setting. From HKSAR Government Press Releases[3], we collect the records of compulsory tests in 81 restricted areas in Hong Kong during the peak

[3] See `https://www.gov.hk/`. As an example, one of the news, as archived in `https://www.info.gov.hk/gia/general/202203/28/P2022032800261.htm`, reports the testing results of the Lei Tim House, Ap Lei Chau Estate in Aberdeen, Hong Kong on March 27, 2022.

Table 2 Model Evaluation on Randomly Drawn Distributions from the Ambiguity Set. Each Cell Shows the Average Cost per Individual (upper) and the Standard Deviation (lower) across All Instances.

(β_n, β_p)	(p, ρ)	**DRO**	Madsen	Morel & Neerchal	Beta-binomial
$(0.05, 0.05)$	$(0.1, 0.25)$	**0.48**	0.48	0.48	0.48
		0.0139	0.0139	0.0199 (+43.2%)	0.0259 (+86.3%)
	$(0.1, 0.5)$	**0.41**	0.41	0.41	0.47 (+14.6%)
		0.0207	0.0207	0.0265 (+28.0%)	0.0689 (+232.9%)
	$(0.1, 0.75)$	**0.33**	0.34 (+3.0%)	0.33	0.38 (+15.2%)
		0.0188	0.0340 (+80.9%)	0.0251 (+33.5%)	0.0529 (+181.4%)
	$(0.2, 0.25)$	**0.66**	0.66	0.66	0.66
		0.0123	0.0123	0.0123	0.0266 (+116.3%)
	$(0.2, 0.5)$	0.58 (+3.6%)	0.66 (+17.9%)	0.56	0.59 (+5.4%)
		0.0125	0.0809 (+547.2%)	0.0269 (+115.2%)	0.0549 (+339.2%)
	$(0.2, 0.75)$	**0.47**	0.53 (+12.8%)	0.47	0.53 (+12.8%)
		0.0222	0.0758 (+241.4%)	0.0325 (+46.4%)	0.0758 (+241.4%)
$(0.1, 0.1)$	$(0.1, 0.25)$	**0.44**	0.44	0.44	0.45 (+2.3%)
		0.0111	0.0111	0.0159 (+43.2%)	0.0253 (+127.9%)
	$(0.1, 0.5)$	0.39 (+2.6%)	0.38	0.38	0.44 (+15.8%)
		0.0165	0.0211 (+27.9%)	0.0256 (+55.2%)	0.0606 (+267.3%)
	$(0.1, 0.75)$	**0.32**	0.33 (+3.1%)	0.32	0.35 (+9.4%)
		0.0150	0.0317 (+111.3%)	0.0224 (+49.3%)	0.0422 (+181.3%)
	$(0.2, 0.25)$	**0.59**	0.59	0.59	0.59
		0.0098	0.0157 (+60.2%)	0.0157 (+60.2%)	0.0212 (+116.3%)
	$(0.2, 0.5)$	**0.51**	0.58 (+13.7%)	0.51	0.53 (+3.9%)
		0.0158	0.0646 (+308.9%)	0.0215 (+36.1%)	0.0471 (+198.1%)
	$(0.2, 0.75)$	0.44 (+2.3%)	0.47 (+9.3%)	0.43	0.47 (+9.3%)
		0.0177	0.0605 (+241.8%)	0.0259 (+46.3%)	0.0605 (+241.8%)

of COVID-19 infection in March and April, 2022. These areas are mainly apartment buildings that have high risk of correlated infection. Our dataset contains the population size and the total infection count for each area. Table 3 provides basic statistics of our dataset, and a full visualization is available in Appendix EC.B.2.1.

Table 3 Statistics of the Dataset

	Minimum	1st Quartile	Median	3rd Quartile	Maximum
Infected Count	0	2	5.5	13	43
Population	110	740.25	1000.5	1660	2210

Using the collected data samples as starting point for each restricted area, we simulate 60-day trajectories by employing an SIR-type epidemiological model with different testing methods. By quantifying the performance differences throughout the disease progression, we aim to elucidate the benefit of our proposed pooling strategy and highlight the importance of incorporating correlation considerations for policymakers.

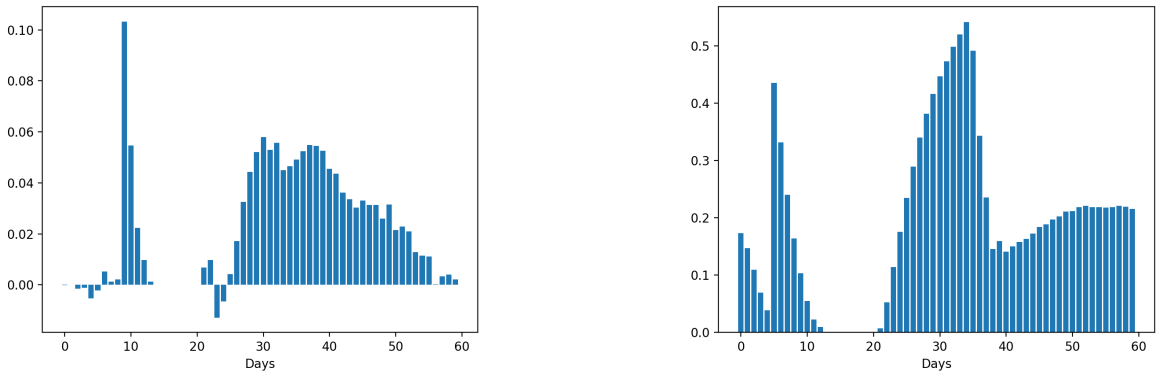
We set up our simulation based on Taghvaei et al. (2020) and Basso et al. (2022). Specifically, for each restricted area and each day, we assume that 20% of the inhabitant exit and re-enter the building. This assumption is corroborated by the fact that none of our studied residential areas was under mandatory isolation. Each individual has three possible states with respect to contagion: susceptible (S), infected (I), and recovered (R). The transitions between states are determined by three random time periods: 1) Upon exposure, a susceptible (S) individual enters an incubation period with length following a lognormal distribution Lognormal$(0.95, 0.46^2)$.[4] 2) The individual then becomes infectious (I) after a period of time uniformly distributed between one-third of the full incubation period and one day less than the full incubation period. 3) The infectious (I) state turns into a recovery state (R) with acquired immunity, after a period that follows a uniform distribution from 7 to 14 days, starting from the end of the incubation period. Infectious individuals could transmit the disease by interaction with susceptibles. Considering the exits and re-enters, there are two main sources of exposure: direct interactions within the community and infections from external encounters. Following Basso et al. (2022), we assume that each individual in the community has a 10% chance of daily contact with others, and if an encounter is between an infected and a susceptible person, there is a 5% risk of transmission per interaction. We posit a 0.1% daily risk of infection outside the studied areas.

The control center conducts a round of all-resident testing on each simulated day.[5] The testing process affects disease propagation in two ways: 1) it directly identifies individuals for quarantine by returning test results to residents, and 2) it provides an estimate for the current propagation status that will be used to plan the next round of testing, thereby indirectly influencing future propagation. The test details are as follows. We start with an outside infection rate of $p = 0.001$ and no correlations $\rho = 0$ as the initial values of p, ρ for the Day-1 testing. For each subsequent day, p, ρ are estimated using the previous day's test results. The average proportion of infections serves as the estimate for p, while ρ, is estimated following a comprehensive review of correlation estimators under similar setups and probabilistic assumptions (Ridout et al. 1999). We adopt two estimators from this review and use their average. Additional details are provided in Appendix EC.B.2.2. All tests are assumed to have a false negative rate $\beta_n = 0.093$ and perfect specificity ($\beta_p = 0$), as supported by Kanji et al. (2021). We set $\lambda_1 = \beta_n, \lambda_2 = \beta_p$ to reflect the inherent test imperfections and help policy makers balance the monetary testing cost and false classifications.

[4] The selection of these parameters follows from Tanaka et al. (2022) that study the distribution of incubation period with the presence of the SARS-CoV-2 Omicron variant. The peak period of Omicron in Hong Kong coincides with the timing of the testing conducted in our study.

[5] Repeated universal testing has been a common practice during COVID-19 pandemic, implemented in multiple countries/regions including Mainland China (Global News 2022), Germany (Financial Times 2020), the United Kingdom (GOV.UK 2020). As our studied areas did not witness repeated universal testing, this simulated study can provide valuable insights in evaluating the counterfactual pros and cons.

We evaluate our proposed grouping strategy against two baselines: the standard independent-based group testing that does not account for correlation across infections, and a heuristic that reflects real-world pooling practices. Specifically, when the infection rate $p < 0.01$, samples are grouped in sets of 5; when $p \in [0, 01, 0.1)$, the group size is 3; and for $p \geq 0.1$, individual tests are adopted. This heuristic is introduced by a local CDC practitioner who led a large-scale COVID-19 testing program. We follow Verdun et al. (2021) and set the maximum testing group size $\Gamma = 10$ for all three methods. For each restrictive area, 20 propagation trajectories are simulated, and the average number of tests per individual across all areas and 20 trajectories is calculated to assess the efficacy of each testing method.



(a) Correlation-based v.s. independent-based testing

(b) Correlation-based testing v.s. real-world practice

Figure 6 **The average per-individual savings in the number of tests with our testing method.**

The resulted performance gaps are depicted in Figure 6. Compared with the independent-based testing method, our approach exhibits a notable reduction in testing costs. As shown in Figure 6(a), the savings in the number of test needed for universal testing is up to 0.1 per individual, suggesting that ignoring correlated infections can lead to pronounced inefficiencies in universal testing. The specific peaks and troughs of this gap are driven by the dynamics of p, ρ associated with each testing method. Notably, while there is no theoretical guarantee for the DRO pooling method to excel in this epidemic setting, our approach consistently outperforms the independent-based method on average, with only two brief intervals where it falls short: Days 3–6 (by about 0.005 test per individual) and Days 24–25 (by about 0.01 test per individual). We relegate the detailed explanations to Appendix EC.B.2.3. Besides the comparison against independent-based pooling strategy, our testing method demonstrates even greater cost savings compared to the heuristic-based COVID-19 testing practices. As shown in Figure 6(b), the reduction in the number of tests per individual can be as large as 0.5.

7. Conclusion

A wealth of practices and lessons have been documented as we navigate our way out of the tough fight against COVID-19, among which an important action taken against the rapid spread is group testing. Recognizing the various shortcomings of traditional group testing schemes, this paper presents an initial attempt to designing an optimal group testing scheme with simultaneous consideration of correlated infections, ambiguous probability distribution and imperfect tests. Specifically, we develop a DRO group testing framework that incorporates information about infection rates and accounts for both single- and multi-cluster scenarios. In the single-cluster case, we justify the optimality of homogeneous testing groups by connecting our analysis to Nash equilibrium principles. This homogeneity is extended to multiple-cluster cases in two ways: within an optimal partition, testing groups maintain a uniform composition of subjects from different clusters; within an optimal testing pool, it should represent each involved cluster equally (with the number of subjects from each differing by at most one). For both scenarios, we provide efficient approaches to determine the optimal grouping strategy and offer managerial insights for test practitioners. We conduct thorough numerical experiments to evaluate the effectiveness, robustness and practicality of our method.

The recurring emergence of various infectious diseases, coupled with their unpredictable nature and resource constraints, presents an ever-increasing challenge in their management while simultaneously presenting an opportunity for operations research to make a critical impact. In view of these, we hope that our study will inspire future research in related areas. For instance, group testing strategies could be explored in dynamic settings that account for disease propagation over time. Additionally, one could consider integrating group testing with other public health initiatives, such as vaccination drives or contact tracing. On a broader scale, it is also valuable to investigate the ethical implications while implementing group testing, such as privacy and equitable access to testing resources. Methodologically, it would be interesting to analyze the effects of ambiguous infection rates and more complex correlation structures on group testing strategies and their efficacy.

Beyond the context of group testing and disease management, we envision our ambiguity set of marginal rates and correlations could inspire research in diverse fields with similar structures of uncertainty. For example, in supply chain management, disruption risks and demand fluctuations at different locations are often correlated. In retail management, consumer behaviors and preferences might be influenced by social networks or demographic commonalities. Similarly, in urban planning, traffic patterns across different city areas may exhibit correlations.

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Online Appendices to "Distributionally Robust Group Testing with Correlation Information"

Our appendices consist of three parts. Section EC.A introduces several theoretical extensions. Section EC.B discusses more details about the numerical experiment. All proofs are listed in Section EC.C.

EC.A. Theoretical Extensions

In this section, we present several theoretical extensions. In Section EC.A.1, we discuss the case where the second-stage cost differs from the first stage. In Section EC.A.2, we provide the approximation bound for the multi-cluster case when N is not the multiple of group size. We finally extend our discussion to the cases with non-homogeneous clusters in Section EC.A.3.

EC.A.1. Cases with general second-stage cost

In Dorfman's group testing scheme, the samples from all subjects will be separated into two parts, where the first parts will be mixed into a group sample, and the second parts will be saved in the case of the second-stage test. However, in practice, the operational costs to save individual samples could be extremely high. For example, for some contagious diseases such as COVID-19, individual samples should be kept properly within surgical boxes.

In this section, we consider the case where the second-stage cost differs from the first stage. Specifically, suppose that once a group sample moves into the second stage, it will induce a cost to contact the certain subjects and collect the individual samples again. Hence, in this case, we will separate the number of tests $T(\mathcal{X},\tilde{\boldsymbol{\xi}})$ by the number of first-stage tests $T^1(\mathcal{X},\tilde{\boldsymbol{\xi}})$ and the number of second-stage tests $T^2(\mathcal{X},\tilde{\boldsymbol{\xi}})$, where each first-stage test has a normalized cost 1 and each second-stage test causes a cost c with $c \geq 1$. For consistency, we keep the total weight of the number of first-stage and second-stage tests as $1-\lambda_1-\lambda_2$. Consequently, the overall testing cost is the weighted sum of the number of false negative classifications (with weight λ_1), the number of false positive classifications (with weight λ_2), the number of first-stage tests (with weight $1-\lambda_1-\lambda_2$) with normalized first-stage cost 1, and the number of second-stage tests with second-stage cost c (with weight $(1-\lambda_1-\lambda_2)c$). Here, if $c=1$, the model reduces to the original one.

It is straightforward that the first stage test number $T^1(\mathcal{X},\tilde{\boldsymbol{\xi}})$ is the number of groups, which we denote as W. For the second stage, we denote $T^2_w(\mathcal{X},\tilde{\boldsymbol{\xi}})$ as the number of second-stage tests of group w for any $w \in [W]$. For any testing group w, we have $T^2_w(\mathcal{X},\tilde{\boldsymbol{\xi}}) = T_w(\mathcal{X},\tilde{\boldsymbol{\xi}})-1$, where $T_w(\mathcal{X},\tilde{\boldsymbol{\xi}})$

is the total testing number of group w where the second-stage cost is 1 (our original model). Thus, the overall objective function is

$$
\begin{aligned}
&\lambda_1\mathbb{E}_{\mathbb{P}}\left[FN(\mathcal{X},\tilde{\boldsymbol{\xi}})\right]+\lambda_2\mathbb{E}_{\mathbb{P}}\left[FP(\mathcal{X},\tilde{\boldsymbol{\xi}})\right]+(1-\lambda_1-\lambda_2)\mathbb{E}_{\mathbb{P}}\left[T^1(\mathcal{X},\tilde{\boldsymbol{\xi}})+cT^2(\mathcal{X},\tilde{\boldsymbol{\xi}})\right]\\
=&\lambda_1\mathbb{E}_{\mathbb{P}}\left[FN(\mathcal{X},\tilde{\boldsymbol{\xi}})\right]+\lambda_2\mathbb{E}_{\mathbb{P}}\left[FP(\mathcal{X},\tilde{\boldsymbol{\xi}})\right]+(1-\lambda_1-\lambda_2)c\mathbb{E}_{\mathbb{P}}\left[\sum_{w=1}^{W}T_w^2(\mathcal{X},\tilde{\boldsymbol{\xi}})\right]+(1-\lambda_1-\lambda_2)W\\
=&\lambda_1\mathbb{E}_{\mathbb{P}}\left[FN(\mathcal{X},\tilde{\boldsymbol{\xi}})\right]+\lambda_2\mathbb{E}_{\mathbb{P}}\left[FP(\mathcal{X},\tilde{\boldsymbol{\xi}})\right]+(1-\lambda_1-\lambda_2)c\mathbb{E}_{\mathbb{P}}\left[\sum_{w=1}^{W}(T_w(\mathcal{X},\tilde{\boldsymbol{\xi}})-1)\right]+(1-\lambda_1-\lambda_2)W\\
=&\lambda_1\mathbb{E}_{\mathbb{P}}\left[FN(\mathcal{X},\tilde{\boldsymbol{\xi}})\right]+\lambda_2\mathbb{E}_{\mathbb{P}}\left[FP(\mathcal{X},\tilde{\boldsymbol{\xi}})\right]+(1-\lambda_1-\lambda_2)c\mathbb{E}_{\mathbb{P}}\left[\sum_{w=1}^{W}T_w(\mathcal{X},\tilde{\boldsymbol{\xi}})\right]+(1-\lambda_1-\lambda_2)(1-c)W
\end{aligned}
$$

By the similar logic and results in Lemma EC.2, we can reformulate the above objective function regarding partition vector $\boldsymbol{z}$ and distribution vector $\boldsymbol{d}$ as follows,

$$
\begin{aligned}
&obj(\boldsymbol{z},\boldsymbol{d})\\
=&\lambda_1\left[N\beta_n(2-\beta_n)p-\beta_n(1-\beta_n)pz_1\right]\\
&+\lambda_2\left[N(1-\beta_n)\beta_p(1-p)+\beta_n\beta_p(1-p)z_1-\beta_p(1-\beta_n-\beta_p)\sum_{n=2}^{\Gamma}\left(z_n n\left(\sum_{k=0}^{N-n}\tbinom{N-n}{k}d_k\right)\right)\right]\\
&+(1-\lambda_1-\lambda_2)c\left[N(1-\beta_n)+\beta_n z_1+\sum_{n=2}^{\Gamma}\left(z_n-(1-\beta_n-\beta_p)z_n n\left(\sum_{k=0}^{N-n}\tbinom{N-n}{k}d_k\right)\right)\right]\\
&+(1-\lambda_1-\lambda_2)(1-c)\sum_{n=1}^{\Gamma}z_n\\
=&\lambda_1 N\beta_n(2-\beta_n)p+\lambda_2 N(1-\beta_n)\beta_p(1-p)+(1-\lambda_1-\lambda_2)cN(1-\beta_n)\\
&-\lambda_1\beta_n(1-\beta_n)pz_1+\lambda_2\beta_n\beta_p(1-p)z_1+(1-\lambda_1-\lambda_2)c\beta_n z_1+(1-\lambda_1-\lambda_2)c\sum_{n=2}^{\Gamma}z_n+(1-\lambda_1-\lambda_2)(1-c)\sum_{n=1}^{\Gamma}z_n\\
&-\lambda_2\beta_p(1-\beta_n-\beta_p)\sum_{n=2}^{\Gamma}\left(z_n n\left(\sum_{k=0}^{N-n}\tbinom{N-n}{k}d_k\right)\right)-(1-\lambda_1-\lambda_2)c(1-\beta_n-\beta_p)\sum_{n=2}^{\Gamma}\left(z_n n\left(\sum_{k=0}^{N-n}\tbinom{N-n}{k}d_k\right)\right)\\
=&\lambda_1 N\beta_n(2-\beta_n)p+\lambda_2 N(1-\beta_n)\beta_p(1-p)+(1-\lambda_1-\lambda_2)cN(1-\beta_n)\\
&-\lambda_1\beta_n(1-\beta_n)pz_1+\lambda_2\beta_n\beta_p(1-p)z_1+(1-\lambda_1-\lambda_2)c\beta_n z_1+(1-\lambda_1-\lambda_2)\sum_{n=1}^{\Gamma}z_n-(1-\lambda_1-\lambda_2)cz_1\\
&-(\lambda_2\beta_p+(1-\lambda_1-\lambda_2)c)(1-\beta_n-\beta_p)\sum_{n=2}^{\Gamma}\left(z_n n\left(\sum_{k=0}^{N-n}\tbinom{N-n}{k}d_k\right)\right)\\
=&D_0+D_1z_1+D_2\sum_{n=2}^{\Gamma}z_n-D_3\sum_{n=2}^{\Gamma}\left(z_n n\left(\sum_{k=0}^{N-n}\tbinom{N-n}{k}d_k\right)\right),
\end{aligned}
$$

where we define the following constants for convenience,

$$
D_0=\lambda_1 N\beta_n(2-\beta_n)p+\lambda_2 N(1-\beta_n)\beta_p(1-p)+(1-\lambda_1-\lambda_2)cN(1-\beta_n)
$$

$$
\begin{aligned}
D_1 =& -\lambda_1\beta_n(1-\beta_n)p + \lambda_2\beta_n\beta_p(1-p) + (1-\lambda_1-\lambda_2)c\beta_n + (1-\lambda_1-\lambda_2)(1-c)\\
D_2 =& (1-\lambda_1-\lambda_2)\\
D_3 =& (\lambda_2\beta_p + (1-\lambda_1-\lambda_2)c)(1-\beta_n-\beta_p).
\end{aligned}
$$

For any given $\boldsymbol{z}$, the inner maximization problem can be formulated as

$$\max_{\boldsymbol{d}\in\mathcal{D}} \sum_{k=0}^{N-n} \binom{N-n}{k} d_k.$$

Hence, with different second-stage costs, the worst-case distribution is independent from $\boldsymbol{z}$ and n, and thus there exists an optimal group size for the optimal partition. The overall problem is equivalent to derive the optimal group size n to minimize the objective function, where if $n=1$, the objective function is $D_0 + ND_1$, and if $n \geq 2$, the objective is

$$D_0 + D_2\frac{N}{n} - D_3 N \sum_{k=0}^{N-n} \binom{N-n}{k} d_k^*.$$

Hence, to obtain the optimal group size, it remains to solve

$$\min_{n\in[\Gamma]} h'(n, \boldsymbol{d}^*),$$

where

$$h'(n,\boldsymbol{d}^*) = \begin{cases} D_1 & \text{if } n=1,\\ \frac{D_2}{n} - D_3\sum_{k=0}^{N-n}\binom{N-n}{k} d_k^* & \text{if } n\geq 2.\end{cases}$$

and the optimal objective value is $D_0 + Nh'(n^*, \boldsymbol{d}^*)$.

Sensitivity analysis in c

If $n \geq 2$, at optimality, the first order condition is:

$$
\begin{aligned}
0 =& \frac{\partial}{\partial n} h'(n,\boldsymbol{d}^*)\\
=& \frac{\partial}{\partial n}\left\{\frac{(1-\lambda_1-\lambda_2)}{n} - (\lambda_2\beta_p + (1-\lambda_1-\lambda_2)c)(1-\beta_n-\beta_p)\sum_{k=0}^{N-n}\binom{N-n}{k} d_k^*\right\}\\
=& -\frac{(1-\lambda_1-\lambda_2)}{n^2} - (\lambda_2\beta_p + (1-\lambda_1-\lambda_2)c)(1-\beta_n-\beta_p)\frac{\partial}{\partial n}\left\{\sum_{k=0}^{N-n}\binom{N-n}{k} d_k^*\right\}
\end{aligned}
$$

Calculate the derivative of $\frac{\partial}{\partial n}h'(n,\boldsymbol{d}^*)$ on c, we have

$$
\begin{aligned}
&\frac{\partial}{\partial c}\left\{\frac{\partial h'(n,\boldsymbol{d}^*)}{\partial n}\right\}\\
=& \frac{\partial}{\partial c}\left\{-\frac{(1-\lambda_1-\lambda_2)}{n^2} - (\lambda_2\beta_p + (1-\lambda_1-\lambda_2)c)(1-\beta_n-\beta_p)\frac{\partial}{\partial n}\left\{\sum_{k=0}^{N-n}\binom{N-n}{k} d_k^*\right\}\right\}\\
=& -(1-\lambda_1-\lambda_2)(1-\beta_n-\beta_p)\frac{\partial}{\partial n}\left\{\sum_{k=0}^{N-n}\binom{N-n}{k} d_k^*\right\}\\
\geq& 0,
\end{aligned}
$$

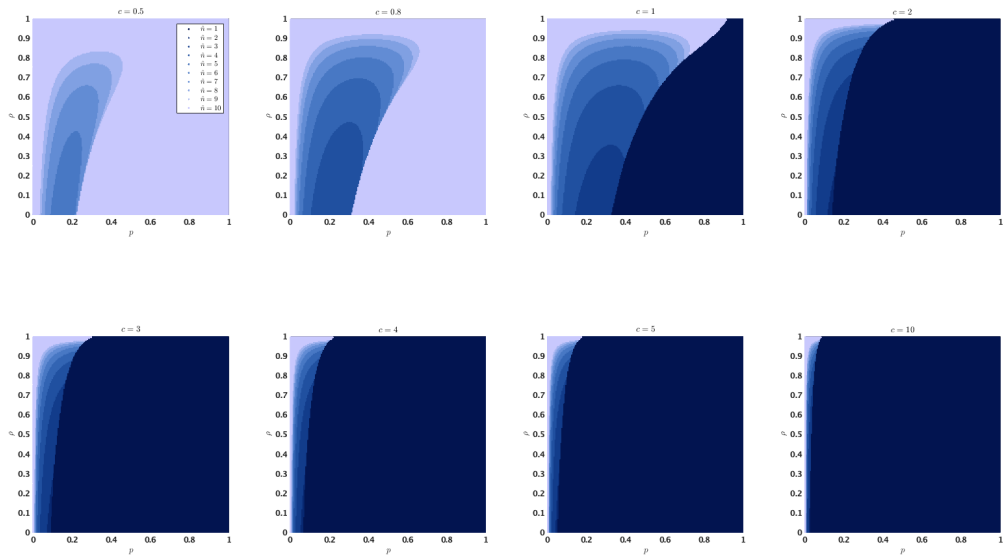


Figure EC.1 **The optimal group size under different second-stage cost: the darker color denotes a smaller group size.**

as $\frac{\partial}{\partial n}\left\{\sum_{k=0}^{N-n}\binom{N-n}{k}d_k^*\right\}\leq 0$.

Consider $c_1 > c_2$. Under c_1 and c_2, we denote the corresponding objective functions as $h_1'(n,\boldsymbol{d}^*), h_2'(n,\boldsymbol{d}^*)$, the optimal group sizes to minimize $h_1'(n,\boldsymbol{d}^*), h_2'(n,\boldsymbol{d}^*)$ when $n\geq 2$ as n_1, n_2. Then, for any $2\leq n<n_1$, we have,

$$h_2'(n_1,\boldsymbol{d}^*)-h_2'(n,\boldsymbol{d}^*)=\int_n^{n_1}\frac{\partial h_2'(k,\boldsymbol{d}^*)}{\partial k}\mathrm{d}k\leq\int_n^{n_1}\frac{\partial h_1'(k,\boldsymbol{d}^*)}{\partial k}\mathrm{d}k=h_1'(n_1,\boldsymbol{d}^*)-h_1'(n,\boldsymbol{d}^*)<0,$$

where the inequalities follow from $\frac{\partial}{\partial c}\left\{\frac{\partial h'(n,\boldsymbol{d}^*)}{\partial n}\right\}\geq 0$ and the optimality of n_1 under c_1, respectively. Thus, we have $n_2\geq n_1$. Hence, the optimal group size n^* decreases in c. Meanwhile, D_1 decreases in c. In other words, with the increase of second-stage cost, the advantages of both individual tests and small-group-size tests increase. Figure EC.1 describes the optimal group size at various second-stage costs c.

EC.A.2. The approximation bound for multi-cluster case when N is not the multiple of group size

For any given M, N, Γ and parameters (p,ρ), we could derive the optimal group structure $\boldsymbol{y}^*$ to minimize $f(\boldsymbol{y})$ by enumeration. Denote $n=\sum_{m=1}^{M}y_m$ as the group size of the optimal group structure. If N is a multiple of n, it is straightforward to construct the feasible and optimal grouping strategy $\boldsymbol{v}^*$, following the logic of Theorem 3.

However, if N is no longer a such multiple, it is infeasible to construct the optimal partition by $\boldsymbol{y}^*$. Next, we focus only on the case where $n \geq 2$, as N is already a multiple of $n = 1$. We construct the following feasible partition $\boldsymbol{v}^0$ such that the deviation of the corresponding total testing cost is not significant from the minimal total testing cost. Under $\boldsymbol{v}^0$, for all clusters $m \in [M]$, we first assign $M\lfloor \frac{N}{n} \rfloor n$ subjects into $M\lfloor \frac{N}{n} \rfloor$ groups according to the optimal group structure $\boldsymbol{y}^*$. Intuitively, by this setting we want to utilize the optimal group structure as much as possible. For the remaining $n' \triangleq N - \lfloor \frac{N}{n} \rfloor n$ subjects in each cluster, we group them by clusters, i.e., the Mn' subjects are assigned to M groups, each group including n' subjects from the same cluster. Hence, the partition $\boldsymbol{v}^0$ includes $M + M\lfloor \frac{N}{n} \rfloor$ groups, resulting in the total testing cost $g^0 \triangleq \sum_{l=1}^{M+M\lfloor \frac{N}{n} \rfloor} g_l(\boldsymbol{v}^0)$.

Next, we discuss $g_l(\boldsymbol{v}^0)$ for $l \in \{1, \ldots, M\lfloor \frac{N}{n} \rfloor\}$, i.e., the groups assigned by the optimal group structure $\boldsymbol{y}^*$, and $l \in \{M\lfloor \frac{N}{n} \rfloor + 1, \ldots, M + M\lfloor \frac{N}{n} \rfloor\}$, i.e., the remaining groups that each group has n' subjects from the same cluster, respectively. Specifically, for $l \in \{1, \ldots, M\lfloor \frac{N}{n} \rfloor\}$, we have

$$g_l(\boldsymbol{v}^0) = nf(\boldsymbol{y}^*),$$

based on Theorem 3 and its proof. For $l \in \{M\lfloor \frac{N}{n} \rfloor + 1, \ldots, M + M\lfloor \frac{N}{n} \rfloor\}$, based on the definition of function $g_l(\cdot)$ in Proposition 4, if $n' = 1$, $g_l(\boldsymbol{v}^0) = C_0 + C_1$; if $n' \geq 2$,

$$g_l(\boldsymbol{v}^0) = C_0 n' + C_2 - C_3 n' \sum_{k=0}^{N-n'} \binom{N-n'}{k} d_k^* \prod_{m=1}^{M-1} \sum_{k=0}^{N} \binom{N}{k} d_k^* = C_0 n' + C_2 - C_3 n' \sum_{k=0}^{N-n'} \binom{N-n'}{k} d_k^*,$$

where the last equality follows from $\sum_{k=0}^{N} \binom{N}{k} d_k^* = 1$. Due to the definition of $h(n, \boldsymbol{d}^*)$ in Equation (6), for $l \in \{M\lfloor \frac{N}{n} \rfloor + 1, \ldots, M + M\lfloor \frac{N}{n} \rfloor\}$, we have $g_l(\boldsymbol{v}^0) = n'(C_0 + h(n', \boldsymbol{d}^*))$.

Hence, we can reformulate and simplify g^0 as follows,

$$g^0 = \sum_{l=1}^{M+M\lfloor \frac{N}{n} \rfloor} g_l(\boldsymbol{v}^0) = M\lfloor \frac{N}{n} \rfloor nf(\boldsymbol{y}^*) + Mn'\left(C_0 + h(n', \boldsymbol{d}^*)\right).$$

Define the optimal total test cost as g^*, i.e., $g^* = \min_{\boldsymbol{v} \in \mathcal{V}} \sum_{l=1}^{MN} g_l(\boldsymbol{v})$. Moreover, according to Equation (EC.18), we have,

$$g^* = \min_{\boldsymbol{v} \in \mathcal{V}} \sum_{l=1}^{MN} g_l(\boldsymbol{v}) \geq MNf(\boldsymbol{y}^*).$$

Then, we bound the gap between the optimal total testing cost g^* and the total testing cost g^0 under the constructed feasible partition $\boldsymbol{v}^0$ as follows,

$$\begin{aligned}
\frac{g^0 - g^*}{g^*} &= \frac{M\lfloor \frac{N}{n} \rfloor nf(\boldsymbol{y}^*) + Mn'(C_0 + h(n', \boldsymbol{d}^*))}{\min_{\boldsymbol{v} \in \mathcal{V}} \sum_{l=1}^{MN} g_l(\boldsymbol{v})} - 1 \leq \frac{M\lfloor \frac{N}{n} \rfloor nf(\boldsymbol{y}^*) + Mn'(C_0 + h(n', \boldsymbol{d}^*))}{MNf(\boldsymbol{y}^*)} - 1 \\
&= \frac{\lfloor \frac{N}{n} \rfloor nf(\boldsymbol{y}^*) + n'(C_0 + h(n', \boldsymbol{d}^*)) - Nf(\boldsymbol{y}^*)}{Nf(\boldsymbol{y}^*)} = \frac{n'(C_0 + h(n', \boldsymbol{d}^*)) - n'f(\boldsymbol{y}^*)}{Nf(\boldsymbol{y}^*)} \\
&= \frac{n'}{N} \frac{C_0 + h(n', \boldsymbol{d}^*) - f(\boldsymbol{y}^*)}{f(\boldsymbol{y}^*)} < \frac{\Gamma}{N} \left(\frac{C_0 + h(n', \boldsymbol{d}^*)}{f(\boldsymbol{y}^*)} - 1 \right),
\end{aligned}$$

where the inequalities follow from the above discussion on g^* and $n' < n \le \Gamma$, respectively, and other equalities follows from algebra.

Next, it suffices to find an upper bound of the numerator $C_0 + h(n', \boldsymbol{d}^*)$ and a lower bound of the denominator $f(\boldsymbol{y}^*)$.

We first discuss the upper bound of $C_0 + h(n', \boldsymbol{d}^*)$. Following from the proof of Proposition 3, it is straightforward that

$$C_0 + h(n', \boldsymbol{d}^*) \le C_0 + \max\{C_1, C_2/2\}.$$

Here we concentrate on the lower bound of $f(\boldsymbol{y}^*) = C_0 + \frac{C_2}{n} - C_3 \prod_{m=1}^{M} \sum_{k=0}^{N-y_m^*} \binom{N-y_m^*}{k} d_k^*$. Above all, $\frac{C_2}{n}$ decreases in n and hence, $\frac{C_2}{n} \ge \frac{C_2}{\Gamma}$. We next discuss the upper bound of $\prod_{m=1}^{M} \sum_{k=0}^{N-y_m^*} \binom{N-y_m^*}{k} d_k^*$ with the constraint that $2 \le \sum_{m=1}^{M} y_m \le \Gamma$. Note that $\sum_{k=0}^{N-y_m^*} \binom{N-y_m^*}{k} d_k^*$ is non-negative and decreasing in y_m^*. Hence, we have,

$$0 \le \sum_{k=0}^{N-\Gamma} \binom{N-\Gamma}{k} d_k^* \le \sum_{k=0}^{N-y_m^*} \binom{N-y_m^*}{k} d_k^* \le \sum_{k=0}^{N-1} \binom{N-1}{k} d_k^* = 1 - p \le \sum_{k=0}^{N} \binom{N}{k} d_k^* = 1.$$

Note that there are at least one non-zero elements in $\boldsymbol{y}^*$. Hence, we derive

$$\prod_{m=1}^{M} \sum_{k=0}^{N-y_m^*} \binom{N-y_m^*}{k} d_k^* \le (1-p)^{|\boldsymbol{y}^*|} \le 1 - p,$$

where $|\boldsymbol{y}^*|$ represents the number of non-zero elements in vector $\boldsymbol{y}^*$. Then, the overall lower bound is,

$$f(\boldsymbol{y}^*) = C_0 + \frac{C_2}{n} - C_3 \prod_{m=1}^{M} \sum_{k=0}^{N-y_m^*} \binom{N-y_m^*}{k} d_k^* \ge C_0 + \frac{C_2}{\Gamma} - C_3(1-p),$$

where the inequality follows from the above discussion.

It remains to show that the lower bound is non-negative, as follows,

$$\begin{aligned}
C_0 + \frac{C_2}{\Gamma} - C_3(1-p) =& \lambda_1 \beta_n (2-\beta_n) p + \lambda_2 (1-\beta_n)\beta_p (1-p) + (1-\lambda_1-\lambda_2)(1-\beta_n) + \frac{1-\lambda_1-\lambda_2}{\Gamma} \\
& - (1-\lambda_1-\lambda_2(1-\beta_p))(1-\beta_n-\beta_p)(1-p) \\
=& \lambda_1 \beta_n (2-\beta_n) p + \lambda_2 (1-\beta_n)\beta_p (1-p) + (1-\lambda_1-\lambda_2)(1-\beta_n) + \frac{1-\lambda_1-\lambda_2}{\Gamma} \\
& - \lambda_2 \beta_p (1-\beta_n-\beta_p)(1-p) - (1-\lambda_1-\lambda_2)(1-\beta_n-\beta_p)(1-p) \\
>& 0,
\end{aligned}$$

where the inequality holds as $1-\beta_n > 1-\beta_n-\beta_p$ and $1-p \le 1$.

Therefore, based on the above discussion on the upper bound of $C_0 + h(n, \boldsymbol{d}^*)$ and the lower bound of $f(\boldsymbol{y}^*)$, we conclude that

$$\frac{g^0 - g^*}{g^*} < \frac{\Gamma}{N}\left[\frac{C_0 + h(n, \boldsymbol{d}^*)}{f(\boldsymbol{y}^*)} - 1\right] \le \frac{\Gamma}{N}\left[\frac{C_0 + \max\{C_1, \frac{C_2}{2}\}}{C_0 + \frac{C_2}{\Gamma} - C_3(1-p)} - 1\right] = \frac{\Gamma}{N} \frac{\max\{C_1, \frac{C_2}{2}\} - \frac{C_2}{\Gamma} + C_3(1-p)}{C_0 + \frac{C_2}{\Gamma} - C_3(1-p)}.$$

EC.A.3. Cases of non-homogeneous clusters

In this section, we analyze cases where clusters are non-homogeneous, i.e., sizes of clusters are different, subjects from different clusters have different prevalence rates and correlation coefficients, called cluster-specific sizes, cluster-specific prevalence rates and cluster-specific correlation coefficients. In particular, for cluster m, we denote the size, prevalence rate and correlation coefficient for subjects as N^m, p^m and ρ^m for all $m \in [M]$, respectively. For convenience, we denote $\mathcal{N} = \sum_{m=1}^{M} N^m$.

In this generalized case, the group testing problem can be formulated as,

$$\min_{\boldsymbol{w} \in \mathcal{W}^{\text{Gen}}} \sup_{\mathbb{P}^{\mathcal{M}} \in \mathcal{P}^{\mathcal{M},\text{Gen}}} \lambda_1 \mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[FN(\boldsymbol{w}, \tilde{\boldsymbol{\xi}})] + \lambda_2 \mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[FP(\boldsymbol{w}, \tilde{\boldsymbol{\xi}})] + (1 - \lambda_1 - \lambda_2) \mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[T(\boldsymbol{w}, \tilde{\boldsymbol{\xi}})], \quad \text{(EC.1)}$$

where $\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[FN(\boldsymbol{w}, \tilde{\boldsymbol{\xi}})]$, $\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[FP(\boldsymbol{w}, \tilde{\boldsymbol{\xi}})]$ and $\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[T(\boldsymbol{w}, \tilde{\boldsymbol{\xi}})]$ denote the expected number of false positive classifications, false negative classifications, and total tests needed under partition $\boldsymbol{w}$, respectively. Similar to Equation (2), the joint distribution is assumed to be within the following ambiguity set,

$$\mathcal{P}^{\mathcal{M},\text{Gen}} = \left\{ \mathbb{P}^{\mathcal{M}} \left| \begin{array}{ll} \mathbb{P}^m\left(\tilde{\xi}_{m,n} = 1\right) = p^m & \forall n \in [N^m], m \in [M], \\ \mathbb{P}^m\left(\tilde{\xi}_{m,n_1} = 1, \tilde{\xi}_{m,n_2} = 1\right) = (p^m)^2 + \rho^m p^m (1 - p^m) & \forall n_1, n_2 \in [N^m], n_1 \neq n_2, m \in [M], \\ \mathbb{P}^m\left(\tilde{\boldsymbol{\xi}}^m = P_\pi \boldsymbol{\xi}^m\right) = \mathbb{P}^m\left(\tilde{\boldsymbol{\xi}}^m = \boldsymbol{\xi}^m\right) & \forall P_\pi, \boldsymbol{\xi}^m \in \{0,1\}^{N^m}, m \in [M], \\ \mathbb{P}^{\mathcal{M}}\left(\tilde{\boldsymbol{\xi}}^1 = \boldsymbol{\xi}^1, \ldots, \tilde{\boldsymbol{\xi}}^M = \boldsymbol{\xi}^M\right) = \prod_{m=1}^{M} \mathbb{P}^m\left(\tilde{\boldsymbol{\xi}}^m = \boldsymbol{\xi}^m\right) & \forall \boldsymbol{\xi}^m \in \{0,1\}^{N^m}, m \in [M]. \end{array} \right. \right\},$$

where the first three constraints follow exactly the same arguments as the constraints in Equation (2). Specifically, for each cluster, we specify the prevalence probability for each individual, the Pearson correlation coefficient for each pair, and that the subjects are exchangeable as described in Assumption 1. Moreover, the feasible region $\mathcal{W}^{\text{Gen}}$ is defined as,

$$\mathcal{W}^{\text{Gen}} = \left\{ \boldsymbol{w} \left| \begin{array}{ll} \sum_{l=1}^{\mathcal{N}} w_{m,n,l} = 1 & \forall m \in [M], n \in [N^m] \\ \sum_{m=1}^{M} \sum_{n=1}^{N^m} w_{m,n,l} \leq \Gamma & \forall l \in [\mathcal{N}] \\ w_{m,n,l} \in \{0,1\} & \forall m \in [M], n \in [N^m], l \in [\mathcal{N}] \end{array} \right. \right\}.$$

Then, we can generalize the results in Proposition 4 as the following corollary shows.

Corollary EC.1. *Under cluster-specific sizes, prevalence rates and correlation coefficients, the worst-case distribution to Problem (EC.1), denoted by $\mathbb{P}^{\mathcal{M}*,Gen}$, satisfies $\mathbb{P}^{\mathcal{M}*,Gen}\left(\tilde{\boldsymbol{\xi}}^m = \boldsymbol{\xi}\right) = d^{m*}_{\mathbf{1}^\top \boldsymbol{\xi}}$ for any cluster $m \in [M]$, where $\boldsymbol{d}^{m*} = (d_0^{m*}, d_1^{m*}, \ldots, d_N^{m*})$ is defined in Lemma 2 under infection characteristics (p^m, ρ^m) and cluster size N^m for any $m \in [M]$. Problem (EC.1) is equivalent to*

$$\min_{\boldsymbol{v} \in \mathcal{V}^{Gen}} \sum_{l=1}^{\mathcal{N}} g_l^{Gen}(\boldsymbol{v}), \quad \text{(EC.2)}$$

where

$$\mathcal{V}^{Gen} = \left\{ \boldsymbol{v} \;\middle|\; \begin{array}{ll} \sum_{l=1}^{\mathcal{N}} v_{m,l} = N^m & \forall m \in [M], \\ \sum_{m=1}^{M} v_{m,l} \le \Gamma & \forall l \in [\mathcal{N}], \\ v_{m,l} \in \{0,1,\ldots,\Gamma\} & \forall m \in [M], l \in [\mathcal{N}] \end{array} \right\},$$

$$g_l^{Gen}(\boldsymbol{v}) = \begin{cases} 0, & \text{if } \sum_{m=1}^{M} v_{m,l} = 0, \\ \lambda_1 \beta_n \sum_{m=1}^{M} p^m v_{m,l} + \lambda_2 \beta_p \sum_{m=1}^{M} (1-p^m) v_{m,l} + (1-\lambda_1-\lambda_2), & \text{if } \sum_{m=1}^{M} v_{m,l} = 1, \\ (1-\lambda_1-\lambda_2) + \lambda_1 \beta_n (2-\beta_n) \sum_{m=1}^{M} p^m v_{m,l} + \lambda_2 (1-\beta_n) \beta_p \sum_{m=1}^{M} (1-p^m) v_{m,l} & \\ +(1-\lambda_1-\lambda_2)(1-\beta_n) \sum_{m=1}^{M} v_{m,l} & \\ -\left(1-\lambda_1-\lambda_2(1-\beta_p)\right)(1-\beta_n-\beta_p) \left(\sum_{m=1}^{M} v_{m,l}\right) \prod_{m=1}^{M} \left(\sum_{k=0}^{N^m - v_{m,l}} \binom{N^m - v_{m,l}}{k} d_k^{m*} \right), & \text{otherwise.} \end{cases}$$

for any $\boldsymbol{v} \in \mathcal{V}^{Gen}$.

Hence, we reformulate the problem from a min-max problem involving high-dimensional decision variables $\boldsymbol{w}$ and $\mathbb{P}^{\mathcal{M}}$ to a convex integer optimization problem with reduced complexity.

EC.B. Numerical Supplements

In this section, we provide supplementary numerical results to support and extend the results in main article. In Section EC.B.1, we provide a new numerical study on the robustness of parameters p, ρ, and in Section EC.B.2, we discuss details in the case study.

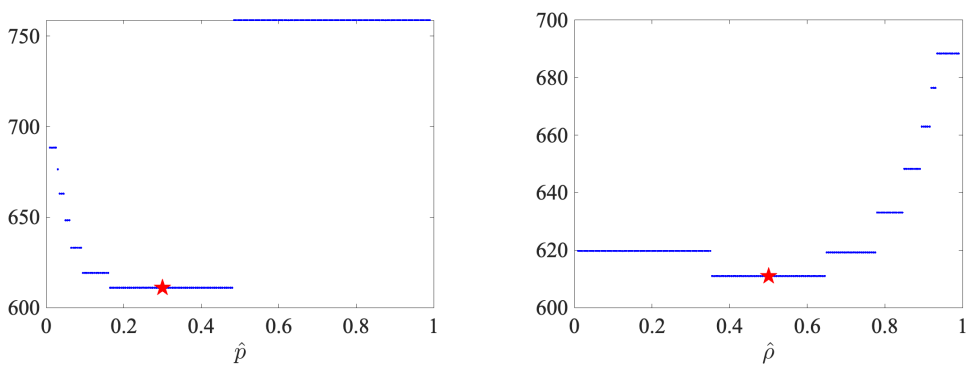
EC.B.1. Robustness on parameters p, ρ

In this experiment, we discuss the performance when the model parameters (prevalence p and correlation information ρ) are estimated with errors. For example, with historical data of the infected population in a certain area, the infection parameters (p, ρ) can be estimated using statistical methods, which are discussed in Appendix EC.B.2.2 for details. However, the decision maker could overestimate or underestimate the parameters.

To address this concern, we build up the robustness of our model on infection parameters (p, ρ) by measuring the cost of mis-specification. For true values of prevalence rate p and correlation coefficient ρ, the parameters could be estimated or misspecified as $\hat{p}$ and $\hat{\rho}$. In reality, the estimated parameters $(\hat{p}, \hat{\rho})$ are available to decision makers, which results in the optimal common group

size $n^*(\hat{p},\hat{\rho})$. Next, we measure the misspecification effect by comparing the overall testing costs of $n^*(\hat{p},\hat{\rho})$ and the true optimal common group size $n^*(p,\rho)$ derived under true parameters (p,ρ). Specifically, we compare the values of $f(n^*(p,\rho),p,\rho)$ and $f(n^*(\hat{p},\hat{\rho}),p,\rho)$, where $f(n,p,\rho) \triangleq NC_0 + Nh(n,\boldsymbol{d}^*(p,\rho))$ measures the total testing cost of the grouping strategy n under true infection parameters (p,ρ).

We conduct numerical experiments to check the robustness of p and ρ in the subsequent setting. Consider a population with $N = 1,500$, and set the group size limit $\Gamma = 10$. Assuming the true parameter $p = 0.30$ and $\rho = 0.50$, we derive the numerical results under $(\lambda_1,\lambda_2) = (0.3,0.2)$ and $(\beta_n,\beta_p) = (0.05,0.01)$. Based on the numerical results in Figure EC.2, we then elaborate on the robustness of the total testing cost with respect to p and ρ, respectively. In Figure EC.2(a), the red star labels the true value of prevalence p, and the blue line refers to the total testing cost when the prevalence rate is mis-specified as the values in horizontal axis. Figure EC.2(a) reveals that small deviation from the true value of p, i.e., $\hat{p} \in [0.16,0.48]$, results in no deviation of the overall testing cost. Similarly, Figure EC.2(b) reveals that a small deviation from the true value of ρ, i.e., $\hat{\rho} \in [0.35,0.64]$, also results in no deviation of the overall testing cost.



(a) Robustness on p (b) Robustness on ρ

Figure EC.2 Robustness on p and ρ

Intuitively, a slight deviation will not affect the real performance, but the performance gap may increase if the estimated parameters further deviate. Therefore, with the estimated parameters $(\hat{p},\hat{\rho})$, decision makers can be confident of the solution and the corresponding performances. We next conduct experiments on different sets of true parameters (p,ρ) and introduce more results as follows.

We further generalize the above experiment considering more scenarios: the true parameters $p \in \{0.01,0.1,0.3\}$ and $\rho \in \{0.1,0.3,0.8\}$. For each combination of p and ρ, we consider the corresponding confidence level by $\pm 30\%$. The numerical results reveal the robustness on p and ρ, respectively,

displayed in Figure EC.3 and EC.4. Figure EC.3 illustrates the impact of mis-specified prevalence on the overall test cost for different prevalence values $p \in [0.01, 0.1, 0.3]$. For the low-prevalence case ($p = 0.01$ in the first column of Figure EC.3), any mis-specified prevalence $\hat{p}$ within the range $[0.007, 0.013]$ results in no deviation of the overall testing cost because of the small estimation set. As the prevalence increases to $p = 0.1$ in the second column of the figure, the estimation $\hat{p}$ achieves the same testing cost when it is close to the true prevalence p. However, if the estimation deviates significantly from the true prevalence $\dot{p}$, it can lead to an increase in the overall testing cost. For the high-prevalence case ($p = 0.3$ in the third column of Figure EC.3), the degree of deviation of the estimated prevalence value $\hat{p}$ from the true prevalence p demonstrates the robustness of corresponding total testing costs. In summary, a small deviation from the true prevalence p has negligible influence on the total testing cost, while a significant deviation leads to a substantial increase in the total testing cost. This demonstrates the robustness of p in its vicinity and the sensitivity of p when it deviates significantly from the true value. Regarding the correlation coefficient ρ, our numerical results in Figure EC.4 also demonstrate the robustness of the estimated correlation $\hat{\rho}$ across various correlation values: low-correlation ($\rho = 0.1$), mid-correlation ($\rho = 0.3$), and high-correlation scenarios ($\rho = 0.8$). Similar to the results for mis-specifying the prevalence rate p, a small deviation from the estimated $\hat{\rho}$ to the true ρ results in an ignorable influence on the total testing cost, while a significant deviation leads to a substantial increase in the total testing cost. This yields the robustness of ρ near the true value and the sensitivity when it deviates significantly from the true value.

Consequently, by integrating the aforementioned experiments with varying true prevalence values p and correlation coefficients ρ, we effectively establish the robustness of p and ρ against the estimated values $\hat{p}$ and $\hat{\rho}$. This demonstrates the resilience of our research against estimation inaccuracies and substantially enhances the practical significance of our model and findings.

EC.B.2. Details of the case study

EC.B.2.1. Data visualization. We provide a full visualization of our dataset in Figure EC.5, where each point captures the population and infection count in one studied area.

EC.B.2.2. Estimating ρ. Given several clusters of binary variables, Ridout et al. (1999) review a variety of methods for estimating the unknown "success rate" p (infection rate in our context) and the "intraclass correlation" ρ (the identical pairwise Pearson correlation coefficient in our context), under the assumptions of i) different clusters can have different sizes; ii) only the population size and the count of 1's (positive individuals in our context) are known for each group; and iii) p and ρ do not vary with the cluster size. Two estimators are adopted for our study: the AOV and KPR methods. We formulate the two estimators as follows.

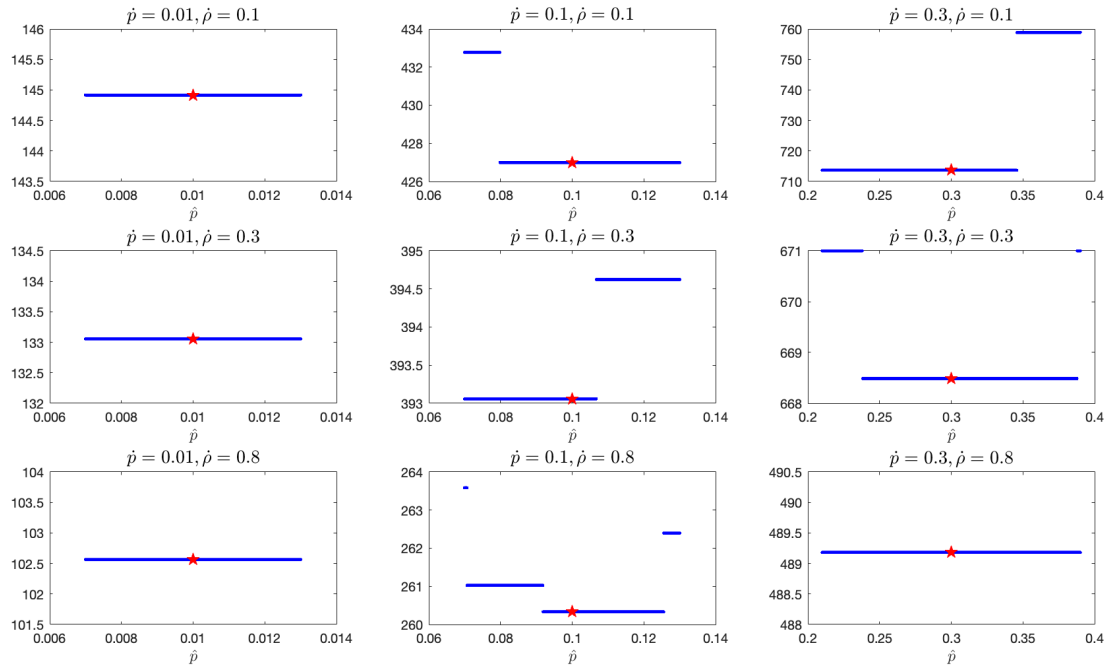


Figure EC.3 **Real total costs of the p-deviated grouping strategy and the optimal grouping strategy where** $N = 1500$, $\Gamma = 10$, $\dot{p} \in \{0.01, 0.1, 0.3\}$, $\dot{\rho} \in \{0.1, 0.3, 0.8\}$, $\varepsilon = 0.3$, $(\lambda_1, \lambda_2) = (0.3, 0.2)$ **and** $(\beta_n, \beta_p) = (0.05, 0.01)$

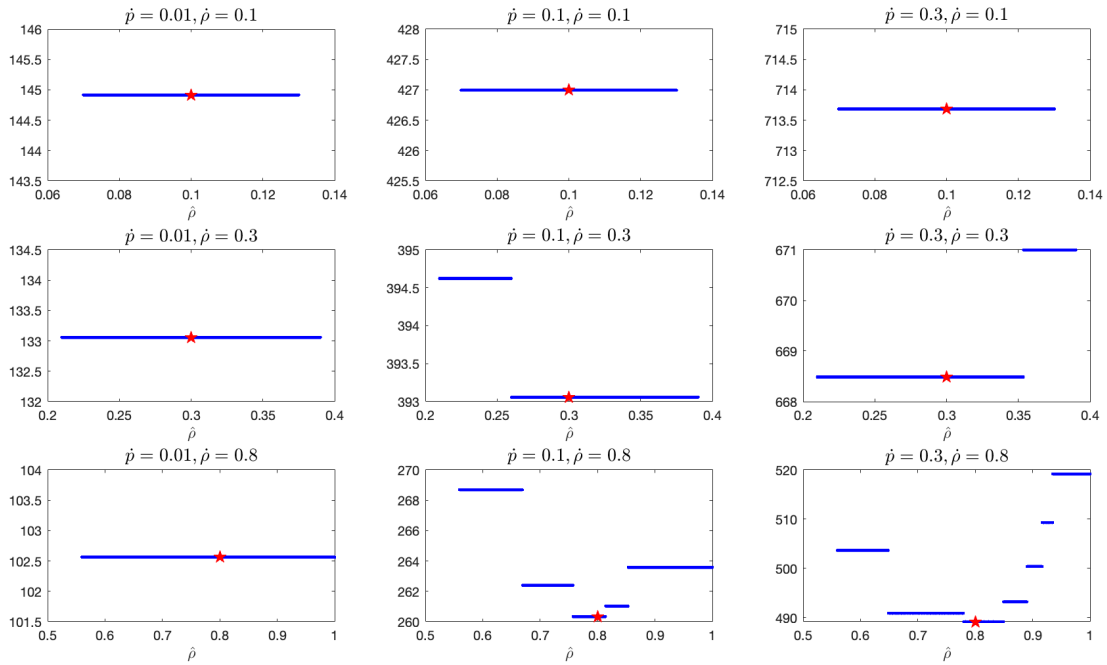


Figure EC.4 **Real total costs of the ρ-deviated grouping strategy and the optimal grouping strategy where** $N = 1500$, $\Gamma = 10$, $\dot{p} \in \{0.01, 0.1, 0.3\}$, $\dot{\rho} \in \{0.1, 0.3, 0.8\}$, $\varepsilon = 0.3$, $(\lambda_1, \lambda_2) = (0.3, 0.2)$ **and** $(\beta_n, \beta_p) = (0.05, 0.01)$

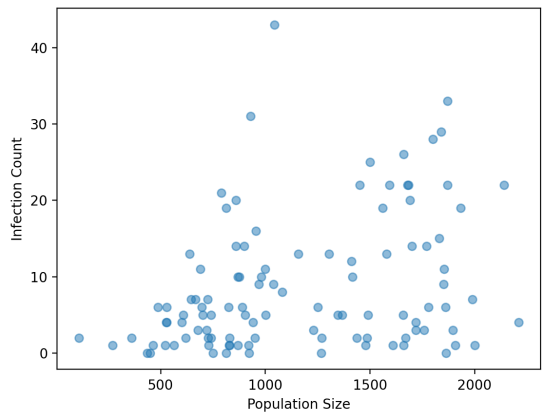


Figure EC.5 Population sizes and infection counts for 81 compulsory testing areas.

Denote by $K = 81$ the total number of restrictive areas. For each area, denote the data by $(N_j, C_j)_{j\in[K]}$, with N_j and C_j being the population size and the total infection counts, respectively, in the j-th area.

- The AOV estimator: the Analysis of Variance Estimator, given by

$$\hat{\rho}_{AOV} = \frac{MS_b - MS_w}{MS_b + (N_0 - 1)MS_w},$$

where $N_0 = \frac{1}{K-1}\left(\sum_{i=1}^K N_i - \frac{\sum_{i=1}^K N_i^2}{\sum_{i=1}^K N_i}\right)$, $MS_b = \frac{1}{K-1}\left(\sum_{i=1}^K \frac{C_i^2}{N_i} - \frac{\left(\sum_{i=1}^K C_i\right)^2}{\sum_{i=1}^K N_i}\right)$, $MS_w = \frac{1}{\sum_{i=1}^K (N_i-1)}\left(\sum_{i=1}^K C_i - \sum_{i=1}^K \frac{C_i^2}{N_i}\right)$.

- The KPR estimator: a proportion-based moment estimator, defined as

$$\hat{\rho}_{KPR} = \frac{S_w - \tilde{\pi}_w\left(1-\tilde{\pi}_w\right)\sum_{i=1}^K \frac{w_i(1-w_i)}{N_i}}{\tilde{\pi}_w\left(1-\tilde{\pi}_w\right)\left(\sum_{i=1}^K w_i\left(1-w_i\right) - \sum_{i=1}^K \frac{w_i(1-w_i)}{N_i}\right)},$$

where $S_w = \sum_{i=1}^K w_i(\tilde{\pi}_i - \tilde{\pi}_w)^2$, $\tilde{\pi}_w = \sum_{i=1}^K w_i\tilde{\pi}_i$, with $w_i = \frac{C_i}{N_i}$ and $\tilde{\pi}_i = \frac{N_i}{\sum_{j=1}^K N_j}$ is proportional to the group size.

EC.B.2.3. Result analysis. Figure EC.6 captures the dynamics of the true p, ρ based on our epidemic model, and the estimated ones based on test results. Estimation gaps are caused by false negatives and one-day delay in updating estimates. We observe significant fluctuations in p and ρ, which inevitably play a crucial role in explaining the performance differences between the three testing methods (as shown in Figure 6). Specifically, during periods of rapidly increasing p (early in the outbreak, around Days 1–10), the heuristic-based approach struggles to adapt due to its simplistic assumption of fixed group sizes, thereby leading to inefficient pooling. During Days 10–20, the infection rates remain steadily high while the correlations are relatively low, which

results in similar performance across all three testing methods. From Day 20 onward, the infection rate drops significantly while correlations begin to rise, causing the independent-based method to perform worse due to its inability to account for ρ. In contrast, our correlation-based DRO method dynamically adjusts to the changing p and ρ, effectively capturing correlations and minimizing the number of tests.

That said, there are two brief intervals where the correlated testing method falls short: Days 3–6 and Days 24–25. During Days 3–6, both p and ρ are very low, putting the system in a state highly sensitive to small variations in ρ (as shown in Figure 3). As there are small differences in ρ between the trajectories of correlated and independent-based testing methods, this sensitivity results in slight, unpredictable gaps in testing costs, causing our method to be marginally outperformed. Similarly, on Days 24–25, although ρ becomes significant, the specific combination of p and ρ places the system at the threshold between pooled testing and individual testing (see Figure 3[6]), making it more sensitive to slight perturbations and misspecifications in p and ρ.

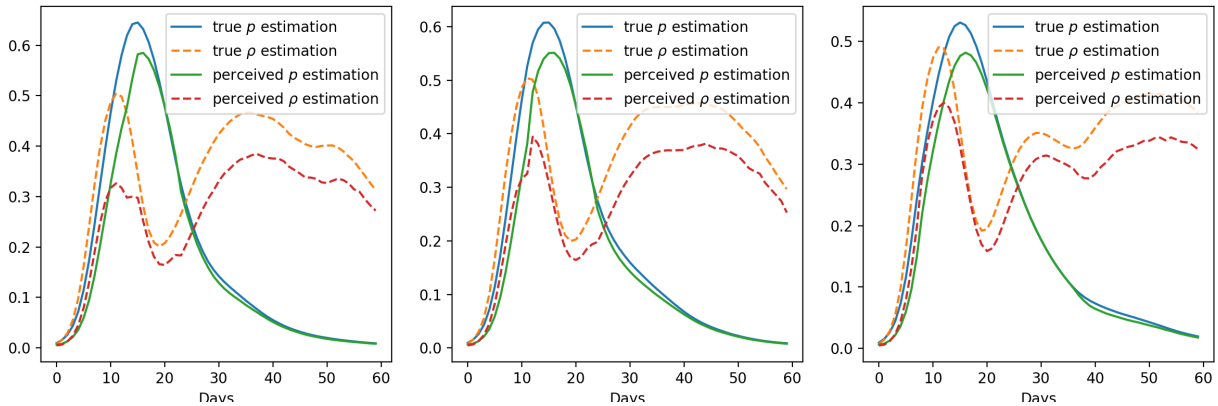


Figure EC.6 **Average p, ρ trajectories for three testing methods over 20 random instances—correlation-based group testing via DRO (left), independent-based group testing via stochastic programming (middle) and the heuristic-based current practice (right).**

Apart from providing insights into the performance gaps between different testing approaches, Figure EC.6 also helps generate policy insights for group-testing designs in general localized outbreaks. While infection rates typically rise to a peak before gradually declining, correlation tends to be high during active chain infections and exhibits two peaks—one before and one after the infection peak. Correlation drops at the infection peak itself, likely due to fewer susceptible individuals that limit chain infections. As a result, the different stages of p, ρ combination correspond

[6] While the subfigures in Figure 3 do not perfectly match the parameters in this case study, they provide an intuitive understanding of the system's behavior.

loosely to the upper-left and lower-right corners of Figure 3. This implies that test practitioners should consider larger pools at the early and late stages of the outbreak, when correlation is high and prevalence is low. Conversely, smaller pools or individual testing are preferable during the infection peak.

EC.C. Proofs

In this section, we provide proofs of the theoretical results.

EC.C.1. Proof of Lemma 1

In this proof, under any random classification vector $\tilde{\boldsymbol{\xi}} = (\tilde{\xi}_1, \ldots, \tilde{\xi}_N)$ and partition decision $\mathcal{X} = \{\Omega_1, \ldots, \Omega_W\}$, we calculate the number of false negative classifications $FN_i(\mathcal{X}, \tilde{\boldsymbol{\xi}})$, the number of false positive classifications $FP_i(\mathcal{X}, \tilde{\boldsymbol{\xi}})$ and the number of tests $T_w(\mathcal{X}, \tilde{\boldsymbol{\xi}})$ for any subject $i \in [N]$ and group $w \in [W]$ in the following steps.

- Number of false negative classifications for subject i: $FN_i(\mathcal{X}, \tilde{\boldsymbol{\xi}})$

Note that false negative classification is non-zero only when subject i is positive, i.e., $\tilde{\xi}_i = 1$. Hence, if $\tilde{\xi}_i = 0$, $FN_i(\mathcal{X}, \tilde{\boldsymbol{\xi}}) = 0$. Then, we discuss two cases where $\tilde{\xi}_i = 1$.

If subject i is tested individually, then $FN_i(\mathcal{X}, \tilde{\boldsymbol{\xi}})$ is exactly the false negative probability β_n.

If subject i is first tested in a group with size $n \geq 2$, then the pooled sample is positive, and subject i would be classified as false negative in two cases. The first one is that the test result for the pooled sample is correct, i.e., positive, but the subject itself is then tested as false negative in the second stage. The second one is when the test result for the pooled sample is wrong, i.e., false negative, and hence this subject is classified as negative without going to the second stage. Hence, $FN_i(\mathcal{X}, \tilde{\boldsymbol{\xi}}) = (1 - \beta_n) \cdot \beta_n + \beta_n = \beta_n(2 - \beta_n)$.

To sum up,

$$FN_i(\mathcal{X}, \tilde{\boldsymbol{\xi}}) = \begin{cases} \beta_n & \text{if } \tilde{\xi}_i = 1 \text{ and } i \text{ is tested individually in } \mathcal{X}, \\ \beta_n(2 - \beta_n) & \text{if } \tilde{\xi}_i = 1 \text{ and } i \text{ is tested in a group in } \mathcal{X}, \\ 0 & \text{if } \tilde{\xi}_i = 0. \end{cases}$$

- Number of false positive classifications for subject i: $FP_i(\mathcal{X}, \tilde{\boldsymbol{\xi}})$

Note that false positive classification is non-zero only when subject i is negative, i.e., $\tilde{\xi}_i = 0$. Hence, if $\tilde{\xi}_i = 1$, $FN_i(\mathcal{X}, \tilde{\boldsymbol{\xi}}) = 0$. Now we consider the two cases where the subject i is non-infected.

If subject i is tested individually, $FP_i(\mathcal{X}, \tilde{\boldsymbol{\xi}})$ is exactly the false positive probability β_p.

If subject i is first tested in the group $\Omega \in \mathcal{X}$ with group size $n \geq 2$, then the above conditional expectation depends on whether the pooled sample of Ω is negative or not (i.e., all subjects in Ω are negative or at least one of them is positive). Conditional on there being no positive in the group, the false positive classification of the subject i is non-zero only if both the pooled sample of Ω (in the first stage) and the individual sample i itself (in the second stage) are tested as false

positive, i.e., $FP_i(\mathcal{X},\tilde{\boldsymbol{\xi}})=\beta_p^2$. If there is an infected subject in the group, the negative subject i is first tested with some positive subjects in the group and then tested individually if the first stage result is true positive. Hence, $FP_i(\mathcal{X},\tilde{\boldsymbol{\xi}})=(1-\beta_n)\beta_p$.

To sum up,

$$FP_i(\mathcal{X},\tilde{\boldsymbol{\xi}})=\begin{cases}\beta_p & \text{if } \tilde{\xi}_i=0 \text{ and } i \text{ is tested individually in } \mathcal{X},\\ \beta_p^2 & \text{if } \tilde{\xi}_i=0 \text{ and } i \text{ is tested in a non-infected group in } \mathcal{X},\\ (1-\beta_n)\beta_p & \text{if } \tilde{\xi}_i=0 \text{ and } i \text{ is tested in an infected group in } \mathcal{X},\\ 0 & \text{if } \tilde{\xi}_i=1.\end{cases}$$

- Number of tests: $T_w(\mathcal{X},\tilde{\boldsymbol{\xi}})$.

For any given group $\Omega_w\in\mathcal{X}$ whose size is denoted by $|\Omega_w|$, we calculate $T_w(\mathcal{X},\tilde{\boldsymbol{\xi}})$ by the following two cases.

If $|\Omega_w|=1$, then this group includes only one subject and the number of tests $T_w(\mathcal{X},\tilde{\boldsymbol{\xi}})=1$.

If $|\Omega_w|\geq 2$, then the expected number of tests depends on whether the group includes positive subjects. When $\tilde{\xi}_j=0$ for any $j\in\Omega_w$, the number of tests is

$$T_w(\mathcal{X},\tilde{\boldsymbol{\xi}})=(1-\beta_p)\cdot 1+\beta_p\cdot(1+|\Omega_w|)=1+\beta_p|\Omega_w|$$

When there exists $j\in\Omega_w$ such that $\tilde{\xi}_j=1$, the number of tests is

$$T_w(\mathcal{X},\tilde{\boldsymbol{\xi}})=(1-\beta_n)(1+|\Omega_w|)+\beta_n\cdot 1=1+(1-\beta_n)|\Omega_w|.$$

To sum up, we have

$$T_w(\mathcal{X},\tilde{\boldsymbol{\xi}})=\begin{cases}1 & \text{if } |\Omega_w|=1,\\ 1+\beta_p|\Omega_w| & \text{if } |\Omega_w|\geq 2 \text{ and all subjects in } \Omega_w \text{ are non-infected},\\ 1+(1-\beta_n)|\Omega_w| & \text{if } |\Omega_w|\geq 2 \text{ and } \Omega_w \text{ includes infected subjects.}\end{cases}$$

Therefore, we conclude the results. Q.E.D.

EC.C.2. Proof of Proposition 1

We conclude the results by the following three steps: 1) construct the equivalence between $\mathbb{P}\in\mathcal{P}$ and $\boldsymbol{d}\in\mathcal{D}$; 2) construct the equivalence between $\mathcal{X}\in\mathcal{B}$ and $\boldsymbol{z}\in\mathcal{Z}$; 3) formulate the objective terms (total counts of false negatives, false positives and tests) under newly introduces $\boldsymbol{d}$ and $\boldsymbol{z}$, which results in Problem (4); 4) simplify Problem (4) to Problem (5) by algebra.

For the above steps, we will first accomplish Step 1) by Lemma EC.1; then finish Step 2) and 3) by Lemma EC.2; in the remaining part of this proof, we will simplify $\lambda_1 FN_{\boldsymbol{d}}(\boldsymbol{z})+\lambda_2 FP_{\boldsymbol{d}}(\boldsymbol{z})+(1-\lambda_1-\lambda_2)T_{\boldsymbol{d}}(\boldsymbol{z})$.

We first establish a rigorous equivalence between $\mathbb{P}\in\mathcal{P}$ and $\boldsymbol{d}_{\mathbb{P}}\in\mathcal{D}$ in our model.

LEMMA EC.1. *Given any vector* $\boldsymbol{d}=(d_0,d_1,\ldots,d_N)\in\mathcal{D}$*, we define a probability distribution* $\mathbb{Q}_{\boldsymbol{d}}$ *by*

$$\mathbb{Q}_{\boldsymbol{d}}\left(\tilde{\boldsymbol{\xi}}=\boldsymbol{\xi}\right)=d_{\mathbf{1}^\top\boldsymbol{\xi}},\qquad \forall\boldsymbol{\xi}\in\{0,1\}^N.$$

Then, given any $\mathbb{P}\in\mathcal{P}$*, we have* $\boldsymbol{d}_{\mathbb{P}}\in\mathcal{D}$ *and* $\mathbb{Q}_{\boldsymbol{d}_{\mathbb{P}}}=\mathbb{P}$*. Moreover,* $\mathcal{P}=\{\mathbb{Q}_{\boldsymbol{d}}:\boldsymbol{d}\in\mathcal{D}\}$*.*

With the reformulations into $\boldsymbol{d}_{\mathbb{P}}\in\mathcal{D}$ and $\boldsymbol{z}\in\mathcal{Z}$ in place, to completely represent the original model as a problem of $\boldsymbol{z}\in\mathcal{Z}$ and $\boldsymbol{d}\in\mathcal{D}$, now it remains to redefine our objectives as functions of the updated variables. By Lemma EC.1, for any distribution $\mathbb{P}\in\mathcal{P}$, there exists $\boldsymbol{d}_{\mathbb{P}}\in\mathcal{D}$ such that $\boldsymbol{d}_{\mathbb{P}}$ captures the information from $\mathbb{P}$. Correspondingly, for any $\boldsymbol{d}\in\mathcal{D}$ and $\boldsymbol{z}\in\mathcal{Z}$, we define $FN_{\boldsymbol{d}}(\boldsymbol{z})$, $FP_{\boldsymbol{d}}(\boldsymbol{z})$ and $T_{\boldsymbol{d}}(\boldsymbol{z})$ to reformulate the objectives $\mathbb{E}_{\mathbb{P}}[T(\mathcal{X},\tilde{\boldsymbol{\xi}})],\mathbb{E}_{\mathbb{P}}[FN(\mathcal{X},\tilde{\boldsymbol{\xi}})],\mathbb{E}_{\mathbb{P}}[FP(\mathcal{X},\tilde{\boldsymbol{\xi}})]$ for any $\mathbb{P}\in\mathcal{P}$ and $\mathcal{X}\in\mathcal{B}$, as presented in the following lemma.

LEMMA EC.2. *The total counts of false negatives, false positives and tests have the following reformulations,*

$$\begin{aligned}
&\mathbb{E}_{\mathbb{P}}[FN(\mathcal{X},\tilde{\boldsymbol{\xi}})]=FN_{\boldsymbol{d}}(\boldsymbol{z}):=N\beta_n(2-\beta_n)p-\beta_n(1-\beta_n)pz_1,\\
&\mathbb{E}_{\mathbb{P}}[FP(\mathcal{X},\tilde{\boldsymbol{\xi}})]=FP_{\boldsymbol{d}}(\boldsymbol{z}):=N(1-\beta_n)\beta_p(1-p)+\beta_n\beta_p(1-p)z_1\\
&\qquad\qquad -\beta_p(1-\beta_n-\beta_p)\textstyle\sum_{n=2}^{\Gamma}\left(z_n n\left(\sum_{k=0}^{N-n}\binom{N-n}{k}d_k\right)\right),\\
&\mathbb{E}_{\mathbb{P}}[T(\mathcal{X},\tilde{\boldsymbol{\xi}})]=T_{\boldsymbol{d}}(\boldsymbol{z}):=N(1-\beta_n)+\beta_n z_1+\textstyle\sum_{n=2}^{\Gamma}\left(z_n-(1-\beta_n-\beta_p)z_n n\left(\sum_{k=0}^{N-n}\binom{N-n}{k}d_k\right)\right),
\end{aligned}$$

where $\boldsymbol{d}$ *captures the joint distribution* $\mathbb{P}$*, and* $\boldsymbol{z}$ *is associated to the partition* $\mathcal{X}$*.*

The proof of Lemma EC.1 and EC.2 is included in Section EC.C.14 and EC.C.15. Now, it remains to simplify $\lambda_1 FN_{\boldsymbol{d}}(\boldsymbol{z})+\lambda_2 FP_{\boldsymbol{d}}(\boldsymbol{z})+(1-\lambda_1-\lambda_2)T_{\boldsymbol{d}}(\boldsymbol{z})$ as follows.

$$\begin{aligned}
&\lambda_1 FN_{\boldsymbol{d}}(\boldsymbol{z})+\lambda_2 FP_{\boldsymbol{d}}(\boldsymbol{z})+(1-\lambda_1-\lambda_2)T_{\boldsymbol{d}}(\boldsymbol{z})\\
=&\lambda_1\left(N\beta_n(2-\beta_n)p-\beta_n(1-\beta_n)pz_1\right)\\
&+\lambda_2\left(N(1-\beta_n)\beta_p(1-p)+\beta_n\beta_p(1-p)z_1-\beta_p(1-\beta_n-\beta_p)\sum_{n=2}^{\Gamma}\left(z_n n\left(\sum_{k=0}^{N-n}\binom{N-n}{k}d_k\right)\right)\right)\\
&+(1-\lambda_1-\lambda_2)\left(N(1-\beta_n)+\beta_n z_1+\sum_{n=2}^{\Gamma}\left(z_n-(1-\beta_n-\beta_p)z_n n\left(\sum_{k=0}^{N-n}\binom{N-n}{k}d_k\right)\right)\right)\\
=&\lambda_1 N\beta_n(2-\beta_n)p+\lambda_2 N(1-\beta_n)\beta_p(1-p)+(1-\lambda_1-\lambda_2)N(1-\beta_n)\\
&+\left(-\lambda_1\beta_n(1-\beta_n)p+\lambda_2\beta_n\beta_p(1-p)+(1-\lambda_1-\lambda_2)\beta_n\right)z_1+(1-\lambda_1-\lambda_2)\sum_{n=2}^{\Gamma}z_n\\
&-\left(\lambda_2\beta_p(1-\beta_n-\beta_p)+(1-\lambda_1-\lambda_2)(1-\beta_n-\beta_p)\right)\sum_{n=2}^{\Gamma}\left(z_n n\left(\sum_{k=0}^{N-n}\binom{N-n}{k}d_k\right)\right)\\
=&NC_0+C_1z_1+C_2\sum_{n=2}^{\Gamma}z_n-C_3\sum_{n=2}^{\Gamma}\left(z_n n\left(\sum_{k=0}^{N-n}\binom{N-n}{k}d_k\right)\right)
\end{aligned}$$

$$= NC_0 + \sum_{n=1}^{\Gamma} n z_n h(n, \boldsymbol{d}),$$

where the first equality follows from Lemma EC.2, and other equalities follow from algebra and definitions of C_0, C_1, C_2, C_3 and $h(n, \boldsymbol{d})$ in Equation (6) and (7).

Therefore, the proof is completed. Q.E.D.

EC.C.3. Proof of Lemma 2

In this proof, we derive the worst-case distribution $\boldsymbol{d}^*$ by solving the optimization problem $\max_{\boldsymbol{d}\in\mathcal{D}} h(n, \boldsymbol{d})$ for an arbitrary n. We will discuss the scenarios: a) $n = 1$, which is straightforward; b) $n \in \{2, \ldots, \Gamma\}$. Under the second scenario, the maximization problem reduces to a linear programming, and we use strong duality to prove the optimality of $\boldsymbol{d}^*$ by the following steps: 1) reformulate the maximization problem to an equivalent maximization problem regarding $\boldsymbol{v}$; 2) define $\boldsymbol{v}^*$ by $\boldsymbol{d}^*$ based on the relationship between two equivalent maximization problems; 3) show that $\boldsymbol{v}^*$ is feasible for primal maximization problem; 4) construct the dual problem and initialize a solution $\boldsymbol{s}^*$; 5) show the equivalent objective values of primal problem under solution $\boldsymbol{d}^*$ and dual problem under solution $\boldsymbol{s}^*$; 6) prove the feasibility of $\boldsymbol{s}^*$ for the dual problem.

Recall the definition of $h(n, \boldsymbol{d})$:

$$h(n, \boldsymbol{d}) = \begin{cases} C_1 & \text{if } n = 1, \\ \dfrac{C_2}{n} - C_3 \displaystyle\sum_{k=0}^{N-n} \binom{N-n}{k} d_k & \text{if } n \geq 2. \end{cases}$$

If $n = 1$, the function $h(n, \boldsymbol{d})$ takes a constant value of C_1, hence is independent of $\boldsymbol{d}$. Therefore, we have $\sup_{\boldsymbol{d}\in\mathcal{D}} h(n, \boldsymbol{d}) = h(n, \boldsymbol{d}^*)$.

Now we consider only the case of $n \geq 2$. For any $n \in \{2, \ldots, \Gamma\}$, we have,

$$\sup_{\boldsymbol{d}\in\mathcal{D}} h(n, \boldsymbol{d}) = \sup_{\boldsymbol{d}\in\mathcal{D}} \left\{ \frac{C_2}{n} - C_3 \sum_{k=0}^{N-n} \binom{N-n}{k} d_k \right\} = \frac{C_2}{n} - C_3 \min_{\boldsymbol{d}\in\mathcal{D}} \left\{ \sum_{k=0}^{N-n} \binom{N-n}{k} d_k \right\}.$$

Hence, it remains to show that, for any $n \in \{2, \ldots, \Gamma\}$, $\boldsymbol{d}^*$ is the optimal solution of following problem,

$$\begin{aligned} \min \quad & \sum_{k=0}^{N-n} \binom{N-n}{k} d_k \\ \text{s.t.} \quad & \sum_{k=0}^{N} \binom{N}{k} d_k = 1 \\ & \sum_{k=1}^{N} \binom{N-1}{k-1} d_k = p \\ & \sum_{k=2}^{N} \binom{N-2}{k-2} d_k = p^2 + \rho p(1-p) \\ & \boldsymbol{d} \geq \boldsymbol{0} \end{aligned} \tag{EC.3}$$

For Problem (EC.3), the constraints are complex because of combinatorial numbers. To simplify the constraints, we change the decision variables from $\boldsymbol{d}$ to $\boldsymbol{v}$ by choosing $v_k = \binom{N}{k} d_k$. In this case, we have

$$\begin{aligned}
&\tbinom{N-n}{k} d_k = \frac{\binom{N-n}{k}\binom{N}{k} d_k}{\binom{N}{k}} = \frac{\binom{N-n}{k}}{\binom{N}{k}} v_k = \frac{(N-n)!}{N!}\frac{(N-k)!}{(N-n-k)!} v_k && \forall k \in \{0,\dots,N-n\},\\
&\tbinom{N-1}{k-1} d_k = \frac{(N-1)!}{(k-1)!(N-k)!} d_k = \frac{k}{N}\frac{N!}{k!(N-k)!} d_k = \frac{k}{N}\tbinom{N}{k} d_k = \frac{k}{N} v_k && \forall k \in \{1,\dots,N\},\\
&\tbinom{N-2}{k-2} d_k = \frac{(N-2)!}{(k-2)!(N-k)!} d_k = \frac{k(k-1)}{N(N-1)}\frac{N!}{k!(N-k)!} d_k = \frac{k(k-1)}{N(N-1)}\tbinom{N}{k} d_k = \frac{k(k-1)}{N(N-1)} v_k && \forall k \in \{2,\dots,N\}.
\end{aligned}$$

Therefore, the optimal $\boldsymbol{d}$ for Problem (EC.3) can be obtained via solving from the following problem,

$$\begin{aligned}
&\min\ \boldsymbol{c}^\top \boldsymbol{v}\\
&\text{s.t. } \boldsymbol{A}\boldsymbol{v} = \boldsymbol{b},\ \boldsymbol{v} \geq \boldsymbol{0}
\end{aligned} \tag{EC.4}$$

where $\boldsymbol{c} \in \mathbb{R}^{N+1}$ with $c_k = \sum_{k=0}^{N-n} \frac{(N-k)!}{(N-n-k)!}$ for $k \in \{0,\dots,N-n\}$, $\boldsymbol{b} \in \mathbb{R}^3$ and $\boldsymbol{A} \in \mathbb{R}^{3\times(N+1)}$ as follows,

$$\boldsymbol{A} = \begin{pmatrix} 1 & 1 & 1 & 1 & \cdots & 1 \\ 0 & 1 & 2 & 3 & \cdots & N \\ 0 & 0 & 2 & 6 & \cdots & N(N-1) \end{pmatrix},\ \boldsymbol{b} = \begin{pmatrix} 1 \\ Np \\ N(N-1)(p^2+\rho p(1-p)) \end{pmatrix}.$$

Problem (EC.4) is a standard LP with full rank 3. Hence, an optimal extreme point can be found in a solution $\hat{\boldsymbol{v}}$ that includes at most 3 components corresponding to a basis matrix $\boldsymbol{B}$, where $\boldsymbol{B}$ is comprises of 3 linearly independent columns of $\boldsymbol{A}$. We select column $\bar{k}$, $\bar{k}+1$ and N to define $\boldsymbol{v}^*$, i.e., $\begin{pmatrix} v^*_{\bar{k}} \\ v^*_{\bar{k}+1} \\ v^*_N \end{pmatrix} = \boldsymbol{B}^{-1}\boldsymbol{b}$ and $v_k = 0$ for any $k \notin \{\bar{k}, \bar{k}+1, N\}$ where $\bar{k}$ is defined in Lemma 2 and

$$\boldsymbol{B} = \begin{pmatrix} 1 & 1 & 1 \\ \bar{k} & \bar{k}+1 & N \\ \bar{k}(\bar{k}-1) & \bar{k}(\bar{k}+1) & N(N-1) \end{pmatrix}.$$

It is straightforwards that $v^*_k = \binom{N}{k} d^*_k$ for any $k \in \{0,1,\dots,N\}$, where $\boldsymbol{d}^*$ is defined in Equation (8). Specifically, we have,

$$\begin{aligned}
&v^*_{\bar{k}} = \frac{N(1-p)}{N-\bar{k}}\left(\bar{k}+1-(N-1)p(1-\rho)\right)\\
&v^*_{\bar{k}+1} = \frac{N(1-p)}{N-\bar{k}-1}\left(-\bar{k}+(N-1)p(1-\rho)\right)\\
&v^*_N = \frac{\bar{k}^2+(1-2Np)\bar{k}+N(N-1)(p^2+\rho p(1-p))}{(N-\bar{k})(N-\bar{k}-1)}\\
&v^*_k = 0 \qquad \forall k \in \{0,1,...,N\}\backslash\{\bar{k},\bar{k}+1,N\}
\end{aligned} \tag{EC.5}$$

and the objective value under $\boldsymbol{v}^*$ is $\boldsymbol{c}^\top \boldsymbol{v}^* = \boldsymbol{c}_{\boldsymbol{B}}^\top \boldsymbol{B}^{-1}\boldsymbol{b}$, where $\boldsymbol{c}_{\boldsymbol{B}} = \begin{pmatrix} c_{\bar{k}} \\ c_{\bar{k}+1} \\ c_N \end{pmatrix}$.

Then, to discuss the optimal solution to Problem (EC.3), it remains to show that for any $n \in \{2,\dots,\Gamma\}$, $\boldsymbol{v}^*$ is the optimal solution of Problem (EC.4) by the following four steps. We first prove

that the $\boldsymbol{v}^*$ defined by (EC.5) always exists (step 1) and is feasible for Problem (EC.4) (step 2). Then, we write down the dual of Problem (EC.4) to find a dual solution and prove the dual problem under the constructed dual solution has the same objective value of Problem (EC.4) under solution $\boldsymbol{v}^*$ (step 3). Finally, we prove the feasibility of the dual solution (step 4). Based these four steps, we prove the optimality of $\boldsymbol{v}^*$ by strong duality of linear programming.

(1) The existecne of $\bar{k}$

It suffices to show that $\bar{k} = \lfloor (N-1)p(1-\rho) \rfloor \in \{0, 1, \ldots, N-2\}$. According to $p \in (0,1)$ and $\rho \in [0,1)$, we have $0 \le \bar{k} \le N-2$ due to $0 < (N-1)p(1-\rho) < N-1$.

(2) Primal feasibility: $\boldsymbol{v}^*$ defined by (EC.5) is feasible for Problem (EC.4)

Note that $\boldsymbol{A}\boldsymbol{v}^* = \boldsymbol{b}$ holds due to the definition of $\boldsymbol{v}^*$. It remains to check the non-negativity constraint, i.e., $\boldsymbol{v} \ge \boldsymbol{0}$. $v^*_{\bar{k}} \ge 0$ is true as $\bar{k} > (N-1)p(1-\rho) - 1$, and $v^*_{\bar{k}+1} \ge 0$ follows from $\bar{k} \le (N-1)p(1-\rho)$.

Hence, it remains to show $v^*_N \ge 0$, i.e.,

$$\frac{\bar{k}^2 + (1-2Np)\bar{k} + N(N-1)(p^2 + \rho p(1-p))}{(N-\bar{k})(N-\bar{k}-1)} \ge 0.$$

Define $f(\bar{k}) = \bar{k}^2 + (1-2Np)\bar{k} + N(N-1)(p^2 + \rho p(1-p))$. Since $(N-\bar{k})(N-\bar{k}-1) \ge 0$ (which is due to $\bar{k} \le N-2$), now it suffices to show $\min_{\bar{k} \in \{0,1,\ldots,N-2\}} f(\bar{k}) \ge 0$. Note that $f(\bar{k})$ is a convex quadratic function in $\bar{k}$. Hence, denoting $\bar{k}^* = \arg\min_{\bar{k} \in \mathbb{R}} f(\bar{k})$, by first-order-condition, we can easily get $\bar{k}^* = Np - \frac{1}{2}$. We discuss the non-negativity of $\min_{\bar{k} \in \{0,1,\ldots,N-2\}} f(\bar{k})$ by 3 cases.

Case 1: $\bar{k}^* \le 0$, i.e., $p \le \frac{1}{2N}$, then,

$$\min_{\bar{k} \in \{0,1,\ldots,N-2\}} f(\bar{k}) = f(0) = N(N-1)(p^2 + \rho p(1-p)) \ge 0.$$

Case 2: $\bar{k}^* \ge N-1$, i.e., $p \ge 1 - \frac{1}{2N}$, then,

$$\begin{aligned}
\min_{\bar{k} \in \{0,1,\ldots,N-2\}} f(\bar{k}) &= f(N-2) \\
&\ge f(N-1) \\
&= (N-1)^2 + (1-2Np)(N-1) + N(N-1)(p^2 + \rho p(1-p)) \\
&= N(N-1)(p^2 - 2p + 1 + \rho p(1-p)) \\
&= N(N-1)((p-1)^2 + \rho p(1-p)) \\
&\ge 0,
\end{aligned}$$

where the first inequality follows from properties of convex quadratic functions.

Case 3: $0 < \bar{k}^* < N-1$, i.e., $\frac{1}{2N} < p < 1 - \frac{1}{2N}$, then,

$$\begin{aligned}\min_{\bar{k}\in\{0,1,\ldots,N-2\}} f(\bar{k}) &\geq f(\bar{k}^*)\\ &= -\left(Np - \frac{1}{2}\right)^2 + N(N-1)[p^2 + \rho p(1-p)]\\ &= -Np^2 + Np - \frac{1}{4} + N(N-1)\rho p(1-p)\\ &\geq -Np^2 + Np - \frac{1}{4}.\end{aligned}$$

Define $h(p) = -Np^2 + Np - \frac{1}{4}$. Then, as $h(p)$ is concave in p, we have,

$$\min_{\frac{1}{2N}<p<1-\frac{1}{2N}} h(p) > \min\left\{h\left(\frac{1}{2N}\right), h\left(1 - \frac{1}{2N}\right)\right\}.$$

Moreover, by simple algebra,

$$h\left(\frac{1}{2N}\right) = h\left(1 - \frac{1}{2N}\right) = \frac{1}{4} - \frac{1}{4N} > 0.$$

Then, $\min_{\bar{k}\in\{0,1,\ldots,N-2\}} f(\bar{k}) \geq \min_{\frac{1}{2N}<p<1-\frac{1}{2N}} h(p) > 0$.

Summarizing the three cases above, $\min_{\bar{k}\in\{0,1,\ldots,N-2\}} f(\bar{k}) \geq 0$ always holds, which implies $v_N^* \geq 0$.

(3) Dual of Problem (EC.4) with a solution $\boldsymbol{s}^*$ and same objective values

The dual problem of Problem (EC.4) is as follows,

$$\max\left\{\boldsymbol{b}^\top \boldsymbol{s}\right\} \quad \text{s.t.} \quad \boldsymbol{A}^\top \boldsymbol{s} \leq \boldsymbol{c}. \tag{EC.6}$$

Note that the optimal solution to Problem (EC.6) satisfies the complementary slackness for primal-dual pair. Then, we construct a solution $\boldsymbol{s}^*$ such that

$$\boldsymbol{A}_k^\top \boldsymbol{s}^* = c_k \quad \forall k \in \{\bar{k}, \bar{k}+1, N\} \Leftrightarrow \boldsymbol{B}^\top \boldsymbol{s}^* = \begin{pmatrix} c_{\bar{k}} \\ c_{\bar{k}+1} \\ c_N \end{pmatrix} = \boldsymbol{c}_{\boldsymbol{B}} \Leftrightarrow \boldsymbol{s}^* = (\boldsymbol{B}^{-1})^\top \boldsymbol{c}_{\boldsymbol{B}},$$

where $\boldsymbol{A}_k$ denotes the k-th column of matrix $\boldsymbol{A}$. Specifically, as $c_N = 0$ always holds, we have the following closed form of $\boldsymbol{s}^*$,

$$\begin{aligned} s_1^* &= \frac{N(\bar{k}+1)}{N-\bar{k}} c_{\bar{k}} - \frac{N\bar{k}}{N-\bar{k}-1} c_{\bar{k}+1},\\ s_2^* &= -\frac{N+\bar{k}}{N-\bar{k}} c_{\bar{k}} + \frac{N+\bar{k}-1}{N-\bar{k}-1} c_{\bar{k}+1},\\ s_3^* &= \frac{1}{N-\bar{k}} c_{\bar{k}} - \frac{1}{N-\bar{k}-1} c_{\bar{k}+1},\end{aligned} \tag{EC.7}$$

and the corresponding objective value for Problem (EC.6) is $\boldsymbol{b}^\top \boldsymbol{s}^* = \boldsymbol{b}^\top (\boldsymbol{B}^{-1})^\top \boldsymbol{c}_{\boldsymbol{B}}$. Recall the objective value of Problem (EC.4) under $\boldsymbol{v}^*$ is $\boldsymbol{c}_{\boldsymbol{B}}^\top \boldsymbol{B}^{-1} \boldsymbol{b}$. Then, it is obvious that the objective value of Problem (EC.4) under $\boldsymbol{v}^*$ is equivalent to the objective value of Problem (EC.6) under $\boldsymbol{s}^*$.

(4) Feasibility of s^* in the dual problem (EC.6)

Define $g(k) = RHS(k) - LHS(k)$, where $RHS(k)$ and $LHS(k)$ denote the right-hand-side and the left-hand-side values of the k-th constraint in Problem (EC.6) under dual variable $\boldsymbol{s}^*$, respectively. That is,

$$g(k) = c_k - (s_1^* + ks_2^* + k(k-1)s_3^*) \qquad \forall k \in \{0, 1, \cdots, N\}. \tag{EC.8}$$

To prove the feasibility of $\boldsymbol{s}^*$, we will show $g(k) \geq 0$ for $k \in \{0, 1, \ldots, N\}$ in three different scenarios.

(4.1) Case 1: $\bar{k} \in \{0, \cdots, N-n-1\}$

In this case, $c_{\bar{k}} = \frac{(N-\bar{k})!}{(N-n-\bar{k})!}$, $c_{\bar{k}+1} = \frac{(N-\bar{k}-1)!}{(N-n-\bar{k}-1)!}$, and

$$\begin{aligned}
s_1^* &= \frac{N(\bar{k}+1)}{N-\bar{k}} c_{\bar{k}} - \frac{N\bar{k}}{N-\bar{k}-1} c_{\bar{k}+1} = \frac{(N-\bar{k}-2)!}{(N-n-\bar{k})!} N(N-1+(n-2)\bar{k}), \\
s_2^* &= -\frac{N+\bar{k}}{N-\bar{k}} c_{\bar{k}} + \frac{N+\bar{k}-1}{N-\bar{k}-1} c_{\bar{k}+1} = -\frac{(N-\bar{k}-2)!}{(N-n-\bar{k})!}((N-1)n+\bar{k}(n-2)), \\
s_3^* &= \frac{1}{N-\bar{k}} c_{\bar{k}} - \frac{1}{N-\bar{k}-1} c_{\bar{k}+1} = \frac{(N-\bar{k}-2)!}{(N-n-\bar{k})!}(n-1).
\end{aligned}$$

First, simplify $LHS(k), RHS(k)$ by the updated values of $c_{\bar{k}}$ and $c_{\bar{k}+1}$ as follows,

$$\begin{aligned}
LHS(k) &= \frac{(N-\bar{k}-2)!}{(N-n-\bar{k})!} \left(N(N-1+(n-2)\bar{k}) - k((N-1)n+\bar{k}(n-2)) + k(k-1)(n-1)\right) \\
&= \frac{(N-\bar{k}-2)!}{(N-n-\bar{k})!} \left((n-1)k^2 - (Nn-1+\bar{k}(n-2))k + N(N-1+(n-2)\bar{k})\right) \qquad \forall k \in \{0, 1, \ldots, N\}, \\
RHS(k) &= \begin{cases} \dfrac{(N-k)!}{(N-n-k)!} & \text{if } k \in \{0, 1, \ldots, N-n\}, \\ 0 & \text{if } k \in \{N-n+1, \ldots, N\}. \end{cases}
\end{aligned}$$

Next, we show $g(k) = RHS(k) - LHS(k) \geq 0$ for any k in $\{0, 1, \cdots, N\}$.

For any k in $\{N-n+1, \cdots, N\}$, we have

$$g(k) = \frac{(N-\bar{k}-2)!}{(N-n-\bar{k})!} \left(-(n-1)k^2 + (Nn-1+\bar{k}(n-2))k - N(N-1+(n-2)\bar{k})\right), \tag{EC.9}$$

which is concave in k. By concavity, for any k in $\{N-n+1, \cdots, N\}$, we have,

$$g(k) \geq \min\{g(N-n+1), g(N)\}.$$

Hence, if suffices to show $g(N-n+1) \geq 0$ and $g(N) \geq 0$.

For $g(N-n+1)$, in (EC.9), replacing k by $(N-n+1)$ and multiplying $\frac{(N-n-\bar{k})!}{(N-\bar{k}-2)!}$, it has the same sign as

$$-(n-1)(N-n+1)^2 + (Nn-1+\bar{k}(n-2))(N-n+1) - N(N-1+(n-2)\bar{k})$$

$$
\begin{aligned}
&=\bar{k}((n-2)(N-n+1)-N(n-2))-(n-1)(N-n+1)^2+(Nn-1)(N-n+1)-N(N-1)\\
&=-\bar{k}(n-1)(n-2)+(N-n)(n-1)(n-2)\\
&=(N-n-\bar{k})(n-1)(n-2)\\
&\geq 0,
\end{aligned}
$$

where the inequality holds since $n \geq 2$ and $\bar{k} \in \{0, \cdots, N-n-1\}$.

For $g(N)$, in (EC.9), replacing k by N and multiplying $\frac{(N-n-\bar{k})!}{(N-\bar{k}-2)!}\frac{1}{N}$, it has the same sign as

$$
-(n-1)N+(Nn-1+\bar{k}(n-2))-(N-1+(n-2)\bar{k})=0.
$$

Therefore, $g(k) \geq 0$ for any k in $\{N-n+1, \cdots, N\}$.

For any k in $\{0, \cdots, N-n\}$, we have,

$$
\begin{aligned}
g(k) =& RHS(k)-LHS(k)\\
=& \frac{(N-k)!}{(N-n-k)!}-\frac{(N-\bar{k}-2)!}{(N-\bar{k}-n)!}\left((n-1)k^2-(Nn-1+\bar{k}(n-2))k+N(N-1+(n-2)\bar{k})\right)\\
=& \frac{(N-k)(N-k-1)(N-k-2)!}{(N-n-k)!}\frac{(N-\bar{k}-n)!}{(N-\bar{k}-2)!}\frac{(N-\bar{k}-2)!}{(N-\bar{k}-n)!}\\
&-\frac{(N-\bar{k}-2)!}{(N-\bar{k}-n)!}\left[(n-1)k^2-(Nn-1+\bar{k}(n-2))k+N(N-1+(n-2)\bar{k}).\right]
\end{aligned}
$$

Define $t_k=\frac{(N-k-2)!}{(N-k-n)!}\frac{(N-\bar{k}-n)!}{(N-\bar{k}-2)!}=\frac{(N-k-2)!}{(N-k-n)!}/\frac{(N-\bar{k}-2)!}{(N-\bar{k}-n)!}$, then,

$$
\begin{aligned}
g(k) =& (N-k)(N-k-1)t_k\frac{(N-\bar{k}-2)!}{(N-\bar{k}-n)!}\\
&-\frac{(N-\bar{k}-2)!}{(N-\bar{k}-n)!}\left((n-1)k^2-(Nn-1+\bar{k}(n-2))k+N(N-1+(n-2)\bar{k})\right)\\
=& \frac{(N-\bar{k}-2)!}{(N-\bar{k}-n)!}((N-k)(N-k-1)t_k-(n-1)k^2+(Nn-1+\bar{k}(n-2))k-N(N-1+(n-2)\bar{k}))\\
=& \frac{(N-\bar{k}-2)!}{(N-\bar{k}-n)!}(N-k)\left((N-k-1)(t_k-1)+(n-2)(k-\bar{k})\right).
\end{aligned}
$$

Define $u(k)=(N-k-1)(t_k-1)+(n-2)(k-\bar{k})$. Then, it is equivalent to show $u(k) \geq 0$ for any k in $\{0, \cdots, N-n\}$.

For any k in $\{0, \cdots, N-n-1\}$, we calculate the difference between $u(k)$ and $u(k+1)$ as follows,

$$
\begin{aligned}
&u(k)-u(k+1)\\
=&(N-k-1)(t_k-1)+(n-2)(k-\bar{k})-(N-k-2)(t_{k+1}-1)-(n-2)(k+1-\bar{k})\\
=&(N-k-1)t_k-(N-k-2)t_{k+1}-(n-1)\\
=&(N-k-1)\frac{(N-k-2)!}{(N-k-n)!}\frac{(N-\bar{k}-n)!}{(N-\bar{k}-2)!}-(N-k-2)\frac{(N-k-3)!}{(N-k-n-1)!}\frac{(N-\bar{k}-n)!}{(N-\bar{k}-2)!}-(n-1)
\end{aligned}
$$

$$
\begin{aligned}
&=\frac{(N-k-1)!}{(N-k-n)!}\frac{(N-\bar{k}-n)!}{(N-\bar{k}-2)!}-\frac{(N-k-2)!}{(N-k-n-1)!}\frac{(N-\bar{k}-n)!}{(N-\bar{k}-2)!}-(n-1)\\
&=\frac{(N-k-2)!}{(N-k-n)!}\frac{(N-\bar{k}-n)!}{(N-\bar{k}-2)!}((N-k-1)-(N-k-n))-(n-1)\\
&=(n-1)\left(\frac{(N-k-2)!}{(N-k-n)!}\frac{(N-\bar{k}-n)!}{(N-\bar{k}-2)!}-1\right)\\
&=(n-1)(t_k-1).
\end{aligned}
$$

Note that $t_k=\frac{(N-k-2)!}{(N-k-n)!}\frac{(N-\bar{k}-n)!}{(N-\bar{k}-2)!}$ decreases in k and $t_{\bar{k}}=1$. Hence, $t_k\geq t_{\bar{k}}=1$ for k in $\{0,\cdots,\bar{k}\}$, and $t_k\leq t_{\bar{k}}=1$ for k in $\{\bar{k}+1,\ldots,N-n\}$. Then, we derive that, $u(k)-u(k+1)\geq 0$ for k in $\{0,\cdots,\bar{k}\}$, and $u(k)-u(k+1)\leq 0$ for k in $\{\bar{k}+1,\cdots,N-n\}$. That is,

$$u(0)\geq u(1)\geq\cdots\geq u(\bar{k})=u(\bar{k}+1)\leq\cdots\leq u(N-n),$$

where $u(\bar{k})=u(\bar{k}+1)$ follows from $t_k=1$. Then, for any k in $\{0,1,\ldots,N-n\}$, we have,

$$u(k)\geq u(\bar{k}+1)=u(\bar{k})=0,$$

where $u(\bar{k})=0$ follows from $t_{\bar{k}}=1$ and the definition of $u(k)$; hence, $g(k)\geq 0$ for any k in $\{0,1,\ldots,N-n\}$.

To sum up, in the case that $\bar{k}\in\{0,\cdots,N-n-1\}$, we have $g(k)\geq 0$, for any $k\in\{0,\ldots,N\}$, which implies that $\boldsymbol{s}^*$ defined by (EC.7) is feasible for Problem (EC.6).

(4.2) Case 2: $\bar{k}=N-n$

In this case, $c_{\bar{k}}=n!$, $c_{\bar{k}+1}=0$ and

$$
\begin{aligned}
s_1^*&=N(N-n+1)(n-1)!,\\
s_2^*&=-(2N-n)(n-1)!,\\
s_3^*&=(n-1)!.
\end{aligned}
$$

First, we simplify $LHS(k),RHS(k)$ by updated $c_{\bar{k}}$ and $c_{\bar{k}+1}$ as follows,

$$
\begin{aligned}
&LHS(k)=(n-1)!(k^2-(2N-n+1)k+N(N-n+1)) && \forall k\in\{0,\cdots,N\},\\
&RHS(k)=\begin{cases}\dfrac{(N-k)!}{(N-n-k)!} & \text{if } k\in\{0,\cdots,N-n\},\\ 0, & \text{if } k\in\{N-n+1,\cdots,N\}.\end{cases}
\end{aligned}
$$

For $k\in\{N-n+1,\cdots,N\}$,

$$g(k)=-(n-1)!(k^2-(2N-n+1)k+N(N-n+1)),$$

which is concave in k. At the two extreme points, $N-n+1$ and N, by simple algebra we can easily have $g(N-n+1)=g(N)=0$. Therefore, the concavity implies $\min_{k\in\{N-n+1,\cdots,N\}}g(k)=\min\{g(N-n+1),g(N)\}=0$.

For k in $\{0,\cdots,N-n\}$, then,

$$g(k)=\frac{(N-k)!}{(N-n-k)!}-(n-1)!(k^2-(2N-n+1)k+N(N-n+1)). \tag{EC.10}$$

Given any k in $\{0,\cdots,N-n-1\}$, we now show that $g(k)-g(k+1)\geq 0$ as follows,

$$\begin{aligned}
&g(k)-g(k+1)\\
=&\frac{(N-k)!}{(N-n-k)!}-(n-1)!(k^2-(2N-n+1)k+N(N-n+1))\\
&-\frac{(N-k-1)!}{(N-n-k-1)!}+(n-1)!((k+1)^2-(2N-n+1)(k+1)+N(N-n+1))\\
=&\frac{n(N-k-1)!}{(N-n-k)!}+(n-1)!(2k+n-2N)\\
\geq&n(N-k-1)(n-1)!+(n-1)!(2k+n-2N)\\
=&(n-1)!(n-2)(N-k)\\
\geq&0
\end{aligned}$$

where the first inequality follows from $\frac{(N-k-2)!}{(N-k-n)!}\geq(n-1)!$, which is proved by the following steps. If $n=2$, then, $\frac{(N-k-2)!}{(N-k-n)!}=1=(n-1)!$. If $n\geq 3$, then,

$$\frac{(N-k-2)!}{(N-k-n)!}=(N-k-2)(N-k-3)\cdots(N-k-n+1)\geq(n-1)(n-2)\cdots 2=(n-1)!$$

since $N-k-2\geq n-1$, $N-k-3\geq n-2$, $\cdots$, and $N-k-n+1\geq 2$ (due to $k\leq N-n-1$).

Then, as $g(k)-g(k+1)\geq 0$ for any k in $\{0,\cdots,N-n-1\}$, we can conclude

$$g(0)\geq g(1)\geq\cdots\geq g(N-n)=0,$$

where $g(N-n)=0$ follows from simply algebra by replacing k with $(N-n)$ in (EC.10).

Therefore, when $\bar{k}=N-n$, we have $g(k)\geq 0$, which implies $\boldsymbol{s}^*$ is feasible for Problem (EC.6).

(4.3) Case 3: $\bar{k}\in\{N-n+1,\cdots,N\}$

In this case, $c_{\bar{k}}=c_{\bar{k}+1}=0$ and $\boldsymbol{s}^*=\mathbf{0}$. Hence, for any $k\in\{0,1,\ldots,N\}$, we have $g(k)=b_k-0\geq 0$, and thus, dual feasibility holds.

In summary, $\boldsymbol{v}^*$ exists and is feasible for the primal problem (EC.4), $\boldsymbol{s}^*$ is feasible for the dual problem (EC.6); moreover, the objective value of Problem (EC.4) under solution $\boldsymbol{v}^*$ is equal to the objective value of Problem (EC.6) under solution $\boldsymbol{s}^*$. According to strong duality, $\boldsymbol{v}^*$ is optimal for Problem (EC.4), and hence $\boldsymbol{d}^*$ is optimal for Problem (EC.3). Q.E.D.

EC.C.4. Proof of Theorem 1

In this part, we justify the optimal pair $(\boldsymbol{z}^*, \boldsymbol{d}^*)$ by showing that

$$\min_{\boldsymbol{z}\in\mathcal{Z}} \max_{\boldsymbol{d}\in\mathcal{D}} \sum_{n=1}^{\Gamma} n z_n h(n, \boldsymbol{d}) = \sum_{n=1}^{\Gamma} n z_n^* h(n, \boldsymbol{d}^*) = N h(n^*, \boldsymbol{d}^*),$$

where $\boldsymbol{d}^*$ is defined in Lemma 2, $n^* \in \arg\min_{n\in[\Gamma]} h(n, \boldsymbol{d}^*)$, and $z_{n^*}^* = \frac{N}{n^*}$ and $z_n^* = 0$ for any $n \neq n^*$.

For any $\boldsymbol{z} \in \mathcal{Z}$, we have

$$\max_{\boldsymbol{d}\in\mathcal{D}} \sum_{n=1}^{\Gamma} n z_n h(n, \boldsymbol{d}) \leq \sum_{n=1}^{\Gamma} n z_n \max_{\boldsymbol{d}\in\mathcal{D}} h(n, \boldsymbol{d}) = \sum_{n=1}^{\Gamma} n z_n h(n, \boldsymbol{d}^*) \leq \max_{\boldsymbol{d}\in\mathcal{D}} \sum_{n=1}^{\Gamma} n z_n h(n, \boldsymbol{d}),$$

where the inequalities follow from the nature of maximization problem; the equality follows from $\boldsymbol{d}^* \in \arg\max_{\boldsymbol{d}\in\mathcal{D}} h(n, \boldsymbol{d})$ for an arbitrary $n \in [\Gamma]$, as shown in Lemma 2. Hence, it is straightforward that $\max_{\boldsymbol{d}\in\mathcal{D}} \sum_{n=1}^{\Gamma} n z_n h(n, \boldsymbol{d}) = \sum_{n=1}^{\Gamma} n z_n h(n, \boldsymbol{d}^*)$ for any $\boldsymbol{z} \in \mathcal{Z}$. Then, for any $\boldsymbol{z} \in \mathcal{Z}$,

$$\max_{\boldsymbol{d}\in\mathcal{D}} \sum_{n=1}^{\Gamma} n z_n h(n, \boldsymbol{d}) = \sum_{n=1}^{\Gamma} n z_n h(n, \boldsymbol{d}^*) \geq \sum_{n=1}^{\Gamma} n z_n h(n^*, \boldsymbol{d}^*) = N h(n^*, \boldsymbol{d}^*),$$

where the inequality follows from the definition of n^*. Next, we obtain a lower bound of $\max_{\boldsymbol{d}\in\mathcal{D}} \sum_{n=1}^{\Gamma} n z_n h(n, \boldsymbol{d})$ for any $\boldsymbol{z} \in \mathcal{Z}$ and the bound is tight for $(\boldsymbol{z}^*, \boldsymbol{d}^*)$ where $z_{n^*}^* = \frac{N}{n^*}$ and $z_n = 0$ for any $n \neq n^*$.

Therefore, according to the definition of $\boldsymbol{z}^*$,

$$\min_{\boldsymbol{z}\in\mathcal{Z}} \max_{\boldsymbol{d}\in\mathcal{D}} \sum_{n=1}^{\Gamma} n z_n h(n, \boldsymbol{d}) = \sum_{n=1}^{\Gamma} n z_n^* h(n, \boldsymbol{d}^*) = N h(n^*, \boldsymbol{d}^*),$$

which concludes the optimal solution $(\boldsymbol{z}^*, \boldsymbol{d}^*)$ to $\min_{\boldsymbol{z}\in\mathcal{Z}} \max_{\boldsymbol{d}\in\mathcal{D}} \sum_{n=1}^{\Gamma} n z_n h(n, \boldsymbol{d})$. According to the definition of $(\boldsymbol{z}^*, \boldsymbol{d}^*)$, $\boldsymbol{z}^*$ has one non-zero element and $\boldsymbol{d}^*$ has three non-zero components.

We further build up minimax theorem to Problem (5), i.e.,

$$\min_{\boldsymbol{z}\in\mathcal{Z}} \max_{\boldsymbol{d}\in\mathcal{D}} \sum_{n=1}^{\Gamma} n z_n h(n, \boldsymbol{d}) = \max_{\boldsymbol{d}\in\mathcal{D}} \min_{\boldsymbol{z}\in\mathcal{Z}} \sum_{n=1}^{\Gamma} n z_n h(n, \boldsymbol{d}),$$

which will be used to analyze the worst-case scenario in Section 4.2. For any $\boldsymbol{d} \in \mathcal{D}$, we denote $n_{\boldsymbol{d}}^* \in \arg\min_{n\in[\Gamma]} h(n, \boldsymbol{d})$ and have the following results,

$$\sum_{n=1}^{\Gamma} n z_n h(n, \boldsymbol{d}) \geq \sum_{n=1}^{\Gamma} n z_n h(n_{\boldsymbol{d}}^*, \boldsymbol{d}) = N h(n_{\boldsymbol{d}}^*, \boldsymbol{d}) \qquad \forall \boldsymbol{z} \in \mathcal{Z},$$

where the first inequality follows from the definition of $n_{\boldsymbol{d}}^*$. Note that $n_{\boldsymbol{d}}^*$ is equivalent to $\boldsymbol{z}_{\boldsymbol{d}}^*$ where $z_{\boldsymbol{d},n_{\boldsymbol{d}}^*}^* = \frac{N}{n_{\boldsymbol{d}}^*}$ and $z_{\boldsymbol{d},n}^* = 0$ for $n \neq n_{\boldsymbol{d}}^*$. Hence, it is straightforward that for any $\boldsymbol{d} \in \mathcal{D}$, we have $\min_{\boldsymbol{z}\in\mathcal{Z}} \sum_{n=1}^{\Gamma} n z_n h(n, \boldsymbol{d}) = N h(n_{\boldsymbol{d}}^*, \boldsymbol{d})$. Next, we have

$$\max_{\boldsymbol{d}\in\mathcal{D}} \min_{\boldsymbol{z}\in\mathcal{Z}} \sum_{n=1}^{\Gamma} n z_n h(n, \boldsymbol{d}) = \max_{\boldsymbol{d}\in\mathcal{D}} N h(n_{\boldsymbol{d}}^*, \boldsymbol{d}) \geq N h(n_{\boldsymbol{d}^*}^*, \boldsymbol{d}^*) = N h(n^*, \boldsymbol{d}^*),$$

where the inequality follows from $\boldsymbol{d}^* \in \mathcal{D}$ and the second equality follows from the definition of n^* and $n^*_{\boldsymbol{d}}$. Combining the above results, we have

$$\max_{\boldsymbol{d}\in\mathcal{D}} \min_{\boldsymbol{z}\in\mathcal{Z}} \sum_{n=1}^{\Gamma} n z_n h(n,\boldsymbol{d}) \geq N h(n^*,\boldsymbol{d}^*) = \min_{\boldsymbol{z}\in\mathcal{Z}} \max_{\boldsymbol{d}\in\mathcal{D}} \sum_{n=1}^{\Gamma} n z_n h(n,\boldsymbol{d}).$$

Note that according to max-min inequality (see, for instance Equation (5.46) in Boyd et al. (2004)),

$$\max_{\boldsymbol{d}\in\mathcal{D}} \min_{\boldsymbol{z}\in\mathcal{Z}} \sum_{n=1}^{\Gamma} n z_n h(n,\boldsymbol{d}) \leq \min_{\boldsymbol{z}\in\mathcal{Z}} \max_{\boldsymbol{d}\in\mathcal{D}} \sum_{n=1}^{\Gamma} n z_n h(n,\boldsymbol{d}).$$

Therefore, we conclude minimax theorem in this case. Q.E.D.

EC.C.5. Proof of Equation (9)

According to Lemma 2, we have

$$\max_{\boldsymbol{d}\in\mathcal{D}} \sum_{n=1}^{\Gamma} n z_n h(n,\boldsymbol{d}) = \sum_{n=1}^{\Gamma} n z_n h(n,\boldsymbol{d}^*), \quad \forall \boldsymbol{z}\in\mathcal{Z}.$$

Then, we have,

$$\min_{\boldsymbol{z}\in\mathcal{Z}} \max_{\boldsymbol{d}\in\mathcal{D}} \sum_{n=1}^{\Gamma} n z_n h(n,\boldsymbol{d}) = \min_{\boldsymbol{z}\in\mathcal{Z}} \sum_{n=1}^{\Gamma} n z_n h(n,\boldsymbol{d}^*) \leq \max_{\boldsymbol{d}\in\mathcal{D}} \min_{\boldsymbol{z}\in\mathcal{Z}} \sum_{n=1}^{\Gamma} n z_n h(n,\boldsymbol{d}),$$

where the equality follows from Lemma 2, and the inequality holds as $\boldsymbol{d}^* \in \mathcal{D}$.

Due to the max-min inequality (see, for instance, Equation (5.46) in Boyd et al. (2004)), we have,

$$\min_{\boldsymbol{z}\in\mathcal{Z}} \max_{\boldsymbol{d}\in\mathcal{D}} \sum_{n=1}^{\Gamma} n z_n h(n,\boldsymbol{d}) \geq \max_{\boldsymbol{d}\in\mathcal{D}} \min_{\boldsymbol{z}\in\mathcal{Z}} \sum_{n=1}^{\Gamma} n z_n h(n,\boldsymbol{d}).$$

Therefore, we conclude Equation (9).

Q.E.D.

EC.C.6. Proof of Proposition 2

Leveraging the results in Lemma 2, the following reformulation holds for Problem (5):

$$NC_0 + \min_{\boldsymbol{z}\in\mathcal{Z}} \max_{\boldsymbol{d}\in\mathcal{D}} \sum_{n=1}^{\Gamma} n z_n h(n,\boldsymbol{d}) = NC_0 + \min_{n\in[\Gamma]} \max_{\boldsymbol{d}\in\mathcal{D}} N h(n,\boldsymbol{d}) = NC_0 + N \min_{n\in[\Gamma]} h(n,\boldsymbol{d}^*). \tag{EC.11}$$

Here, the first equality follows from Theorem 1 — to identify the optimal partition, it suffices to only choose an $n \in [\Gamma]$ such that $z_n = N/n$ and $z_i = 0$ for all $i \neq n$. The second equality follows from the worst-case nature of $\boldsymbol{d}^*$ in Lemma 2. Consequently, solving Problem (5) is equivalent to finding an optimal group size n^* such that

$$n^* \in \arg\min_{n\in[\Gamma]} h(n,\boldsymbol{d}^*),$$

with the optimal objective value $NC_0 + N h(n,\boldsymbol{d}^*)$.

Therefore, we conclude the results. Q.E.D.

EC.C.7. Proof of Proposition 3

In this proof, we compare the objective values under partition decisions $\boldsymbol{z}^0$ and $\boldsymbol{z}^*$ by the following steps: 1) formulate the corresponding objective values under $\boldsymbol{z}^*$ and $\boldsymbol{z}^0$, i.e., f^* and f^0; 2) formulate the ratio of $\frac{f^0-f^*}{f^*}$; 3) discuss the upper bound of $\frac{f^0-f^*}{f^*}$ by deriving an upper bound for numerator and a lower bound for denominator.

We first formulate the optimal objective value f^* based on Problem (5), i.e., the objective value under optimal decision $\boldsymbol{z}^*$,

$$f^* = NC_0 + \min_{\boldsymbol{z}\in\mathcal{Z}} \max_{\boldsymbol{d}\in\mathcal{D}} \left\{ \sum_{n=1}^{\Gamma} n z_n h(n,\boldsymbol{d}) \right\} = NC_0 + \min_{\boldsymbol{z}\in\mathcal{Z}} \left\{ \sum_{n=1}^{\Gamma} n z_n h(n,\boldsymbol{d}^*) \right\}$$

where the equality holds since the worst-case distribution vector $\boldsymbol{d}^*$ is independent from partition decision, as shown in Lemma 2. The objective value f^0 under partition decision $\boldsymbol{z}^0$ is

$$\begin{aligned} f^0 =& NC_0 + \lfloor \frac{N}{n^*} \rfloor n^* h(n^*,\boldsymbol{d}^*) + \left(N - \lfloor \frac{N}{n^*} \rfloor n^* \right) \cdot 1 \cdot h\left(N - \lfloor \frac{N}{n^*} \rfloor n^*, \boldsymbol{d}^* \right) \\ =& NC_0 + \lfloor \frac{N}{n^*} \rfloor n^* h(n^*,\boldsymbol{d}^*) + \left(N - \lfloor \frac{N}{n^*} \rfloor n^* \right) h\left(N - \lfloor \frac{N}{n^*} \rfloor n^*, \boldsymbol{d}^* \right). \end{aligned}$$

It is straightforward that $f^* \le f^0$ as $\boldsymbol{z}^0$ is a feasible partition under population size N and $\boldsymbol{z}^*$ is the optimal decision to minimize the overall objective value. Then we have,

$$\begin{aligned} & \frac{f^0 - f^*}{f^*} \\ =& \frac{NC_0 + \lfloor \frac{N}{n^*} \rfloor n^* h(n^*,\boldsymbol{d}^*) + \left(N - \lfloor \frac{N}{n^*} \rfloor n^* \right) h\left(N - \lfloor \frac{N}{n^*} \rfloor n^*, \boldsymbol{d}^* \right)}{NC_0 + \min_{\boldsymbol{z}\in\mathcal{Z}} \left\{ \sum_{n=1}^{\Gamma} n z_n h(n,\boldsymbol{d}^*) \right\}} - 1 \\ \le& \frac{NC_0 + \lfloor \frac{N}{n^*} \rfloor n^* h(n^*,\boldsymbol{d}^*) + \left(N - \lfloor \frac{N}{n^*} \rfloor n^* \right) h\left(N - \lfloor \frac{N}{n^*} \rfloor n^*, \boldsymbol{d}^* \right)}{NC_0 + Nh(n^*,\boldsymbol{d}^*)} - 1 \\ =& \frac{\lfloor \frac{N}{n^*} \rfloor n^* C_0 + \lfloor \frac{N}{n^*} \rfloor n^* h(n^*,\boldsymbol{d}^*) + \left(N - \lfloor \frac{N}{n^*} \rfloor n^* \right) C_0 + \left(N - \lfloor \frac{N}{n^*} \rfloor n^* \right) h\left(N - \lfloor \frac{N}{n^*} \rfloor n^*, \boldsymbol{d}^* \right)}{NC_0 + Nh(n^*,\boldsymbol{d}^*)} - 1 \\ =& \frac{\lfloor \frac{N}{n^*} \rfloor n^*}{N} + \frac{N - \lfloor \frac{N}{n^*} \rfloor n^*}{N} \frac{C_0 + h\left(N - \lfloor \frac{N}{n^*} \rfloor n^*, \boldsymbol{d}^* \right)}{C_0 + h(n^*,\boldsymbol{d}^*)} - 1 \\ =& \frac{N - \lfloor \frac{N}{n^*} \rfloor n^*}{N} \left(\frac{C_0 + h\left(N - \lfloor \frac{N}{n^*} \rfloor n^*, \boldsymbol{d}^* \right)}{C_0 + h(n^*,\boldsymbol{d}^*)} - 1 \right), \end{aligned}$$

where the equalities follows from algebra and the inequality holds as

$$\min_{\boldsymbol{z}\in\mathcal{Z}} \left\{ \sum_{n=1}^{\Gamma} n z_n h(n,\boldsymbol{d}^*) \right\} = \sum_{n=1}^{\Gamma} n z_n^* h(n,\boldsymbol{d}^*) \ge \sum_{n=1}^{\Gamma} n z_n^* h(n^*,\boldsymbol{d}^*) = Nh(n^*,\boldsymbol{d}^*),$$

where the inside inequality holds from the definition of n^*.

Next, as C_0 is a constant defined in Equation (7), we further bound $\frac{f^0-f^*}{f^*}$ by obtaining an upper bound for $C_0 + h\left(N - \lfloor \frac{N}{n^*} \rfloor n^*, \boldsymbol{d}^* \right)$ and a lower bound for $C_0 + h(n^*,\boldsymbol{d}^*)$. Recall that

$$h(n,\boldsymbol{d}^*) = \begin{cases} C_1 & \text{if } n = 1, \\ \frac{C_2}{n} - C_3 \sum_{k=0}^{N-n} \binom{N-n}{k} d_k^* & \text{if } n \ge 2. \end{cases}$$

Then, we focus on the **upper bound** for $C_0 + h\left(N - \lfloor \frac{N}{n^*} \rfloor n^*, \boldsymbol{d}^*\right)$. It suffices to find an upper bound of $h(n, \boldsymbol{d}^*)$ for any $n \in [\Gamma]$. When $n \geq 2$, it is straightforward that

$$h(n, \boldsymbol{d}^*) = \frac{C_2}{n} - C_3 \sum_{k=0}^{N-n} \binom{N-n}{k} d_k^* \leq \frac{C_2}{n} \leq \frac{C_2}{2}.$$

For $n = 1$, $h(n, \boldsymbol{d}^*) = C_1$. Hence, we derive $h(n, \boldsymbol{d}^*) \leq \max\{C_1, \frac{C_2}{2}\}$, and thus, $C_0 + h\left(N - \lfloor \frac{N}{n^*} \rfloor n^*, \boldsymbol{d}^*\right) \leq C_0 + \max\{C_1, \frac{C_2}{2}\}$.

It remains to discuss the **lower bound** for $C_0 + h\left(n^*, \boldsymbol{d}^*\right)$. As $n^* \geq 2$, it suffices to find out the lower bound of $h(n, \boldsymbol{d}^*)$ for any $n \geq 2$. Note that $\sum_{k=0}^{N-n} \binom{N-n}{k} d_k^*$ decreases in n, and thus, we have,

$$0 \leq \sum_{k=0}^{N-\Gamma} \binom{N-\Gamma}{k} d_k^* \leq \sum_{k=0}^{N-n} \binom{N-n}{k} d_k^* \leq \sum_{k=0}^{N-2} \binom{N-2}{k} d_k^* = 1 - p - p(1-p)(1-\rho),$$

where the equalities follows from the definition of feasible region $\mathcal{D}$ and $\boldsymbol{d}^* \in \mathcal{D}$. For more details, $\sum_{k=0}^{N-2} \binom{N-2}{k} d_k^*$ represents the probability that the first two subjects are non-infected from a population N. To discuss all the scenarios of these two subjects, based on prevalence rate p and correlation ρ, it is straightforward that the probability for the (infected, infected) scenario is $p^2 + \rho p(1-p)$, which is coincides with the second constraint to define $\mathcal{D}$. Hence, due to the definition of p, the probability for (infected, non-infected) scenario would be $p - p^2 - \rho p(1-p)$. Based on the exchangeable nature among subjects, the probability for (non-infected, infected) scenario is also $p - p^2 - \rho p(1-p)$. Then, we derive that the probability for (non-infected, non-infected) scenario, i.e., $\sum_{k=0}^{N-2} \binom{N-2}{k} d_k^*$, is $1 - p - (p - p^2 - \rho p(1-p))$, which could be further simplified as $1 - p - p(1-p)(1-\rho)$.

Then, for any $n \geq 2$, we build up the lower bound as follows,

$$h(n, \boldsymbol{d}^*) \geq \frac{C_2}{\Gamma} - C_3(1 - p - p(1-p)(1-\rho)),$$

where the inequality follows from monotonicity of $\sum_{k=0}^{N-n} \binom{N-n}{k} d_k^*$ and $C_2 \geq 0$. Thus, we have

$$C_0 + h(n^*, \boldsymbol{d}^*) \geq C_0 + \frac{C_2}{\Gamma} - C_3(1 - p - p(1-p)(1-\rho)).$$

Note that $C_0 + h(n, \boldsymbol{d}^*)$ represents the testing cost per subject and thus are positive. Hence, the upper bound of $C_0 + h\left(N - \lfloor \frac{N}{n^*} \rfloor n^*, \boldsymbol{d}^*\right)$ we derived is positive. So, to construct the overall bound, it remains to show that the lower bound of $C_0 + h(n^*, \boldsymbol{d}^*)$ is positive, as follows,

$$\begin{aligned}
C_0 + h(n^*, \boldsymbol{d}^*) \geq & C_0 + \frac{C_2}{\Gamma} - C_3(1 - p - p(1-p)(1-\rho)) \\
= & \lambda_1 \beta_n (2 - \beta_n) p + \lambda_2 (1 - \beta_n) \beta_p (1-p) + (1 - \lambda_1 - \lambda_2)(1 - \beta_n) + \frac{1 - \lambda_1 - \lambda_2}{\Gamma} \\
& - [\lambda_2 \beta_p (1 - \beta_n - \beta_p) + (1 - \lambda_1 - \lambda_2)(1 - \beta_n - \beta_p)](1 - p - p(1-p)(1-\rho)) \\
= & \lambda_1 \beta_n (2 - \beta_n) p + \frac{1 - \lambda_1 - \lambda_2}{\Gamma} \\
& + \lambda_2 \beta_p (1 - \beta_n)(1-p) - \lambda_2 \beta_p (1 - \beta_n - \beta_p)(1 - p - p(1-p)(1-\rho)) \\
& + (1 - \lambda_1 - \lambda_2)(1 - \beta_n) - (1 - \lambda_1 - \lambda_2)(1 - \beta_n - \beta_p)(1 - p - p(1-p)(1-\rho)) \\
> & 0,
\end{aligned}$$

where the last inequality follows from $1-\beta_n \geq 1-\beta_n-\beta_p$ and $1 \geq 1-p \geq 1-p-p(1-p)(1-\rho)$.

Therefore, we conclude the results as follows,

$$\begin{aligned}
\frac{f^0-f^*}{f^*} \leq & \frac{N-\lfloor \frac{N}{n^*} \rfloor n^*}{N}\left(\frac{C_0+h\left(N-\lfloor \frac{N}{n^*} \rfloor n^*, \boldsymbol{d}^*\right)}{C_0+h(n^*,\boldsymbol{d}^*)}-1\right) \\
\leq & \frac{\Gamma}{N}\left(\frac{C_0+\max\{C_1,\frac{C_2}{2}\}}{C_0+\frac{C_2}{\Gamma}-C_3(1-p-p(1-p)(1-\rho))}-1\right) \\
= & \frac{\Gamma}{N}\frac{\max\{C_1,\frac{C_2}{2}\}-\frac{C_2}{\Gamma}+C_3(1-p-p(1-p)(1-\rho))}{C_0+\frac{C_2}{\Gamma}-C_3(1-p-p(1-p)(1-\rho))} \\
= & \frac{\Gamma}{N}\frac{\max\{C_1,C_2/2\}\Gamma-C_2+C_3\Gamma(1-p-p(1-p)(1-\rho))}{C_0\Gamma+C_2-C_3\Gamma(1-p-p(1-p)(1-\rho))} \\
\triangleq & C\frac{\Gamma}{N}.
\end{aligned}$$

where the first and second inequalities follow from the above discussion, $N-\lfloor \frac{N}{n^*} \rfloor n^* < n^* \leq \Gamma$, the derived upper and lower bounds correspondingly. Q.E.D.

EC.C.8. Proof of Theorem 2

In this proof, we first propose the formulation of objectives under worst-case distribution $\boldsymbol{d}^*$, i.e., $FN_{\boldsymbol{d}^*}(n)$, $FP_{\boldsymbol{d}^*}(n)$ and $T_{\boldsymbol{d}^*}(n)$, followed by sequentially delving into each statement. To discuss the sensitivity of optimal group size with respect to weight parameters (λ_1,λ_2) and accuracy parameters (β_n,β_p), we allow the group size to be continuous.

Throughout this proof, we denote $\boldsymbol{d}^*$ the worst-case distribution vector as specified in Lemma 2 and define n as the candidate of optimal common group size in Theorem 1. Denote the optimal group size as n^*. According to Lemma EC.2, we can reformulate the objectives under different candidates of optimal group size as follows. If $n=1$, we have $z_1=N$ and,

$$\begin{aligned}
FN_{\boldsymbol{d}^*}(n) =& N\beta_n(2-\beta_n)p-\beta_n(1-\beta_n)pz_1 = N\beta_n p, \\
FP_{\boldsymbol{d}^*}(n) =& N(1-\beta_n)\beta_p(1-p)+\beta_n\beta_p(1-p)z_1-\beta_p(1-\beta_n-\beta_p)\sum_{t=2}^{\Gamma}\left(z_t t\left(\sum_{k=0}^{N-t}\binom{N-t}{k}d_k^*\right)\right) \\
=& N(1-\beta_n)\beta_p(1-p)+\beta_n\beta_p(1-p)N \\
=& N\beta_p(1-p), \\
T_{\boldsymbol{d}^*}(n) =& N.
\end{aligned}$$

If $n \geq 2$, we have $z_n=\frac{N}{n}$ and,

$$\begin{aligned}
FN_{\boldsymbol{d}^*}(n) =& N\beta_n(2-\beta_n)p-\beta_n(1-\beta_n)pz_1 = N\beta_n(2-\beta_n)p, \\
FP_{\boldsymbol{d}^*}(n) =& N(1-\beta_n)\beta_p(1-p)+\beta_n\beta_p(1-p)z_1-\beta_p(1-\beta_n-\beta_p)\sum_{t=2}^{\Gamma}\left(z_t t\left(\sum_{k=0}^{N-t}\binom{N-t}{k}d_k^*\right)\right)
\end{aligned}$$

$$
\begin{aligned}
&=N(1-\beta_n)\beta_p(1-p)-\beta_p(1-\beta_n-\beta_p)\left(z_n n\left(\sum_{k=0}^{N-n}\binom{N-n}{k}d_k^*\right)\right)\\
&=N(1-\beta_n)\beta_p(1-p)-N\beta_p(1-\beta_n-\beta_p)\left(\sum_{k=0}^{N-n}\binom{N-n}{k}d_k^*\right),\\
T_{\boldsymbol{d}^*}(n)&=N(1-\beta_n)+\beta_n z_1+\sum_{t=2}^{\Gamma}\left(z_t-(1-\beta_n-\beta_p)z_t t\left(\sum_{k=0}^{N-t}\binom{N-t}{k}d_k^*\right)\right)\\
&=N(1-\beta_n)+\left(z_n-(1-\beta_n-\beta_p)z_n n\left(\sum_{k=0}^{N-n}\binom{N-n}{k}d_k^*\right)\right)\\
&=N(1-\beta_n)+\frac{N}{n}-N(1-\beta_n-\beta_p)\left(\sum_{k=0}^{N-n}\binom{N-n}{k}d_k^*\right)
\end{aligned}
$$

Then, we prove the four statements as follows.

Statement (i).

If $\lambda_1=1$, the overall objective function is $FN_{\boldsymbol{d}^*}(n)$. In this case, $FN_{\boldsymbol{d}^*}(n)=N\beta_n p$ if $n=1$, and $FN_{\boldsymbol{d}^*}(n)=N\beta_n(2-\beta_n)p$ if $n\geq 2$. Then, it is straightforward that individually testing all subjects can minimize the expected number of false negative classifications, i.e., $n^*=1$.

If $\lambda_2=1$, the overall objective function is $FP_{\boldsymbol{d}^*}(n)$. In this case, $FP_{\boldsymbol{d}^*}(n)=N\beta_p(1-p)$ if $n=1$, and if $n\geq 2$, we have

$$
\begin{aligned}
FP_{\boldsymbol{d}^*}(n)&=N(1-\beta_n)\beta_p(1-p)-N\beta_p(1-\beta_n-\beta_p)\sum_{k=0}^{N-n}\binom{N-n}{k}d_k^*\\
&=N\beta_p(1-p)-N\beta_n\beta_p(1-p)-N\beta_p(1-\beta_n-\beta_p)\sum_{k=0}^{N-n}\binom{N-n}{k}d_k^*\\
&\leq N\beta_p(1-p).
\end{aligned}
$$

Note that $\sum_{k=0}^{N-n}\binom{N-n}{k}d_k^*$ decreases in n. Hence, to minimize $FP_{\boldsymbol{d}^*}(n)$, we always have $n^*=2$.

Statement (ii).

In this statement, we consider $\lambda_2=0$. The overall objective function is $\lambda_1 FN_{\boldsymbol{d}^*}(n)+(1-\lambda_1)T_{\boldsymbol{d}^*}(n)$. If $n\geq 2$, the objective function is,

$$
\begin{aligned}
G(n)&\triangleq\lambda_1 FN_{\boldsymbol{d}^*}(n)+(1-\lambda_1)T_{\boldsymbol{d}^*}(n)\\
&=\lambda_1 N\beta_n(2-\beta_n)p+(1-\lambda_1)\left(N(1-\beta_n)+\frac{N}{n}-N(1-\beta_n-\beta_p)\left(\sum_{k=0}^{N-n}\binom{N-n}{k}d_k^*\right)\right),
\end{aligned}
$$

with the first-order derivative as follows,

$$
\begin{aligned}
\frac{\partial G(n)}{\partial n}&=(1-\lambda_1)\left(-\frac{N}{n^2}-N(1-\beta_n-\beta_p)\frac{\partial}{\partial n}\left(\sum_{k=0}^{N-n}\binom{N-n}{k}d_k^*\right)\right)\\
&=N(1-\lambda_1)\left(-\frac{1}{n^2}+(\beta_n+\beta_p-1)\frac{\partial}{\partial n}\left(\sum_{k=0}^{N-n}\binom{N-n}{k}d_k^*\right)\right).
\end{aligned}
$$

Note that $\sum_{k=0}^{N-n}\binom{N-n}{k}d_k^*$ decreases in n, and thus, $\frac{\partial}{\partial n}\left(\sum_{k=0}^{N-n}\binom{N-n}{k}d_k^*\right)\leq 0$. Consequently, $\frac{\partial G(n)}{\partial n}$ strictly decreases in $(\beta_n+\beta_p)$.

Consider $\beta_{n,1}+\beta_{p,1}<\beta_{n,2}+\beta_{p,2}$. Under $\beta_{n,1},\beta_{p,1}$ and $\beta_{n,2},\beta_{p,2}$, we denote the corresponding objective functions as $G^1(n),G^2(n)$, the optimal group sizes to minimize $G(n)$ when $n\geq 2$ as $\bar{n}^1,\bar{n}^2$. Then, for any $0\leq n\leq\bar{n}^1$, we have

$$G^2(\bar{n}^1)-G^2(n)=\int_n^{\bar{n}^1}\frac{\partial G^2(k)}{\partial k}dk<\int_n^{\bar{n}^1}\frac{\partial G^1(k)}{\partial k}dk=G^1(\bar{n}^1)-G^1(n)\leq 0, \tag{EC.12}$$

where the first inequality holds as $\frac{\partial G(n)}{\partial n}$ decreases in $(\beta_n+\beta_p)$, and the second inequality follows from the optimality of $\bar{n}^1$. Therefore, any $n\leq\bar{n}^1$ cannot outperform $\bar{n}^1$ in minimizing $G^2(\cdot)$, and we have $\bar{n}^2\geq\bar{n}^1$.

Hence, the minimizer $\bar{n}$ of $G(n)$ when $n\geq 2$ increases in $\beta_n+\beta_p$. This proves the second argument in Statement (ii).

Next, it remains to we compare $G(\bar{n})$ and $G(1)$, where $G(1)$ denotes the objective function if $n=1$ as follows,

$$G(1)=\lambda_1 N\beta_n p+(1-\lambda_1)N.$$

Then, we have

$$\begin{aligned}
n^*=1 &\Leftrightarrow\ G(1)\leq\min_n G(n)\\
&\Leftrightarrow\ \lambda_1 N\beta_n p+(1-\lambda_1)N\leq\min_{\boldsymbol{z}:z_1=0}\{\lambda_1 N\beta_n(2-\beta_n)p+(1-\lambda_1)T_{\boldsymbol{d}^*}(\boldsymbol{z})\}\\
&\Leftrightarrow\ -\lambda_1 N\beta_n p(1-\beta_n)\leq\min_{\boldsymbol{z}:z_1=0}\{(1-\lambda_1)T_{\boldsymbol{d}^*}(\boldsymbol{z})\}-(1-\lambda_1)N\\
&\Leftrightarrow\ \lambda_1 N\beta_n p(1-\beta_n)\geq(1-\lambda_1)\max_{\boldsymbol{z}:z_1=0}\{N-T_{\boldsymbol{d}^*}(\boldsymbol{z})\},
\end{aligned}$$

where the LHS of the last line increases in λ_1 from 0 and the RHS decreases linearly in λ_1 to 0. Define $\underline{\lambda}_1$ such that $\underline{\lambda}_1 N\beta_n p(1-\beta_n)=(1-\underline{\lambda}_1)\max_{\boldsymbol{z}:z_1=0}\{N-T_{\boldsymbol{d}^*}(\boldsymbol{z})\}$. Then, $n^*=1$ if $\lambda_1\geq\underline{\lambda}_1$.

To sum up, if $\lambda_1\geq\underline{\lambda}_1$, $n^*=1$; otherwise, $n^*\geq 2$ and increases in $\beta_n+\beta_p$.

Statement (iii).

Consider $n\geq 2$. According to Equation (11), the objective function under different n is,

$$h(n,\boldsymbol{d}^*)=\frac{1-\lambda_1-\lambda_2}{n}-(1-\lambda_1-\lambda_2(1-\beta_p))(1-\beta_n-\beta_p)\sum_{k=0}^{N-n}\binom{N-n}{k}d_k^*.$$

with the following first-order derivative,

$$\frac{\partial h(n,\boldsymbol{d}^*)}{\partial n}=-\frac{1-\lambda_1-\lambda_2}{n^2}-(1-\lambda_1-\lambda_2(1-\beta_p))(1-\beta_n-\beta_p)\frac{\partial}{\partial n}\left(\sum_{k=0}^{N-n}\binom{N-n}{k}d_k^*\right).$$

Note that $\frac{\partial}{\partial n}\left(\sum_{k=0}^{N-n}\binom{N-n}{k}d_k^*\right)\le 0$ and is independent of β_n and β_p. Then, we discuss the monotonicity of $\frac{\partial h(n,\boldsymbol{d}^*)}{\partial n}$ on β_n and β_p, as follows,

$$\begin{aligned}
&\frac{\partial}{\partial\beta_n}\left((1-\lambda_1-\lambda_2(1-\beta_p))(1-\beta_n-\beta_p)\right)=-(1-\lambda_1-\lambda_2(1-\beta_p))\le 0,\\
&\frac{\partial}{\partial\beta_p}\left((1-\lambda_1-\lambda_2(1-\beta_p))(1-\beta_n-\beta_p)\right)\\
=&\frac{\partial}{\partial\beta_p}\left((1-\lambda_1-\lambda_2)(1-\beta_n-\beta_p)+\lambda_2(1-\beta_n)\beta_p-\lambda_2\beta_p^2\right)\\
=&-(1-\lambda_1-\lambda_2)+\lambda_2(1-\beta_n)-2\lambda_2\beta_p.
\end{aligned}$$

Then, $\frac{\partial h(n,\boldsymbol{d}^*)}{\partial n}$ always decreases in β_n. For β_p, if $\lambda_2=0$, $\frac{\partial h(n,\boldsymbol{d}^*)}{\partial n}$ always decreases in β_p; if $\lambda_2>0$, $\frac{\partial h(n,\boldsymbol{d}^*)}{\partial n}$ decreases in β_p when

$$-(1-\lambda_1-\lambda_2)+\lambda_2(1-\beta_n)-2\lambda_2\beta_p\le 0\Leftrightarrow\beta_p\ge\frac{-(1-\lambda_1-\lambda_2)+\lambda_2(1-\beta_n)}{2\lambda_2}=\frac{1-\beta_n}{2}-\frac{1-\lambda_1-\lambda_2}{2\lambda_2}.$$

Hence, following the similar logic as Proof of Statement (ii), n^* increases in β_n and increases in β_p if $\lambda_2=0$ or $\beta_p\ge\frac{1-\beta_n}{2}-\frac{1-\lambda_1-\lambda_2}{2\lambda_2}$.

Statement (iv).

If $n\ge 2$, the overall objective function is $\lambda_1 FN_{\boldsymbol{d}^*}(n)+\lambda_2 FP_{\boldsymbol{d}^*}(n)+(1-\lambda_1-\lambda_2)T_{\boldsymbol{d}^*}(n)$ and n^* satisfies

$$\begin{aligned}
n^*\in&\arg\min_{n\in\{2,\dots,\Gamma\}}\{\lambda_1 FN_{\boldsymbol{d}^*}(n)+\lambda_2 FP_{\boldsymbol{d}^*}(n)+(1-\lambda_1-\lambda_2)T_{\boldsymbol{d}^*}(n)\}\\
=&\arg\min_{n\in\{2,\dots,\Gamma\}}\left\{\lambda_1 FN_{\boldsymbol{d}^*}(n)+(1-\lambda_1)\left(\frac{\lambda_2}{1-\lambda_1}FP_{\boldsymbol{d}^*}(n)+\frac{1-\lambda_1-\lambda_2}{1-\lambda_1}T_{\boldsymbol{d}^*}(n)\right)\right\}\\
=&\arg\min_{n\in\{2,\dots,\Gamma\}}\left\{\lambda_1 FN_{\boldsymbol{d}^*}(n)+(1-\lambda_1)\left(\frac{\lambda_2}{1-\lambda_1}FP_{\boldsymbol{d}^*}(n)+\left(1-\frac{\lambda_2}{1-\lambda_1}\right)T_{\boldsymbol{d}^*}(n)\right)\right\}\\
=&\arg\min_{n\in\{2,\dots,\Gamma\}}\left\{\frac{\lambda_2}{1-\lambda_1}FP_{\boldsymbol{d}^*}(n)+\left(1-\frac{\lambda_2}{1-\lambda_1}\right)T_{\boldsymbol{d}^*}(n)\right\},
\end{aligned}$$

where the last equality holds as $FN_{\boldsymbol{d}^*}(n)$ keeps unchanged for any $n\ge 2$. Consequently, n^* depends on (λ_1,λ_2) through $\frac{\lambda_2}{1-\lambda_1}$.

Therefore, we concludes all the statements. Q.E.D.

EC.C.9. Proof of Proposition 4

This proof includes two steps: reformulate Problem (13) by simplifying the decision $\boldsymbol{w}$ and derive the worst-case distribution. Specifically, in the first step, we reformulate the original minimax problem regarding partition $\boldsymbol{w}$ and joint distribution $\mathbb{P}^{\mathcal{M}}$ to a minimax problem regarding partition $\boldsymbol{v}$ (with less partition information than $\boldsymbol{w}$) and joint distribution $\mathbb{P}^{\mathcal{M}}$. In the second step, we achieve the worst-case distribution and reformulate the objective function under this worst-case distribution to derive the overall minimization problem regarding partition $\boldsymbol{v}$.

- **Step 1. Reformulation.**

We first simplify the decision vector $\boldsymbol{w}$ to $\boldsymbol{v}$ by reformulate Problem (13) to the following problem (EC.13).

$$\min_{\boldsymbol{v}\in\mathcal{V}} \sup_{\mathbb{P}^{\mathcal{M}}\in\mathcal{P}^{\mathcal{M}}} \sum_{l=1}^{MN} \left(\lambda_1 \mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[FN_l(\boldsymbol{v},\tilde{\boldsymbol{\xi}})] + \lambda_2 \mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[FP_l(\boldsymbol{v},\tilde{\boldsymbol{\xi}})] + (1-\lambda_1-\lambda_2)\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[T_l(\boldsymbol{v},\tilde{\boldsymbol{\xi}})]\right), \quad \text{(EC.13)}$$

where $\boldsymbol{v} = (v_{m,l})_{m\in[M],l\in[MN]}$, and $v_{m,l} = \sum_{n\in[N]} w_{m,n,l}$ stands for the number of subjects who come from cluster m and are scheduled in the l-th testing group. The feasible set for the decision $\mathcal{V}$, defined in Proposition 4, and the objective function is specified by

$$\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[FN_l(\boldsymbol{v},\tilde{\boldsymbol{\xi}})] = \begin{cases} 0, & \text{if } \sum_{m=1}^{M} v_{m,l} = 0, \\ \beta_n p, & \text{if } \sum_{m=1}^{M} v_{m,l} = 1, \\ \beta_n(2-\beta_n)p\sum_{m=1}^{M} v_{m,l}, & \text{if } \sum_{m=1}^{M} v_{m,l} \geq 2. \end{cases}$$

$$\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[FP_l(\boldsymbol{v},\tilde{\boldsymbol{\xi}})] = \begin{cases} 0, & \text{if } \sum_{m=1}^{M} v_{m,l} = 0, \\ \beta_p(1-p), & \text{if } \sum_{m=1}^{M} v_{m,l} = 1, \\ (1-\beta_n)\beta_p(1-p)\sum_{m=1}^{M} v_{m,l} & \\ \quad -\beta_p(1-\beta_n-\beta_p)\sum_{m=1}^{M} v_{m,l}\mathbb{P}^{\mathcal{M}}(l \text{ is a non-infected group}), & \text{if } \sum_{m=1}^{M} v_{m,l} \geq 2. \end{cases}$$

$$\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[T_l(\boldsymbol{v},\tilde{\boldsymbol{\xi}})] = \begin{cases} 0, & \text{if } \sum_{m=1}^{M} v_{m,l} = 0, \\ 1, & \text{if } \sum_{m=1}^{M} v_{m,l} = 1, \\ 1+(1-\beta_n)\sum_{m=1}^{M} v_{m,l} & \\ \quad -(1-\beta_n-\beta_p)\sum_{m=1}^{M} v_{m,l}\mathbb{P}^{\mathcal{M}}\left(l \text{ is a non-infected group}\right), & \text{if } \sum_{m=1}^{M} v_{m,l} \geq 2. \end{cases}$$

We show the equivalence by three parts.

In Part I, we formulate the objective function of Problems (13) and (EC.13). In Part II, we show that for any given $\boldsymbol{w}^0 \in \mathcal{W}$, the defined $\boldsymbol{v}^0$ belongs to $\mathcal{V}$, and Problem (13) under $\boldsymbol{w}^0$ have the same objective value as Problem (EC.13) under $\boldsymbol{v}^0$. In Part III, we show that for any given $\boldsymbol{v}^1 \in \mathcal{V}$, there exists a $\boldsymbol{w}^1$ belongs to $\mathcal{W}$, and Problem (13) under $\boldsymbol{w}^1$ have the same objective value as Problem (EC.13) under $\boldsymbol{v}^1$.

Part I.

For any $\boldsymbol{w} \in \mathcal{W}$, the objective value of Problem (13) is,

$$\lambda_1 \mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[FN(\boldsymbol{w},\tilde{\boldsymbol{\xi}})] + \lambda_2 \mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[FP(\boldsymbol{w},\tilde{\boldsymbol{\xi}})] + (1-\lambda_1-\lambda_2)\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[T(\boldsymbol{w},\tilde{\boldsymbol{\xi}})].$$

Then, we calculate $\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[FN(\boldsymbol{w},\tilde{\boldsymbol{\xi}})]$, $\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[FP(\boldsymbol{w},\tilde{\boldsymbol{\xi}})]$ and $\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[T(\boldsymbol{w},\tilde{\boldsymbol{\xi}})]$ as follows. Following from the proof of Lemma EC.2, for each subject (m,n), the expected number of false negative classifications and number of false positive classifications is,

$$\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[FN_{m,n}(\boldsymbol{w},\tilde{\boldsymbol{\xi}})] = \begin{cases} \beta_n p, & \text{if subject } (m,n) \text{ is tested individually}, \\ \beta_n(2-\beta_n)p, & \text{if subject } (m,n) \text{ is tested in a group}. \end{cases}$$

$$\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[FP_{m,n}(\boldsymbol{w},\tilde{\boldsymbol{\xi}})] = \begin{cases} \beta_p(1-p), & \text{if subject } (m,n) \text{ is tested individually,} \\ (1-\beta_n)\beta_p(1-p) - \beta_p(1-\beta_n-\beta_p) & \\ \quad \cdot\, \mathbb{P}^{\mathcal{M}}\left((m,n) \text{ in a negative group}\right), & \text{if subject } (m,n) \text{ is tested in a group.} \end{cases}$$

where whether subject (m,n) is tested individually or in a group depends on the partition decision $\boldsymbol{w}$. Specifically, for any subject (m,n), individually testing subject (m,n) can be described as $\sum_{l=1}^{MN}\left(w_{m,n,l}\cdot\left(\sum_{i=1}^{M}\sum_{j=1}^{N} w_{i,j,l}\right)\right)=1$. As for the expected number of tests, we have the following formulation for an arbitrary group $l\in[MN]$,

$$\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[T_l(\boldsymbol{w},\tilde{\boldsymbol{\xi}})] = \begin{cases} 0, & \text{if } \sum_{m=1}^{M}\sum_{n=1}^{N} w_{m,n,l}=0, \\ 1, & \text{if } \sum_{m=1}^{M}\sum_{n=1}^{N} w_{m,n,l}=1, \\ 1+(1-\beta_n)\left(\sum_{m=1}^{M}\sum_{n=1}^{N} w_{m,n,l}\right) & \\ \quad -(1-\beta_n-\beta_p)\left(\sum_{m=1}^{M}\sum_{n=1}^{N} w_{m,n,l}\right)\mathbb{P}^{\mathcal{M}}\left(l \text{ is negative}\right), & \text{if } \sum_{m=1}^{M}\sum_{n=1}^{N} w_{m,n,l}\geq 2. \end{cases}$$

where $\sum_{m=1}^{M}\sum_{n=1}^{N} w_{m,n,l}=1$ represents group l contains one subjects, while $\sum_{m=1}^{M}\sum_{n=1}^{N} w_{m,n,l}\geq 2$ represents group l contains two or more subjects, for any $l\in[MN]$. Next, the overall objective value is

$$\lambda_1\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}\left[\sum_{m=1}^{M}\sum_{n=1}^{N} FN_{m,n}(\boldsymbol{w},\tilde{\boldsymbol{\xi}})\right]+\lambda_2\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}\left[\sum_{m=1}^{M}\sum_{n=1}^{N} FP_{m,n}(\boldsymbol{w},\tilde{\boldsymbol{\xi}})\right]+(1-\lambda_1-\lambda_2)\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}\left[\sum_{l=1}^{MN} T_l(\boldsymbol{w},\tilde{\boldsymbol{\xi}})\right].$$

For any $\boldsymbol{v}\in\mathcal{V}$, following from the similar logic of proof of Lemma EC.2, we recalculate the characteristics in the group view, i.e., $\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[FN_l(\boldsymbol{v},\tilde{\boldsymbol{\xi}})],\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[FP_l(\boldsymbol{v},\tilde{\boldsymbol{\xi}})]$ as the expected number of false negative classifications and number of false positive classifications of group l, as follows,

$$\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[FN_l(\boldsymbol{v},\tilde{\boldsymbol{\xi}})] = \begin{cases} 0, & \text{if } \sum_{m=1}^{M} v_{m,l}=0, \\ \beta_n p, & \text{if } \sum_{m=1}^{M} v_{m,l}=1, \\ \beta_n(2-\beta_n)p\sum_{m=1}^{M} v_{m,l}, & \text{if } \sum_{m=1}^{M} v_{m,l}\geq 2. \end{cases}$$

$$\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[FP_l(\boldsymbol{v},\tilde{\boldsymbol{\xi}})] = \begin{cases} 0, & \text{if } \sum_{m=1}^{M} v_{m,l}=0, \\ \beta_p(1-p), & \text{if } \sum_{m=1}^{M} v_{m,l}=1, \\ (1-\beta_n)\beta_p(1-p)\sum_{m=1}^{M} v_{m,l} & \\ \quad -\beta_p(1-\beta_n-\beta_p)\sum_{m=1}^{M} v_{m,l}\mathbb{P}^{\mathcal{M}}(l \text{ is a non-infected group}) & \text{if } \sum_{m=1}^{M} v_{m,l}\geq 2. \end{cases}$$

$$\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[T_l(\boldsymbol{v},\tilde{\boldsymbol{\xi}})] = \begin{cases} 0, & \text{if } \sum_{m=1}^{M} v_{m,l}=0, \\ 1, & \text{if } \sum_{m=1}^{M} v_{m,l}=1, \\ 1+(1-\beta_n)\sum_{m=1}^{M} v_{m,l} & \\ \quad -(1-\beta_n-\beta_p)\sum_{m=1}^{M} v_{m,l}\mathbb{P}^{\mathcal{M}}\left(l \text{ is a non-infected group}\right), & \text{if } \sum_{m=1}^{M} v_{m,l}\geq 2. \end{cases}$$

where group "l is a non-infected group" represents that the subjects assigned to group l are non-infected, and $\sum_{m=1}^{M} v_{m,l}$ calculated the number of subjects assigned to group l, for all group $l\in[MN]$.

Part II. For any given $\boldsymbol{w}^0 \in \mathcal{W}$, the objective value of Problem (13) is,

$$\lambda_1 \mathbb{E}_{\mathbb{P}^{\mathcal{M}}}\left[\sum_{m=1}^{M}\sum_{n=1}^{N} FN_{m,n}(\boldsymbol{w}^0, \tilde{\boldsymbol{\xi}})\right] + \lambda_2 \mathbb{E}_{\mathbb{P}^{\mathcal{M}}}\left[\sum_{m=1}^{M}\sum_{n=1}^{N} FP_{m,n}(\boldsymbol{w}^0, \tilde{\boldsymbol{\xi}})\right] + (1-\lambda_1-\lambda_2)\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}\left[\sum_{l=1}^{MN} T_l(\boldsymbol{w}^0, \tilde{\boldsymbol{\xi}})\right].$$

To define $\boldsymbol{v}^0$, let $v^0_{m,l} = \sum_{n=1}^{N} w^0_{m,n,l}$. It is straightforward that

$$\begin{aligned}
&\lambda_1 \mathbb{E}_{\mathbb{P}^{\mathcal{M}}}\left[\sum_{m=1}^{M}\sum_{n=1}^{N} FN_{m,n}(\boldsymbol{w}^0, \tilde{\boldsymbol{\xi}})\right] + \lambda_2 \mathbb{E}_{\mathbb{P}^{\mathcal{M}}}\left[\sum_{m=1}^{M}\sum_{n=1}^{N} FP_{m,n}(\boldsymbol{w}^0, \tilde{\boldsymbol{\xi}})\right] + (1-\lambda_1-\lambda_2)\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}\left[\sum_{l=1}^{MN} T_l(\boldsymbol{w}^0, \tilde{\boldsymbol{\xi}})\right] \\
=&\lambda_1 \mathbb{E}_{\mathbb{P}^{\mathcal{M}}}\left[\sum_{l=1}^{MN} FN_l(\boldsymbol{v}^0, \tilde{\boldsymbol{\xi}})\right] + \lambda_2 \mathbb{E}_{\mathbb{P}^{\mathcal{M}}}\left[\sum_{l=1}^{MN} FP_l(\boldsymbol{v}^0, \tilde{\boldsymbol{\xi}})\right] + (1-\lambda_1-\lambda_2)\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}\left[\sum_{l=1}^{MN} T_l(\boldsymbol{v}^0, \tilde{\boldsymbol{\xi}})\right],
\end{aligned}$$

as

$$\sum_{m=1}^{M}\sum_{n=1}^{N} \mathbb{P}^{\mathcal{M}}\left((m,n) \text{ in a non-infected group}\right) = \sum_{l=1}^{MN} \mathbb{P}^{\mathcal{M}}\left(l \text{ is a a non-infected group}\right).$$

Part III. For any $\boldsymbol{v}^1 \in \mathcal{V}$, the objective value of Problem (EC.13) is

$$\lambda_1 \mathbb{E}_{\mathbb{P}^{\mathcal{M}}}\left[\sum_{l=1}^{MN} FN_l(\boldsymbol{v}^1, \tilde{\boldsymbol{\xi}})\right] + \lambda_2 \mathbb{E}_{\mathbb{P}^{\mathcal{M}}}\left[\sum_{l=1}^{MN} FP_l(\boldsymbol{v}^1, \tilde{\boldsymbol{\xi}})\right] + (1-\lambda_1-\lambda_2)\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}\left[\sum_{l=1}^{MN} T_l(\boldsymbol{v}^1, \tilde{\boldsymbol{\xi}})\right].$$

Define $\boldsymbol{w}^1$ as an arbitrary element in the set $\left\{\boldsymbol{w} \in \mathcal{W} : \sum_{n=1}^{N} w_{m,n,l} = v^1_{m,l}, \forall m \in [M], l \in [MN]\right\}$. Then, the equivalence of two objective functions is also straightforward.

Hence, we conclude the reformulation.

- **Step 2. Worst-case distribution and objective calculation.**

In this step, we first reformulate the inner max problem in Problem (EC.13), and then show that the worst-case distribution has the marginal distribution defined in Lemma 2. Finally, we calculate the objective value under the worst-case distribution to derive the closed form of $g_l(\boldsymbol{v})$.

Reformulation.

Following from the similar logic of Lemma EC.1, for any joint distribution $\mathbb{P}^{\mathcal{M}} = \mathbb{P}^1 \times \cdots \times \mathbb{P}^M$, we construct probability matrix $\boldsymbol{d}^{\mathcal{M}}_{\mathbb{P}^{\mathcal{M}}}$ to capture the whole information from $\mathbb{P}^{\mathcal{M}}$. Mathematically, we define $\boldsymbol{d}^{\mathcal{M}}_{\mathbb{P}^{\mathcal{M}}} = \boldsymbol{d}^1_{\mathbb{P}^{\mathcal{M}}} \times \cdots \times \boldsymbol{d}^M_{\mathbb{P}^{\mathcal{M}}}$, where $\boldsymbol{d}^m_{\mathbb{P}^{\mathcal{M}}} = (d^m_{\mathbb{P}^{\mathcal{M}},0}, d^m_{\mathbb{P}^{\mathcal{M}},1}, \ldots, d^m_{\mathbb{P}^{\mathcal{M}},N})$ with $d^m_{\mathbb{P}^{\mathcal{M}},k} = \mathbb{P}^m\left(\tilde{\boldsymbol{\xi}}^m = \sum_{i=1}^{k} \boldsymbol{e}_i\right)$ for any $k \in \{0, 1, \ldots, N\}$ and $m \in [M]$, and we denote $\sum_{i=1}^{0} \boldsymbol{e}_i = \boldsymbol{0}$ for notational simplicity.

By observation, it suffices to derive an equivalent reformulation for $\mathbb{P}^{\mathcal{M}}\left(l \text{ is a non-infected group}\right)$, and construct an equivalent ambiguity set for $\boldsymbol{d}^{\mathcal{M}}$.

For any distribution $\mathbb{P}^{\mathcal{M}}$, we have the following equivalence,

$$\begin{aligned}
&\mathbb{P}^{\mathcal{M}}\left(l \text{ is a non-infected group}\right) \\
=&\mathbb{P}^1\left(v_{1,l} \text{ subjects assigned from cluster 1 into group } l \text{ are non-infected}\right)\cdots \\
&\mathbb{P}^M\left(v_{M,l} \text{ subjects assigned from cluster } M \text{ into group } l \text{ are non-infected}\right) \\
=&\prod_{m=1}^{M}\sum_{k=0}^{N-v_{m,l}} \binom{N-v_{m,l}}{k} d^m_{\mathbb{P}^{\mathcal{M}},k},
\end{aligned} \tag{EC.14}$$

where the first equality holds due to the independence from different clusters and definition of $\mathbb{P}^m$ for any $m \in [M]$, and the second equality follows from the exchangeability assumption and the definition of $d^m_{\mathbb{P}^{\mathcal{M}},k}$ for any $m \in [M]$. Mathematically, for any cluster $m \in [M]$, we have

$$
\begin{aligned}
&\mathbb{P}^m\left(v_{m,l} \text{ subjects assigned from cluster } m \text{ into group } l \text{ are non-infected}\right)\\
=&\mathbb{P}^m\left(\text{subjects } \{1,\ldots,v_{m,l}\} \text{ in cluster } m \text{ are non-infected}\right)\\
=&\sum_{k=0}^{N-v_{m,l}} \mathbb{P}^m\left(\text{subjects } \{1,\ldots,v_{m,l}\} \text{ in cluster } m \text{ are non-infected; } \{v_{m,l}+1,\ldots,N\} \text{ includes } k \text{ infected subjects}\right)\\
=&\sum_{k=0}^{N-v_{m,l}} \tbinom{N-v_{m,l}}{k}\mathbb{P}^m\left(\text{subjects } \{1,\ldots,v_{m,l}\} \text{ in cluster } m \text{ are non-infected; } \{v_{m,l}+1,\ldots,v_{m,l}+k\} \text{ are infected}\right)\\
=&\sum_{k=0}^{N-v_{m,l}} \tbinom{N-v_{m,l}}{k} d^m_{\mathbb{P}^{\mathcal{M}},k},
\end{aligned}
$$

where the second to the last equalities follow from exchangeability of $v_{m,l}$ subjects within cluster m, probability theory, and exchangeability of k subjects from the remaining $N - v_{m,l}$ subjects within cluster m, respectively.

Next, since the joint distribution $\mathbb{P}^{\mathcal{M}}$ is unknown and belongs to the ambiguity set $\mathcal{P}^{\mathcal{M}}$, the probability matrix $\boldsymbol{d}^{\mathcal{M}}_{\mathbb{P}^{\mathcal{M}}}$ is also unknown. For notational simplicity, we denote the unknown probability matrix as $\boldsymbol{d}^{\mathcal{M}}$ with the ambiguity set $\mathcal{D}^M = \left\{\boldsymbol{d}^M_{\mathbb{P}^{\mathcal{M}}} : \mathbb{P}^{\mathcal{M}} \in \mathcal{P}^{\mathcal{M}}\right\}$.

Recall the $\boldsymbol{d}_{\mathbb{P}}$ defined in Section 4.1 and the definition of $d^m_{\mathbb{P}^{\mathcal{M}},k} = \mathbb{P}^m\left(\tilde{\boldsymbol{\xi}}^m = \sum_{i=1}^k \boldsymbol{e}_i\right)$ for any $k \in \{0,1,\ldots,N\}$. We observe that $\mathbb{P}^m$, the joint distribution of $\tilde{\boldsymbol{\xi}}^m$, satisfies the same settings as the joint distribution $\mathbb{P}$ defined in Section 3. Correspondingly, the ambiguity set of $\boldsymbol{d}^m$ is exactly $\mathcal{D}$ defined in Equation (3). Thus, as $\boldsymbol{d}^{\mathcal{M}}_{\mathbb{P}^{\mathcal{M}}} = \boldsymbol{d}^1_{\mathbb{P}^{\mathcal{M}}} \times \cdots \times \boldsymbol{d}^M_{\mathbb{P}^{\mathcal{M}}}$, we conclude that $\mathcal{D}^{\mathcal{M}} = \mathcal{D} \times \cdots \times \mathcal{D}$.

Hence, for any partition $\boldsymbol{v}$, we can reformulate the inner max problem in Problem (EC.13) with respect of $\boldsymbol{d}^{\mathcal{M}}$ by Equation (EC.14) as follows,

$$
\max_{\boldsymbol{d}^{\mathcal{M}} \in \mathcal{D}^{\mathcal{M}}} \left\{ \sum_{l=1}^{MN} h_l(\boldsymbol{v}, \boldsymbol{d}^{\mathcal{M}}) \right\}, \tag{EC.15}
$$

where

$$h_l(\boldsymbol{v},\boldsymbol{d}^{\mathcal{M}})=\begin{cases}0, & \text{if } \sum_{m=1}^{M} v_{m,l}=0,\\ \lambda_1\beta_n p+\lambda_2\beta_p(1-p)+(1-\lambda_1-\lambda_2), & \text{if } \sum_{m=1}^{M} v_{m,l}=1,\\ (1-\lambda_1-\lambda_2)+(\lambda_1\beta_n(2-\beta_n)p+\lambda_2(1-\beta_n)\beta_p(1-p)+(1-\lambda_1-\lambda_2)(1-\beta_n))\sum_{m=1}^{M} v_{m,l} & \\ -(1-\lambda_1-\lambda_2(1-\beta_p))(1-\beta_n-\beta_p)\left(\sum_{m=1}^{M} v_{m,l}\right)\prod_{m=1}^{M}\left(\sum_{k=0}^{N-v_{m,l}}\binom{N-v_{m,l}}{k}d_k^m\right), & \text{otherwise.}\end{cases} \tag{EC.16}$$

Worst-case distribution to Problem (EC.15).

Now, it suffices to determine the worst-case distribution to Problem (EC.15). By observation, $h_l(\boldsymbol{v},\boldsymbol{d}^{\mathcal{M}})$ is independent of $\boldsymbol{d}^{\mathcal{M}}$ when $\sum_{m=1}^{M} v_{m,l}\le 1$, and the dependence exists on $\prod_{m=1}^{M}\left(\sum_{k=0}^{N-v_{m,l}}\binom{N-v_{m,l}}{k}d_k^m\right)$ when $\sum_{m=1}^{M} v_{m,l}\ge 2$. Hence, it remains to find out the optimal solution to

$$\min_{\boldsymbol{d}^{\mathcal{M}}\in\mathcal{D}^{\mathcal{M}}}\left\{\prod_{m=1}^{M}\left(\sum_{k=0}^{N-v_{m,l}}\binom{N-v_{m,l}}{k}d_k^m\right)\right\}=\min_{\boldsymbol{d}^m\in\mathcal{D},\forall m\in[M]}\left\{\prod_{m=1}^{M}\left(\sum_{k=0}^{N-v_{m,l}}\binom{N-v_{m,l}}{k}d_k^m\right)\right\}, \tag{EC.17}$$

where the equality follows from $\boldsymbol{d}^{\mathcal{M}}=\boldsymbol{d}^1\times\cdots\times\boldsymbol{d}^M$.

Recall that $\boldsymbol{d}^*\in\arg\min_{\boldsymbol{d}\in\mathcal{D}}\sum_{k=0}^{N-n}\binom{N-n}{k}d_k^m$ for any $n\in\{2,\ldots,\Gamma\}$. Moreover, according to the definition of $\mathcal{D}$, for $n=0$, the objective function is $\sum_{k=0}^{N}\binom{N}{k}d_k^m=1$; and for $n=1$, the objective function is $\sum_{k=0}^{N-1}\binom{N-1}{k}d_k^m=1-p$. Hence, we generalize that $\boldsymbol{d}^*\in\arg\min_{\boldsymbol{d}\in\mathcal{D}}\sum_{k=0}^{N-n}\binom{N-n}{k}d_k^m$ for any $n\in\{0,1,\ldots,\Gamma\}$. Next, we use the following inequalities to show that there exists an optimal solution to Problem (EC.17) such that $\boldsymbol{d}^{m*}=\boldsymbol{d}^*$ for any $m\in[M]$,

$$\begin{aligned}\prod_{m=1}^{M}\left(\sum_{k=0}^{N-v_{m,l}}\binom{N-v_{m,l}}{k}d_k^*\right)&\ge\min_{\boldsymbol{d}^m\in\mathcal{D},\forall m\in[M]}\left\{\prod_{m=1}^{M}\left(\sum_{k=0}^{N-v_{m,l}}\binom{N-v_{m,l}}{k}d_k^m\right)\right\}\\ \ge\prod_{m=1}^{M}\left(\min_{\boldsymbol{d}^m\in\mathcal{D}}\left\{\sum_{k=0}^{N-v_{m,l}}\binom{N-v_{m,l}}{k}d_k^m\right\}\right)&=\prod_{m=1}^{M}\left(\sum_{k=0}^{N-v_{m,l}}\binom{N-v_{m,l}}{k}d_k^*\right),\end{aligned}$$

where the first inequality holds as the objective value of any feasible solution is lager than the optimal objective value and $\boldsymbol{d}^m=\boldsymbol{d}^*,\forall m\in[M]$ is a feasible solution to Problem (EC.17); the second inequality holds because the multiplication of separated minimization is smaller than the overall minimization when the separated objective values are non-negative; and the equality follows from $\boldsymbol{d}^*\in\arg\min_{\boldsymbol{d}\in\mathcal{D}}\sum_{k=0}^{N-n}\binom{N-n}{k}d_k^m$ for any $n\in\{0,1,\ldots,\Gamma\}$.

Hence, the worst-case distribution for Problem (EC.15) is $\boldsymbol{d}^{\mathcal{M}*} = \boldsymbol{d}^* \times \cdots \times \boldsymbol{d}^*$.

Objective value under the worst-case distribution.

We replace $\boldsymbol{d}^{\mathcal{M}}$ by $\boldsymbol{d}^{\mathcal{M}*}$ and thus conclude the formulation of $h_l(\boldsymbol{v}, \boldsymbol{d}^{\mathcal{M}*})$. For notational simplicity, we define $g_l(\boldsymbol{v}) = h_l(\boldsymbol{v}, \boldsymbol{d}^{\mathcal{M}*})$. Then, Problem (EC.13) is equivalent to Problem (14).

Therefore, we conclude the results.

Q.E.D.

EC.C.10. Proof of Theorem 3

In this proof, we prove that there exists an optimal partition with same-structure property in two steps: we first focus on the feasibility of same-structure partition $\boldsymbol{v}^*$; then, we conclude the optimality of same-structure partitions by showing that for any partition $\boldsymbol{v}$, there exists a same-structure partition $\boldsymbol{v}^*$ with better objective value.

Step 1. Feasibility.

In this step, we show that $\boldsymbol{v}^* \in \mathcal{V}$ if

$$(v^*_{1,l}, v^*_{2,l}, \ldots, v^*_{M,l}) = \begin{cases} P^i_\pi \boldsymbol{y}^* & \text{if } \exists i \in [M!] \text{ such that } l \in \left\{ \sum_{s=1}^{i-1} k_s + 1, \ldots, \sum_{s=1}^{i} k_s \right\} \\ \boldsymbol{0} & \text{if } l > \boldsymbol{1}^\top \boldsymbol{k} \end{cases}$$

for all $l \in [MN]$, where $\boldsymbol{k} = (k_1, \ldots, k_{M!})$ as any solution of the equation

$$k_1 P^1_\pi \boldsymbol{y}^* + \cdots + k_{M!} P^{M!}_\pi \boldsymbol{y}^* = N\boldsymbol{1},$$

and $P^1_\pi, \ldots, P^{M!}_\pi$ are all the $M!$ distinct $M \times M$ permutation matrices in an arbitrary order.

Recall the definition of $\mathcal{V}$,

$$\mathcal{V} = \left\{ \boldsymbol{v} \in \mathbb{Z}_+^{M \times MN} \;\middle|\; \begin{array}{ll} \sum_{l=1}^{MN} v_{m,l} = N & \forall m \in [M], \\ \sum_{m=1}^{M} v_{m,l} \le \Gamma & \forall l \in [MN], \\ v_{m,l} \in \{0, 1, \ldots, \Gamma\} & \forall m \in [M], l \in [MN] \end{array} \right\}.$$

Due to $\boldsymbol{y}^* \in \mathcal{Y}$, we have $\boldsymbol{y}^* \ge \boldsymbol{0}$, $\sum_{m=1}^{M} y^*_m \le \Gamma$ and $y^*_m \in \{0, 1, \ldots, \Gamma\}$ for all $m \in [M]$. Then, it is straightforward that $\boldsymbol{v}^* \ge \boldsymbol{0}$, $\sum_{m=1}^{M} v^*_{m,l} \le \Gamma$ for all $l \in [MN]$ and $v^*_{m,l} \in \{0, 1, \ldots, \Gamma\}$ for all $m \in [M], l \in [MN]$. It remains to show $\sum_{l=1}^{MN} v^*_{m,l} = N$ for all $m \in [M]$.

For any $m \in [M]$, we have $v^*_{m,l} = (\mathbb{P}^i_\pi \boldsymbol{y}^*)_m$ if there exists $i \in [M!]$ such that $l \in \left\{ \sum_{s=1}^{i-1} k_s + 1, \ldots, \sum_{s=1}^{i} k_s \right\}$, and $v^*_{m,l} = 0$, if $l > \boldsymbol{1}^\top \boldsymbol{k}$, where $\boldsymbol{k}$ satisfies

$$k_1 P^1_\pi \boldsymbol{y}^* + \cdots + k_{M!} P^{M!}_\pi \boldsymbol{y}^* = N\boldsymbol{1}.$$

Then, following from the definition of $\boldsymbol{v}^*$, we have

$$\sum_{l=1}^{MN} v^*_{m,l} = \sum_{l=1}^{\boldsymbol{1}^\top \boldsymbol{k}} v^*_{m,l} = \sum_{i=1}^{M!} \sum_{l=\sum_{s=1}^{i-1} k_s + 1}^{\sum_{s=1}^{i} k_s} v^*_{m,l} = \sum_{i=1}^{M!} k_i (\mathbb{P}^i_\pi \boldsymbol{y}^*)_m = N,$$

where the last equality follows from the equation to define $\boldsymbol{k}$.

Hence, the feasibility of $\boldsymbol{v}^*$ holds.

Step 2. Optimality.

In this step, for any $\boldsymbol{v}^*$ satisfying the definition in Theorem 3, we show that $\sum_{l=1}^{MN} g_l(\boldsymbol{v}) \geq \sum_{l=1}^{MN} g_l(\boldsymbol{v}^*)$ for any arbitrary $\boldsymbol{v} \in \mathcal{V}$.

Recall that $\boldsymbol{y}^* \in \arg\min_{\boldsymbol{y}\in\mathcal{Y}} \ f(\boldsymbol{y})$, where

$$f(\boldsymbol{y}) = \begin{cases} C_0 + C_1, & \text{if } \sum_{m=1}^{M} y_m = 1, \\ C_0 + \frac{C_2}{\sum_{m=1}^{M} y_m} - C_3 \prod_{m=1}^{M} \sum_{k=0}^{N-y_m} \binom{N-y_m}{k} d_k^*, & \text{if } 2 \leq \sum_{m=1}^{M} y_m \leq \Gamma, \end{cases}$$

and C_0, C_2, C_3 are constants defined in Equation (7). The objective value of Problem (14) under any $\boldsymbol{v} \in \mathcal{V}$ is $\sum_{l=1}^{MN} g_l(\boldsymbol{v})$ where

$$g_l(\boldsymbol{v}) = \begin{cases} 0, & \text{if } \sum_{m=1}^{M} v_{m,l} = 0, \\ C_0 + C_1, & \text{if } \sum_{m=1}^{M} v_{m,l} = 1, \\ C_0 \sum_{m=1}^{M} v_{m,l} + C_2 - C_3 \left(\sum_{m=1}^{M} v_{m,l}\right) \prod_{m=1}^{M} \left(\sum_{k=0}^{N-v_{m,l}} \binom{N-v_{m,l}}{k} d_k^*\right), & \text{otherwise.} \end{cases}$$

For any $\boldsymbol{v} \in \mathcal{V}$, define $n_1(\boldsymbol{v})$ as the number of individual tested groups, and the objective value under $\boldsymbol{v}$ satisfies,

$$\begin{aligned}
\sum_{l=1}^{MN} g_l(\boldsymbol{v}) =& n_1(\boldsymbol{v})(C_0 + C_1) + \sum_{\substack{l=1 \\ \sum_{m=1}^{M} v_{m,l} \geq 2}}^{MN} \left(C_0 \sum_{m=1}^{M} v_{m,l} + C_2 - C_3 \sum_{m=1}^{M} v_{m,l} \prod_{m=1}^{M} \sum_{k=0}^{N-v_{m,l}} \binom{N-v_{m,l}}{k} d_k^* \right) \\
=& n_1(\boldsymbol{v})(C_0 + C_1) + \sum_{\substack{l=1 \\ \sum_{m=1}^{M} v_{m,l} \geq 2}}^{MN} \left[\sum_{m=1}^{M} v_{m,l} \left(C_0 + \frac{C_2}{\sum_{m=1}^{M} v_{m,l}} - C_3 \prod_{m=1}^{M} \sum_{k=0}^{N-v_{m,l}} \binom{N-v_{m,l}}{k} d_k^* \right) \right] \\
\geq& n_1(\boldsymbol{v}) f(\boldsymbol{y}^*) + \sum_{\substack{l=1 \\ \sum_{m=1}^{M} v_{m,l} \geq 2}}^{MN} \left[\sum_{m=1}^{M} v_{m,l} f(\boldsymbol{y}^*) \right] \\
=& f(\boldsymbol{y}^*) \left(n_1(\boldsymbol{v}) + \sum_{\substack{l=1 \\ \sum_{m=1}^{M} v_{m,l} \geq 2}}^{MN} \sum_{m=1}^{M} v_{m,l} \right) \\
=& MN f(\boldsymbol{y}^*) \qquad \text{(EC.18)} \\
=& n_1(\boldsymbol{v}^*)(C_0 + C_1) + \sum_{\substack{l=1 \\ \sum_{m=1}^{M} v_{m,l}^* \geq 2}}^{MN} \left[\sum_{m=1}^{M} v_{m,l}^* \left(C_0 + \frac{C_2}{\sum_{m=1}^{M} v_{m,l}^*} - C_3 \prod_{m=1}^{M} \sum_{k=0}^{N-v_{m,l}^*} \binom{N-v_{m,l}^*}{k} d_k^* \right) \right] \\
=& \sum_{l=1}^{MN} g_l(\boldsymbol{v}^*),
\end{aligned}$$

where the inequality follows from

$$f(\boldsymbol{y}^*) \le \left\{ C_0 + C_1, C_0 + \frac{C_2}{\sum_{m=1}^M y_m} - C_3 \prod_{m=1}^M \sum_{k=0}^{N-y_m} \binom{N-y_m}{k} d_k^* \right\} \quad \forall \boldsymbol{y} \in \mathcal{Y},$$

and the penultimate equality holds as in the optimal solution $\boldsymbol{v}^*$, any group l with $\sum_{m=1}^M v_{m,l}^* \ge 2$ satisfies that $(v_{1,l}^*, \ldots, v_{m,l}^*)$ is exactly a permutation of $\boldsymbol{y}^*$.

Therefore, both feasibility and optimality hold for $\boldsymbol{v}^*$ to Problem (14), and we conclude the result. Q.E.D.

EC.C.11. Proof of Theorem 4

To complete this proof, we first propose a claim that the even group outperforms any arbitrary uneven groups. Then, we derive the results by the claim. We finally prove the the claim to conclude this proof.

Claim. *For any $\boldsymbol{y} \in \mathcal{Y}$ with $y_i \ge 1$ and $y_j \ge 1$ where $i \ne j$ and $i, j \in [M]$, we can construct $\boldsymbol{y}'$ such that $\boldsymbol{y}'$ outperforms $\boldsymbol{y}$, i.e., $f(\boldsymbol{y}') \le f(\boldsymbol{y})$, where $\boldsymbol{y}'$ is defined as follows: $n = y_i + y_j$ and,*

(i) if n is odd, $y_i' = \frac{n-1}{2}$, $y_j' = \frac{n+1}{2}$ and $y_m' = y_m$ for any $m \in [M]$ and $m \notin \{i, j\}$;

(ii) if n is even, $y_i' = y_j' = \frac{n}{2}$ and $y_m' = y_m$ for any $m \in [M]$ and $m \notin \{i, j\}$.

To show Theorem 4, we argue that it suffices to prove the above claim. It is straightforward that with the above claim, we can repeatedly equally replace any pair of (y_i, y_j) for any pairs of non-zero elements in $\boldsymbol{y}$, which results in the optimal $\boldsymbol{y}$ satisfies Theorem 4.

Then, we prove the above claim in the remaining proof. For simplicity, we denote (y_1, y_2) with $y_1 + y_2 = n$ as an arbitrary non-zero pair in $\boldsymbol{y}$. Then, as $y_1 + y_2 = y_1' + y_2' = n$, the performances of $\boldsymbol{y}$ and $\boldsymbol{y}'$ can be differentiated by the following $G(y_1, y_2)$ and $G(y_1', y_2')$,

$$G(y_1, y_2) = \sum_{k=0}^{N-y_1} \binom{N-y_1}{k} d_k^* \sum_{k=0}^{N-y_2} \binom{N-y_2}{k} d_k^*.$$

Note that the larger $G(y_1, y_2)$ represents better performance according to the definition of the original objective function $f(\cdot)$. For convenience, we define $g(y) = \sum_{k=0}^{N-y} \binom{N-y}{k} d_k^*$. Then, to find out the optimal $\boldsymbol{y}$ under n with $y_1 \ne 0, y_2 \ne 0$, it remains to discuss

$$y_1^* \in \arg \max_{y_1 \in \{1, \ldots, n-1\}} g(y_1) g(n - y_1).$$

To prove that $y_1^* = \frac{n}{2}$ when n is even and $y_1^* = \frac{n-1}{2}$ when n is odd, we show that it suffices to show $\frac{g(y)}{g(y-1)}$ decreases in $y \in \{2, \ldots, n-1\}$. With this monotonicity result, we have,

$$\frac{g(2)}{g(1)} \ge \frac{g(3)}{g(2)} \ge \cdots \ge \frac{g\left(\lfloor \frac{n}{2} \rfloor - 1\right)}{g\left(\lfloor \frac{n}{2} \rfloor - 2\right)} \ge \frac{g\left(\lfloor \frac{n}{2} \rfloor\right)}{g\left(\lfloor \frac{n}{2} \rfloor - 1\right)} \ge \frac{g\left(\lfloor \frac{n}{2} \rfloor + 1\right)}{g\left(\lfloor \frac{n}{2} \rfloor\right)} \ge \frac{g\left(\lfloor \frac{n}{2} \rfloor + 2\right)}{g\left(\lfloor \frac{n}{2} \rfloor + 1\right)} \ge \cdots \ge \frac{g(n-2)}{g(n-3)} \ge \frac{g(n-1)}{g(n-2)}$$

$$\Rightarrow \begin{cases} g(1)g(n-1) \le g(2)g(n-2) \le \cdots \le g\left(\lfloor \frac{n}{2} \rfloor - 1\right) g\left(\lfloor \frac{n}{2} \rfloor + 1\right) \le g\left(\lfloor \frac{n}{2} \rfloor\right) g\left(\lfloor \frac{n}{2} \rfloor\right) & \text{if } n \text{ is even,} \\ g(1)g(n-1) \le g(2)g(n-2) \le \cdots \le g\left(\lfloor \frac{n}{2} \rfloor - 1\right) g\left(\lfloor \frac{n}{2} \rfloor + 2\right) \le g\left(\lfloor \frac{n}{2} \rfloor\right) g\left(\lfloor \frac{n}{2} \rfloor + 1\right) & \text{if } n \text{ is odd,} \end{cases}$$

$$\Leftrightarrow \begin{cases} y_1^* = \frac{n}{2} & \text{if } n \text{ is even,} \\ y_1^* = \frac{n-1}{2} & \text{if } n \text{ is odd,} \end{cases}$$

where the first line includes $n-2$ terms; the secocnd line follows from $\lfloor \frac{n}{2} \rfloor + \lfloor \frac{n}{2} \rfloor = n$ if n is even and $\lfloor \frac{n}{2} \rfloor + \lfloor \frac{n}{2} \rfloor + 1$ if n is odd.

Hence, we then focus on the monotonicity of $\frac{g(y)}{g(y-1)}$ on $y \in \{2, \ldots, n-1\}$ with the following two cases.

Case 1: $\bar{k}+1 \leq N-y$ and $\bar{k}+1 \leq N-y+1$, i.e., $y \leq N-\bar{k}-1$.

In this case, we have $g(y)$ for any $y \geq 3$ as follows,

$$
\begin{aligned}
g(y) =& \sum_{k=0}^{N-y} \tbinom{N-y}{k} d_k^* = \tbinom{N-y}{\bar{k}} d_{\bar{k}}^* + \tbinom{N-y}{\bar{k}+1} d_{\bar{k}+1}^* \\
=& \tbinom{N-y}{\bar{k}} \frac{(1-p)\left(\bar{k}+1-(N-1)p(1-\rho)\right)}{\binom{N-1}{\bar{k}}} + \tbinom{N-y}{\bar{k}+1} \frac{(1-p)\left(-\bar{k}+(N-1)p(1-\rho)\right)}{\binom{N-1}{\bar{k}+1}} \\
=& (1-p)\left(\bar{k}+1-(N-1)p(1-\rho)\right) \frac{(N-\bar{k}-1)\cdots(N-\bar{k}-y+1)}{(N-1)\cdots(N-y+1)} \\
& + (1-p)\left(-\bar{k}+(N-1)p(1-\rho)\right) \frac{(N-\bar{k}-2)\cdots(N-\bar{k}-y)}{(N-1)\cdots(N-y+1)} \\
=& (1-p)\left(\bar{k}+1-(N-1)p(1-\rho)\right) \frac{(N-\bar{k}-2)\cdots(N-\bar{k}-y+1)}{(N-1)\cdots(N-y+1)}(N-\bar{k}-1) \\
& + (1-p)\left(-\bar{k}+(N-1)p(1-\rho)\right) \frac{(N-\bar{k}-2)\cdots(N-\bar{k}-y+1)}{(N-1)\cdots(N-y+1)}(N-\bar{k}-y) \\
=& (1-p)\frac{(N-\bar{k}-2)\cdots(N-\bar{k}-y+1)}{(N-1)\cdots(N-y+1)} \cdot \\
& \left[(N-\bar{k}-1)\left(\bar{k}+1-(N-1)p(1-\rho)\right) + (N-\bar{k}-y)\left(-\bar{k}+(N-1)p(1-\rho)\right)\right] \\
=& (1-p)\frac{(N-\bar{k}-2)\cdots(N-\bar{k}-y+1)}{(N-1)\cdots(N-y+1)} \cdot \\
& [(N-\bar{k})\left(\bar{k}+1-(N-1)p(1-\rho)\right) + (N-\bar{k})\left(-\bar{k}+(N-1)p(1-\rho)\right) \\
& - \left(\bar{k}+1-(N-1)p(1-\rho)\right) - y\left(-\bar{k}+(N-1)p(1-\rho)\right)] \\
=& (1-p)\frac{(N-\bar{k}-2)\cdots(N-\bar{k}-y+1)}{(N-1)\cdots(N-y+1)} \cdot \\
& [(N-\bar{k}) - \left(\bar{k}+1-(N-1)p(1-\rho)\right) - y\left(-\bar{k}+(N-1)p(1-\rho)\right)].
\end{aligned}
$$

Then, for $g(y-1)$ where $y \geq 4$, we have,

$$
\begin{aligned}
g(y-1) =& (1-p)\frac{(N-\bar{k}-2)\cdots(N-\bar{k}-(y-1)+1)}{(N-1)\cdots(N-(y-1)+1)} \cdot \\
& [(N-\bar{k}) - \left(\bar{k}+1-(N-1)p(1-\rho)\right) - (y-1)\left(-\bar{k}+(N-1)p(1-\rho)\right)] \\
=& (1-p)\frac{(N-\bar{k}-2)\cdots(N-\bar{k}-y+2)}{(N-1)\cdots(N-y+2)} \cdot \\
& [(N-\bar{k}) - \left(\bar{k}+1-(N-1)p(1-\rho)\right) - (y-1)\left(-\bar{k}+(N-1)p(1-\rho)\right)].
\end{aligned}
$$

Next, for $y \geq 4$, we have,

$$
\frac{g(y)}{g(y-1)}
$$

$$
\begin{aligned}
&=\frac{\frac{(N-\bar{k}-2)\cdots(N-\bar{k}-y+1)}{(N-1)\cdots(N-y+1)}}{\frac{(N-\bar{k}-2)\cdots(N-\bar{k}-y+2)}{(N-1)\cdots(N-y+2)}}\cdot \\
&\frac{(N-\bar{k})-\left(\bar{k}+1-(N-1)p(1-\rho)\right)-y\left(-\bar{k}+(N-1)p(1-\rho)\right)}{(N-\bar{k})-\left(\bar{k}+1-(N-1)p(1-\rho)\right)-(y-1)\left(-\bar{k}+(N-1)p(1-\rho)\right)} \\
&=\frac{(N-\bar{k}-y+1)}{(N-y+1)}\frac{(N-\bar{k})-\left(\bar{k}+1-(N-1)p(1-\rho)\right)-y\left(-\bar{k}+(N-1)p(1-\rho)\right)}{(N-\bar{k})-\left(\bar{k}+1-(N-1)p(1-\rho)\right)-(y-1)\left(-\bar{k}+(N-1)p(1-\rho)\right)} \\
&=\left(1-\frac{\bar{k}}{N-y+1}\right)\left(1-\frac{-\bar{k}+(N-1)p(1-\rho)}{(N-\bar{k})-\left(\bar{k}+1-(N-1)p(1-\rho)\right)-(y-1)\left(-\bar{k}+(N-1)p(1-\rho)\right)}\right).
\end{aligned}
\tag{EC.19}
$$

Let $A(y)=1-\frac{\bar{k}}{N-y+1}$ and $B(y)=1-\frac{-\bar{k}+(N-1)p(1-\rho)}{(N-\bar{k})-\left(\bar{k}+1-(N-1)p(1-\rho)\right)-(y-1)\left(-\bar{k}+(N-1)p(1-\rho)\right)}$. Note that both $A(y)$ and $B(y)$ decrease in y, i.e., $A'(y)<0$ and $B'(y)<0$, and $A(y)>0, B(y)>0$. Hence, we have

$$
\frac{\mathrm{d}}{\mathrm{d}y}\frac{g(y)}{g(y-1)}=\frac{\mathrm{d}}{\mathrm{d}y}A(y)B(y)=A'(y)B(y)+A(y)B'(y)\le 0,
$$

and thus, $\frac{g(y)}{g(y-1)}$ decreases in y when $y\ge 4$.

Recall that $\boldsymbol{d}^*\in\mathcal{D}$, and thus $\boldsymbol{d}^*$ satisfies $\sum_{k=1}^{N}\binom{N-1}{k-1}d_k^*=p$ and $\sum_{k=0}^{N}\binom{N}{k}d_k^*=1$. Hence, we have,

$$
1-p=\sum_{k=0}^{N}\binom{N}{k}d_k^*-\sum_{k=1}^{N}\binom{N-1}{k-1}d_k^*=\binom{N}{0}d_0^*+\sum_{k=1}^{N}\left(\binom{N}{k}-\binom{N-1}{k-1}\right)d_k^*=\binom{N}{0}d_0^*+\sum_{k=1}^{N}\binom{N-1}{k}d_k^*=\sum_{k=0}^{N}\binom{N-1}{k}d_k^*.
$$

Next, we check special cases when $y\in\{2,3\}$ as follows,

$$
\begin{aligned}
g(1)&=\sum_{k=0}^{N-1}\binom{N-1}{k}d_k^*=1-p,\\
g(2)&=\sum_{k=0}^{N-2}\binom{N-2}{k}d_k^*=\binom{N-2}{\bar{k}}d_{\bar{k}}^*+\binom{N-2}{\bar{k}+1}d_{\bar{k}+1}^*\\
&=\binom{N-2}{\bar{k}}\frac{(1-p)\left(\bar{k}+1-(N-1)p(1-\rho)\right)}{\binom{N-1}{\bar{k}}}+\binom{N-2}{\bar{k}+1}\frac{(1-p)\left(-\bar{k}+(N-1)p(1-\rho)\right)}{\binom{N-1}{\bar{k}+1}}\\
&=(1-p)\left(\bar{k}+1-(N-1)p(1-\rho)\right)\frac{N-\bar{k}-1}{N-1}+(1-p)\left(-\bar{k}+(N-1)p(1-\rho)\right)\frac{N-\bar{k}-2}{N-1}\\
g(3)&=\binom{N-3}{\bar{k}}\frac{(1-p)\left(\bar{k}+1-(N-1)p(1-\rho)\right)}{\binom{N-1}{\bar{k}}}+\binom{N-3}{\bar{k}+1}\frac{(1-p)\left(-\bar{k}+(N-1)p(1-\rho)\right)}{\binom{N-1}{\bar{k}+1}}\\
&=(1-p)\left(\bar{k}+1-(N-1)p(1-\rho)\right)\frac{(N-\bar{k}-1)(N-\bar{k}-2)}{(N-1)(N-2)}\\
&\quad+(1-p)\left(-\bar{k}+(N-1)p(1-\rho)\right)\frac{(N-\bar{k}-2)(N-\bar{k}-3)}{(N-1)(N-2)}\\
&=(1-p)\left(\bar{k}+1-(N-1)p(1-\rho)\right)\frac{(N-\bar{k}-2)}{(N-1)(N-2)}(N-\bar{k}-1)\\
&\quad+(1-p)\left(-\bar{k}+(N-1)p(1-\rho)\right)\frac{(N-\bar{k}-2)}{(N-1)(N-2)}(N-\bar{k}-3)
\end{aligned}
$$

$$
\begin{aligned}
&=(1-p)\frac{(N-\bar{k}-2)}{(N-1)(N-2)}\cdot[(N-\bar{k})-\left(\bar{k}+1-(N-1)p(1-\rho)\right)-3\left(-\bar{k}+(N-1)p(1-\rho)\right)],\\
g(4)=&(1-p)\frac{(N-\bar{k}-2)\cdots(N-\bar{k}-y+1)}{(N-1)\cdots(N-y+1)}\cdot\\
&[(N-\bar{k})-\left(\bar{k}+1-(N-1)p(1-\rho)\right)-y\left(-\bar{k}+(N-1)p(1-\rho)\right)]\\
=&(1-p)\frac{(N-\bar{k}-2)\cdots(N-\bar{k}-y+1)}{(N-1)\cdots(N-y+1)}\cdot\\
&[(N-\bar{k})-\left(\bar{k}+1-(N-1)p(1-\rho)\right)-y\left(-\bar{k}+(N-1)p(1-\rho)\right)].
\end{aligned}
$$

Next, we argue that $\frac{g(2)}{g(1)}\geq\frac{g(3)}{g(2)}\geq\frac{g(4)}{g(3)}$ by checking that $\frac{g(2)}{g(1)}$ and $\frac{g(3)}{g(2)}$ have the same structure as $\frac{g(y)}{g(y-1)}$ when $y\geq 4$ in Equation (EC.19), as follows,

$$
\begin{aligned}
\frac{g(2)}{g(1)}=&\left(\bar{k}+1-(N-1)p(1-\rho)\right)\frac{N-\bar{k}-1}{N-1}+\left(-\bar{k}+(N-1)p(1-\rho)\right)\frac{N-\bar{k}-2}{N-1}\\
=&\frac{N-\bar{k}-1}{N-1}\left[\left(\bar{k}+1-(N-1)p(1-\rho)\right)+\left(-\bar{k}+(N-1)p(1-\rho)\right)\frac{N-\bar{k}-2}{N-\bar{k}-1}\right]\\
=&\frac{N-\bar{k}-1}{N-1}\left[\left(\bar{k}+1-(N-1)p(1-\rho)\right)+\left(-\bar{k}+(N-1)p(1-\rho)\right)-\left(-\bar{k}+(N-1)p(1-\rho)\right)\frac{1}{N-\bar{k}-1}\right]\\
=&\frac{N-\bar{k}-1}{N-1}\left[1-\frac{-\bar{k}+(N-1)p(1-\rho)}{N-\bar{k}-1}\right],\\
\frac{g(3)}{g(2)}=&\frac{(1-p)\frac{(N-\bar{k}-2)}{(N-1)(N-2)}\cdot[(N-\bar{k})-\left(\bar{k}+1-(N-1)p(1-\rho)\right)-3\left(-\bar{k}+(N-1)p(1-\rho)\right)]}{(1-p)\left(\bar{k}+1-(N-1)p(1-\rho)\right)\frac{N-\bar{k}-1}{N-1}+(1-p)\left(-\bar{k}+(N-1)p(1-\rho)\right)\frac{N-\bar{k}-2}{N-1}}\\
=&\frac{\frac{(N-\bar{k}-2)}{(N-1)(N-2)}\cdot[(N-\bar{k})-\left(\bar{k}+1-(N-1)p(1-\rho)\right)-3\left(-\bar{k}+(N-1)p(1-\rho)\right)]}{\frac{1}{N-1}\left[\left(\bar{k}+1-(N-1)p(1-\rho)\right)(N-\bar{k}-1)+\left(-\bar{k}+(N-1)p(1-\rho)\right)(N-\bar{k}-2)\right]}\\
=&\frac{(N-\bar{k}-2)}{(N-2)}\frac{[(N-\bar{k})-\left(\bar{k}+1-(N-1)p(1-\rho)\right)-3\left(-\bar{k}+(N-1)p(1-\rho)\right)]}{[(N-\bar{k})-\left(\bar{k}+1-(N-1)p(1-\rho)\right)-2\left(-\bar{k}+(N-1)p(1-\rho)\right)]},
\end{aligned}
$$

where both equalities coincide with Equation (EC.19).

To sum up, for any $y\in\{2,\ldots,N-\bar{k}-1\}$, we have $\frac{g(y)}{g(y-1)}$ has the formulation as Equation (EC.19) and decreases in y.

Case 2: $y\geq N-\bar{k}$**.**

In this case, we have

$$
\frac{g(y)}{g(y-1)}=\begin{cases}\frac{\binom{N-y}{\bar{k}}d^*_{\bar{k}}+\binom{N-y}{\bar{k}+1}d^*_{\bar{k}+1}}{\binom{N-(y-1)}{\bar{k}}d^*_{\bar{k}}+\binom{N-(y-1)}{\bar{k}+1}d^*_{\bar{k}+1}} & \text{if } \bar{k}+1\leq N-y \text{ and } \bar{k}+1\leq N-y+1,\\ \frac{\binom{N-y}{\bar{k}}d^*_{\bar{k}}}{\binom{N-(y-1)}{\bar{k}}d^*_{\bar{k}}+\binom{N-(y-1)}{\bar{k}+1}d^*_{\bar{k}+1}} & \text{if } \bar{k}=N-y,\\ 0 & \text{if } \bar{k}=N-y+1,\end{cases}
$$

$$
=\begin{cases}\frac{\binom{N-y}{\bar{k}}d^*_{\bar{k}}+\binom{N-y}{\bar{k}+1}d^*_{\bar{k}+1}}{\binom{N-(y-1)}{\bar{k}}d^*_{\bar{k}}+\binom{N-(y-1)}{\bar{k}+1}d^*_{\bar{k}+1}} & \text{if } y\leq N-\bar{k}-1,\\ \frac{\binom{N-y}{\bar{k}}d^*_{\bar{k}}}{\binom{N-(y-1)}{\bar{k}}d^*_{\bar{k}}+\binom{N-(y-1)}{\bar{k}+1}d^*_{\bar{k}+1}} & \text{if } y=N-\bar{k},\\ 0 & \text{if } y=N-\bar{k}+1,\end{cases}
$$

Note that we have already prove the monotonicity of function $\frac{\binom{N-y}{\bar{k}}d^*_{\bar{k}}+\binom{N-y}{\bar{k}+1}d^*_{\bar{k}+1}}{\binom{N-(y-1)}{\bar{k}}d^*_{\bar{k}}+\binom{N-(y-1)}{\bar{k}+1}d^*_{\bar{k}+1}}$. It is straightforward that we can ignore the $y \geq N-\bar{k}+2$ case with $g(y)=g(y-1)=0$ as y and $y-1$ have the same objective value for the original problem $\max_{y_1\in\{1,\dots,n-1\}} g(y_1)g(n-y_1)$, i.e., $g(y)g(n-y)=g(y-1)g(n-y+1)=0$. Hence, it remains to show that

$$\frac{g(N-\bar{k}-1)}{g(N-\bar{k}-2)} \geq \frac{g(N-\bar{k})}{g(N-\bar{k}-1)} \geq \frac{g(N-\bar{k}+1)}{g(N-\bar{k})}.$$

Note that $\frac{g(N-\bar{k})}{g(N-\bar{k}-1)} \geq \frac{g(N-\bar{k}+1)}{g(N-\bar{k})}$ is straightforward. Then, we focus on $\frac{g(N-\bar{k}-1)}{g(N-\bar{k}-2)} \geq \frac{g(N-\bar{k})}{g(N-\bar{k}-1)}$ as follows,

$$\begin{aligned}
&\frac{g(N-\bar{k}-1)}{g(N-\bar{k}-2)} \geq \frac{g(N-\bar{k})}{g(N-\bar{k}-1)}\\
\Leftrightarrow &\frac{\binom{N-(N-\bar{k}-1)}{\bar{k}}d^*_{\bar{k}}+\binom{N-(N-\bar{k}-1)}{\bar{k}+1}d^*_{\bar{k}+1}}{\binom{N-(N-\bar{k}-2)}{\bar{k}}d^*_{\bar{k}}+\binom{N-(N-\bar{k}-2)}{\bar{k}+1}d^*_{\bar{k}+1}} \geq \frac{\binom{N-(N-\bar{k})}{\bar{k}}d^*_{\bar{k}}}{\binom{N-(N-\bar{k}-1)}{\bar{k}}d^*_{\bar{k}}+\binom{N-(N-\bar{k}-1)}{\bar{k}+1}d^*_{\bar{k}+1}}\\
\Leftrightarrow &\frac{\binom{\bar{k}+1}{\bar{k}}d^*_{\bar{k}}+\binom{\bar{k}+1}{\bar{k}+1}d^*_{\bar{k}+1}}{\binom{\bar{k}+2}{\bar{k}}d^*_{\bar{k}}+\binom{\bar{k}+2}{\bar{k}+1}d^*_{\bar{k}+1}} \geq \frac{\binom{\bar{k}}{\bar{k}}d^*_{\bar{k}}}{\binom{\bar{k}+1}{\bar{k}}d^*_{\bar{k}}+\binom{\bar{k}+1}{\bar{k}+1}d^*_{\bar{k}+1}}\\
\Leftrightarrow &\frac{(\bar{k}+1)d^*_{\bar{k}}+d^*_{\bar{k}+1}}{\frac{(\bar{k}+2)(\bar{k}+1)}{2}d^*_{\bar{k}}+(\bar{k}+2)d^*_{\bar{k}+1}} \geq \frac{d^*_{\bar{k}}}{(\bar{k}+1)d^*_{\bar{k}}+d^*_{\bar{k}+1}}\\
\Leftrightarrow &\left[(\bar{k}+1)d^*_{\bar{k}}+d^*_{\bar{k}+1}\right]^2 \geq d^*_{\bar{k}}\left[\frac{(\bar{k}+2)(\bar{k}+1)}{2}d^*_{\bar{k}}+(\bar{k}+2)d^*_{\bar{k}+1}\right]\\
\Leftrightarrow &\left((\bar{k}+1)^2-\frac{(\bar{k}+2)(\bar{k}+1)}{2}\right)(d^*_{\bar{k}})^2+\bar{k}d^*_{\bar{k}}d^*_{\bar{k}+1}+(d^*_{\bar{k}+1})^2 \geq 0\\
\Leftrightarrow &\frac{\bar{k}(\bar{k}+1)}{2}(d^*_{\bar{k}})^2+\bar{k}d^*_{\bar{k}}d^*_{\bar{k}+1}+(d^*_{\bar{k}+1})^2 \geq 0\\
\Leftrightarrow &\left(d^*_{\bar{k}+1}+\frac{\bar{k}}{2}d^*_{\bar{k}}\right)^2+\left(\frac{\bar{k}(\bar{k}+1)}{2}-\frac{\bar{k}^2}{4}\right)(d^*_{\bar{k}})^2 \geq 0\\
\Leftrightarrow &\left(d^*_{\bar{k}+1}+\frac{\bar{k}}{2}d^*_{\bar{k}}\right)^2+\left(\frac{\bar{k}^2}{4}+\frac{\bar{k}}{2}\right)(d^*_{\bar{k}})^2 \geq 0
\end{aligned}$$

where the last inequality always holds.

Hence, we conclude the monotonicity of $\frac{g(y)}{g(y-1)}$ for any $y \geq N-\bar{k}$.

To sum up, considering the above two cases, $\frac{g(y)}{g(y-1)}$ decreases in y. Therefore, we conclude the results in Theorem 4.

Q.E.D.

EC.C.12. Proof of Proposition 5

We prove Statement (i) and (ii) in the following Step 1 and 2, respectively.

Step 1.

Under given (p,ρ) and group size n, to determine the optimal group structure, it remains to minimize

$$f(\boldsymbol{y})=\begin{cases} -\lambda_1\beta_n(1-\beta_n)p+\lambda_2\beta_n\beta_p(1-p)+(1-\lambda_1-\lambda_2)\beta_n, & \text{if } n=1,\\ \frac{1-\lambda_1-\lambda_2}{n}-(1-\lambda_1-\lambda_2(1-\beta_p))(1-\beta_n-\beta_p)\prod_{m=1}^{M}\sum_{k=0}^{N-y_m}\binom{N-y_m}{k}d^*_k, & \text{if } 2\leq n\leq \Gamma.\end{cases}$$

As $\boldsymbol{d}^*$ is independent of (λ_1, λ_2) and (β_n, β_p), it is straightforward that, the optimal structure under given given (p, ρ) and group size n is independent of (λ_1, λ_2) and (β_n, β_p).

Step 2. It is straightforward by Theorem 2.

Therefore, we conclude the results. Q.E.D.

EC.C.13. Proof of Corollary EC.1

Similar to the proof of Proposition 4, this proof includes two steps: reformulate Problem (EC.1) by simplifying the decision $\boldsymbol{w}$ under cluster-specific prevalence rates and correlation coefficients, and derive the worst-case distribution.

- **Step 1. Reformulation.**

We first simplify the decision vector $\boldsymbol{w}$ to $\boldsymbol{v}$ by reformulate Problem (EC.1) to the following problem (EC.20).

$$\min_{\boldsymbol{v}\in\mathcal{V}^{\text{Gen}}} \sup_{\mathbb{P}^{\mathcal{M}}\in\mathcal{P}^{\mathcal{M},\text{Gen}}} \sum_{l=1}^{\sum_{l=1}^{M} N^m} \left(\lambda_1 \mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[FN_l(\boldsymbol{v},\tilde{\boldsymbol{\xi}})] + \lambda_2 \mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[FP_l(\boldsymbol{v},\tilde{\boldsymbol{\xi}})] + (1-\lambda_1-\lambda_2)\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[T_l(\boldsymbol{v},\tilde{\boldsymbol{\xi}})]\right), \tag{EC.20}$$

where $\boldsymbol{v} = (v_{m,l})_{m\in[M],l\in[\mathcal{N}]}$, and $v_{m,l} = \sum_{n\in[N^m]} w_{m,n,l}$ stands for the number of subjects who come from cluster m and are scheduled in the l-th testing group. The feasible set for the decision $\mathcal{V}^{\text{Gen}}$, defined in Corollary EC.1, and the objective function is specified by

$$\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[FN_l(\boldsymbol{v},\tilde{\boldsymbol{\xi}})] = \begin{cases} 0, & \text{if } \sum_{m=1}^{M} v_{m,l} = 0, \\ \beta_n p, & \text{if } \sum_{m=1}^{M} v_{m,l} = 1, \\ \beta_n(2-\beta_n)p\sum_{m=1}^{M} v_{m,l}, & \text{if } \sum_{m=1}^{M} v_{m,l} \geq 2. \end{cases}$$

$$\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[FP_l(\boldsymbol{v},\tilde{\boldsymbol{\xi}})] = \begin{cases} 0, & \text{if } \sum_{m=1}^{M} v_{m,l} = 0, \\ \beta_p(1-p), & \text{if } \sum_{m=1}^{M} v_{m,l} = 1, \\ (1-\beta_n)\beta_p(1-p)\sum_{m=1}^{M} v_{m,l} & \\ \quad -\beta_p(1-\beta_n-\beta_p)\sum_{m=1}^{M} v_{m,l}\mathbb{P}^{\mathcal{M}}(l \text{ is a non-infected group}), & \text{if } \sum_{m=1}^{M} v_{m,l} \geq 2. \end{cases}$$

$$\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[T_l(\boldsymbol{v},\tilde{\boldsymbol{\xi}})] = \begin{cases} 0, & \text{if } \sum_{m=1}^{M} v_{m,l} = 0, \\ 1, & \text{if } \sum_{m=1}^{M} v_{m,l} = 1, \\ 1+(1-\beta_n)\sum_{m=1}^{M} v_{m,l} & \\ \quad -(1-\beta_n-\beta_p)\sum_{m=1}^{M} v_{m,l}\mathbb{P}^{\mathcal{M}}\left(l \text{ is a non-infected group}\right), & \text{if } \sum_{m=1}^{M} v_{m,l} \geq 2. \end{cases}$$

We show the equivalence by three parts.

In Part I, we formulate the objective function of Problems (EC.1) and (EC.20). In Part II, we show that for any given $\boldsymbol{w}^0 \in \mathcal{W}^{\text{Gen}}$, the defined $\boldsymbol{v}^0$ belongs to $\mathcal{V}^{\text{Gen}}$, and Problem (EC.1) under $\boldsymbol{w}^0$ have the same objective value as Problem (EC.20) under $\boldsymbol{v}^0$. In Part III, we show that for any

given $\boldsymbol{v}^1 \in \mathcal{V}^{\text{Gen}}$, there exists a $\boldsymbol{w}^1$ belongs to $\mathcal{W}^{\text{Gen}}$, and Problem (EC.1) under $\boldsymbol{w}^1$ have the same objective value as Problem (EC.20) under $\boldsymbol{v}^1$.

Part I.

For any $\boldsymbol{w} \in \mathcal{W}^{\text{Gen}}$, the objective value of Problem (EC.1) is,

$$\lambda_1 \mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[FN(\boldsymbol{w}, \tilde{\boldsymbol{\xi}})] + \lambda_2 \mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[FP(\boldsymbol{w}, \tilde{\boldsymbol{\xi}})] + (1-\lambda_1-\lambda_2)\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[T(\boldsymbol{w}, \tilde{\boldsymbol{\xi}})].$$

Then, we calculate $\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[FN(\boldsymbol{w}, \tilde{\boldsymbol{\xi}})]$, $\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[FP(\boldsymbol{w}, \tilde{\boldsymbol{\xi}})]$ and $\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[T(\boldsymbol{w}, \tilde{\boldsymbol{\xi}})]$ as follows. Following from the proof of Lemma EC.2, for each subject (m,n), the expected number of false negative classifications and number of false positive classifications is,

$$\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[FN_{m,n}(\boldsymbol{w}, \tilde{\boldsymbol{\xi}})] = \begin{cases} \beta_n p^m, & \text{if subject } (m,n) \text{ is tested individually,} \\ \beta_n(2-\beta_n)p^m, & \text{if subject } (m,n) \text{ is tested in a group.} \end{cases}$$

$$\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[FP_{m,n}(\boldsymbol{w}, \tilde{\boldsymbol{\xi}})] = \begin{cases} \beta_p(1-p^m), & \text{if subject } (m,n) \text{ is tested individually,} \\ (1-\beta_n)\beta_p(1-p^m) - \beta_p(1-\beta_n-\beta_p) \\ \quad \cdot \mathbb{P}^{\mathcal{M}}\left((m,n) \text{ in a negative group}\right), & \text{if subject } (m,n) \text{ is tested in a group.} \end{cases}$$

where whether subject (m,n) is tested individually or in a group depends on the partition decision $\boldsymbol{w}$. Specifically, for any subject (m,n), individually testing subject (m,n) can be described as $\sum_{l=1}^{\mathcal{N}} \left(w_{m,n,l} \cdot \left(\sum_{i=1}^{M} \sum_{j=1}^{N^m} w_{i,j,l} \right) \right) = 1$. As for the expected number of tests, we have the following formulation for an arbitrary group $l \in [\mathcal{N}]$,

$$\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[T_l(\boldsymbol{w}, \tilde{\boldsymbol{\xi}})] = \begin{cases} 0, & \text{if } \sum_{m=1}^{M} \sum_{n=1}^{N^m} w_{m,n,l} = 0, \\ 1, & \text{if } \sum_{m=1}^{M} \sum_{n=1}^{N^m} w_{m,n,l} = 1, \\ 1 + (1-\beta_n)\left(\sum_{m=1}^{M} \sum_{n=1}^{N^m} w_{m,n,l} \right) \\ \quad -(1-\beta_n-\beta_p)\left(\sum_{m=1}^{M} \sum_{n=1}^{N^m} w_{m,n,l} \right) \mathbb{P}^{\mathcal{M}}\left(l \text{ is negative}\right), & \text{if } \sum_{m=1}^{M} \sum_{n=1}^{N^m} w_{m,n,l} \geq 2. \end{cases}$$

where $\sum_{m=1}^{M} \sum_{n=1}^{N^m} w_{m,n,l} = 1$ represents group l contains one subjects, while $\sum_{m=1}^{M} \sum_{n=1}^{N^m} w_{m,n,l} \geq 2$ represents group l contains two or more subjects, for any $l \in [\mathcal{N}]$. Next, the overall objective value is

$$\lambda_1 \mathbb{E}_{\mathbb{P}^{\mathcal{M}}}\left[\sum_{m=1}^{M} \sum_{n=1}^{N^m} FN_{m,n}(\boldsymbol{w}, \tilde{\boldsymbol{\xi}}) \right] + \lambda_2 \mathbb{E}_{\mathbb{P}^{\mathcal{M}}}\left[\sum_{m=1}^{M} \sum_{n=1}^{N^m} FP_{m,n}(\boldsymbol{w}, \tilde{\boldsymbol{\xi}}) \right] + (1-\lambda_1-\lambda_2)\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}\left[\sum_{l=1}^{\mathcal{N}} T_l(\boldsymbol{w}, \tilde{\boldsymbol{\xi}}) \right].$$

For any $\boldsymbol{v} \in \mathcal{V}^{\text{Gen}}$, following from the similar logic of proof of Lemma EC.2, we recalculate the characteristics in the group view, i.e., $\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[FN_l(\boldsymbol{v}, \tilde{\boldsymbol{\xi}})], \mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[FP_l(\boldsymbol{v}, \tilde{\boldsymbol{\xi}})]$ as the expected number of false negative classifications and number of false positive classifications of group l, as follows,

$$\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[FN_l(\boldsymbol{v}, \tilde{\boldsymbol{\xi}})] = \begin{cases} 0, & \text{if } \sum_{m=1}^{M} v_{m,l} = 0, \\ \beta_n \sum_{m=1}^{M} (p^m v_{m,l}), & \text{if } \sum_{m=1}^{M} v_{m,l} = 1, \\ \beta_n(2-\beta_n) \sum_{m=1}^{M} (p^m v_{m,l}), & \text{if } \sum_{m=1}^{M} v_{m,l} \geq 2. \end{cases}$$

$$\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[FP_l(\boldsymbol{v},\tilde{\boldsymbol{\xi}})]=\begin{cases}0, & \text{if } \sum_{m=1}^{M} v_{m,l}=0,\\ \beta_p\sum_{m=1}^{M}((1-p^m)v_{m,l}), & \text{if } \sum_{m=1}^{M} v_{m,l}=1,\\ (1-\beta_n)\beta_p\sum_{m=1}^{M}((1-p^m)v_{m,l}) & \\ \quad -\beta_p(1-\beta_n-\beta_p)\sum_{m=1}^{M} v_{m,l}\mathbb{P}^{\mathcal{M}}(l \text{ is a non-infected group}) & \text{if } \sum_{m=1}^{M} v_{m,l}\geq 2.\end{cases}$$

$$\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}[T_l(\boldsymbol{v},\tilde{\boldsymbol{\xi}})]=\begin{cases}0, & \text{if } \sum_{m=1}^{M} v_{m,l}=0,\\ 1, & \text{if } \sum_{m=1}^{M} v_{m,l}=1,\\ 1+(1-\beta_n)\sum_{m=1}^{M} v_{m,l} & \\ \quad -(1-\beta_n-\beta_p)\sum_{m=1}^{M} v_{m,l}\mathbb{P}^{\mathcal{M}}\left(l \text{ is a non-infected group}\right), & \text{if } \sum_{m=1}^{M} v_{m,l}\geq 2.\end{cases}$$

where group "l is a non-infected group" represents that the subjects assigned to group l are non-infected, and $\sum_{m=1}^{M} v_{m,l}$ calculated the number of subjects assigned to group l, for all group $l\in[\mathcal{N}]$.

Part II. For any given $\boldsymbol{w}^0\in\mathcal{W}^{\text{Gen}}$, the objective value of Problem (EC.1) is,

$$\lambda_1\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}\left[\sum_{m=1}^{M}\sum_{n=1}^{N^m}FN_{m,n}(\boldsymbol{w}^0,\tilde{\boldsymbol{\xi}})\right]+\lambda_2\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}\left[\sum_{m=1}^{M}\sum_{n=1}^{N^m}FP_{m,n}(\boldsymbol{w}^0,\tilde{\boldsymbol{\xi}})\right]+(1-\lambda_1-\lambda_2)\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}\left[\sum_{l=1}^{\mathcal{N}}T_l(\boldsymbol{w}^0,\tilde{\boldsymbol{\xi}})\right].$$

To define $\boldsymbol{v}^0$, let $v^0_{m,l}=\sum_{n=1}^{N^m} w^0_{m,n,l}$. It is straightforward that

$$\begin{aligned}&\lambda_1\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}\left[\sum_{m=1}^{M}\sum_{n=1}^{N^m}FN_{m,n}(\boldsymbol{w}^0,\tilde{\boldsymbol{\xi}})\right]+\lambda_2\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}\left[\sum_{m=1}^{M}\sum_{n=1}^{N^m}FP_{m,n}(\boldsymbol{w}^0,\tilde{\boldsymbol{\xi}})\right]+(1-\lambda_1-\lambda_2)\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}\left[\sum_{l=1}^{\mathcal{N}}T_l(\boldsymbol{w}^0,\tilde{\boldsymbol{\xi}})\right]\\ =&\lambda_1\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}\left[\sum_{l=1}^{\mathcal{N}}FN_l(\boldsymbol{v}^0,\tilde{\boldsymbol{\xi}})\right]+\lambda_2\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}\left[\sum_{l=1}^{\mathcal{N}}FP_l(\boldsymbol{v}^0,\tilde{\boldsymbol{\xi}})\right]+(1-\lambda_1-\lambda_2)\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}\left[\sum_{l=1}^{\mathcal{N}}T_l(\boldsymbol{v}^0,\tilde{\boldsymbol{\xi}})\right],\end{aligned}$$

as

$$\sum_{m=1}^{M}\sum_{n=1}^{N^m}\mathbb{P}^{\mathcal{M}}\left((m,n) \text{ in a non-infected group}\right)=\sum_{l=1}^{\mathcal{N}}\mathbb{P}^{\mathcal{M}}\left(l \text{ is a a non-infected group}\right).$$

Part III. For any $\boldsymbol{v}^1\in\mathcal{V}^{\text{Gen}}$, the objective value of Problem (EC.20) is

$$\lambda_1\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}\left[\sum_{l=1}^{\mathcal{N}}FN_l(\boldsymbol{v}^1,\tilde{\boldsymbol{\xi}})\right]+\lambda_2\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}\left[\sum_{l=1}^{\mathcal{N}}FP_l(\boldsymbol{v}^1,\tilde{\boldsymbol{\xi}})\right]+(1-\lambda_1-\lambda_2)\mathbb{E}_{\mathbb{P}^{\mathcal{M}}}\left[\sum_{l=1}^{\mathcal{N}}T_l(\boldsymbol{v}^1,\tilde{\boldsymbol{\xi}})\right].$$

Define $\boldsymbol{w}^1$ as an arbitrary element in the set $\left\{\boldsymbol{w}\in\mathcal{W}^{\text{Gen}}:\sum_{n=1}^{N^m} w_{m,n,l}=v^1_{m,l},\forall m\in[M],l\in[\mathcal{N}]\right\}$. Then, the equivalence of two objective functions is also straightforward.

Hence, we conclude the reformulation.

- **Step 2. Worst-case distribution and objective calculation.**

In this proof, we first reformulate the inner max problem in Problem (EC.20), and then show that the worst-case distribution has the marginal distribution for cluster m defined in Lemma 2 under parameters (N^m,p^m,ρ^m), for any $m\in[M]$. Finally, we calculate the objective value under the worst-case distribution to derive the closed form of $g_l^{\text{Gen}}(\boldsymbol{v})$.

Reformulation.

Following from the similar logic of Proposition 4, for any $\boldsymbol{v} \in \mathcal{V}^{\text{Gen}}$, the inner max problem in Problem (EC.20) can be reformulated as,

$$\max_{\boldsymbol{d}^{\mathcal{M}} \in \mathcal{D}^{\mathcal{M},\text{Gen}}} \left\{ \sum_{l=1}^{\mathcal{N}} h_l^{\text{Gen}}(\boldsymbol{v}, \boldsymbol{d}^{\mathcal{M}}) \right\}, \tag{EC.21}$$

where

$$h_l^{\text{Gen}}(\boldsymbol{v}, \boldsymbol{d}^{\mathcal{M}}) = \begin{cases} 0, & \text{if } \sum_{m=1}^{M} v_{m,l} = 0, \\ \lambda_1 \beta_n \sum_{m=1}^{M} p^m v_{m,l} + \lambda_2 \beta_p \sum_{m=1}^{M} (1-p^m) v_{m,l} + (1-\lambda_1-\lambda_2), & \text{if } \sum_{m=1}^{M} v_{m,l} = 1, \\ (1-\lambda_1-\lambda_2) + \lambda_1 \beta_n (2-\beta_n) \sum_{m=1}^{M} p^m v_{m,l} + \lambda_2 (1-\beta_n) \beta_p \sum_{m=1}^{M} (1-p^m) v_{m,l} & \\ +(1-\lambda_1-\lambda_2)(1-\beta_n) \sum_{m=1}^{M} v_{m,l} & \\ -\left(1-\lambda_1-\lambda_2(1-\beta_p)\right)(1-\beta_n-\beta_p) \left(\sum_{m=1}^{M} v_{m,l} \right) \prod_{m=1}^{M} \left(\sum_{k=0}^{N^m - v_{m,l}} \binom{N^m - v_{m,l}}{k} d_k^m \right), & \text{otherwise.} \end{cases} \tag{EC.22}$$

and $\mathcal{D}^{\mathcal{M},\text{Gen}} = \mathcal{D}^1 \times \cdots \times \mathcal{D}^M$, where $\mathcal{D}^m$ is defined in Equation (3) under parameters (N^m, p^m, ρ^m) for any $m \in [M]$.

Worst-case distribution to Problem (EC.21).

The proof of worst-case distribution is straightforward due to the separability of different clusters. Hence, the worst-case distribution for Problem (EC.21) is $\boldsymbol{d}^{\mathcal{M}*,\text{Gen}} = \boldsymbol{d}^{1*} \times \cdots \times \boldsymbol{d}^{M*}$, where $\boldsymbol{d}^{m*} = (d_0^{m*}, d_1^{m*}, \ldots, d_N^{m*})$ is defined in Lemma 2 under infection characteristics (p^m, ρ^m) and size N^m for any $m \in [M]$.

Objective value under the worst-case distribution.

We replace $\boldsymbol{d}^{\mathcal{M}}$ by $\boldsymbol{d}^{\mathcal{M}*,\text{Gen}}$ and thus conclude the formulation of $h_l^{\text{Gen}}(\boldsymbol{v}, \boldsymbol{d}^{\mathcal{M}*,\text{Gen}})$. For notational simplicity, we define $g_l^{\text{Gen}}(\boldsymbol{v}) = h_l(\boldsymbol{v}, \boldsymbol{d}^{\mathcal{M}*,\text{Gen}})$. Then, Problem (EC.20) is equivalent to Problem (EC.2).

Therefore, we conclude the results in Corollary EC.1. Q.E.D.

EC.C.14. Proof of Lemma EC.1

In this proof, we use the following steps to show the equivalence of problems regarding joint distribution $\mathbb{P} \in \mathcal{P}$ and distribution vector $\boldsymbol{d} \in \mathcal{D}$: 1) any joint probability under joint distribution $\mathbb{P}$ can be described by distribution vector $\boldsymbol{d}_{\mathbb{P}}$; 2) any constraints of distribution $\mathbb{P} \in \mathcal{P}$ can be exactly described by constraints to define $\mathcal{D}$; 3) the objective function derived under distribution $\mathbb{P}$ is

equivalent to the one under distribution $\mathbb{Q}_{\boldsymbol{d}_{\mathbb{P}}}$, which is a distribution derived by the distribution vector $\boldsymbol{d}_{\mathbb{P}}$; 4) the joint distribution set generated from distribution vectors $\boldsymbol{d} \in \mathcal{D}$ are equivalent to the original ambiguity set $\mathcal{P}$, i.e., $\mathcal{P} = \{\mathbb{Q}_{\boldsymbol{d}} : \boldsymbol{d} \in \mathcal{D}\}$. Then, we implement the above steps as follows.

Consider any $\mathbb{P} \in \mathcal{P}$ and the corresponding $\boldsymbol{d}_{\mathbb{P}}$. We first show that $\boldsymbol{d}_{\mathbb{P}} \in \mathcal{D}$ as follows. Notice that given any $k \in \{0, 1, \ldots, N\}$, the chance that there are exactly k subjects being positive is

$$\sum_{\boldsymbol{\xi} \in \{0,1\}^N : \mathbf{1}^\top \boldsymbol{\xi} = k} \mathbb{P}\left(\tilde{\boldsymbol{\xi}} = \boldsymbol{\xi}\right) = \sum_{\boldsymbol{\xi} \in \{0,1\}^N : \mathbf{1}^\top \boldsymbol{\xi} = k} \mathbb{P}\left(\tilde{\boldsymbol{\xi}} = \sum_{i=1}^{k} \boldsymbol{e}_i\right) = \sum_{\boldsymbol{\xi} \in \{0,1\}^N : \mathbf{1}^\top \boldsymbol{\xi} = k} d_{\mathbb{P},k} = \binom{N}{k} d_{\mathbb{P},k}, \quad \text{(EC.23)}$$

where the first two equalities arise from exchangeability and the definition of $\boldsymbol{d}_{\mathbb{P}}$, respectively; and the last equality is simply counting the number of such $\boldsymbol{\xi}$. Therefore,

$$1 = \sum_{\boldsymbol{\xi} \in \{0,1\}^N} \mathbb{P}\left(\tilde{\boldsymbol{\xi}} = \boldsymbol{\xi}\right) = \sum_{k=0}^{N} \sum_{\boldsymbol{\xi} \in \{0,1\}^N : \mathbf{1}^\top \boldsymbol{\xi} = k} \mathbb{P}\left(\tilde{\boldsymbol{\xi}} = \boldsymbol{\xi}\right) = \sum_{k=0}^{N} \binom{N}{k} d_{\mathbb{P},k}. \quad \text{(EC.24)}$$

By $\mathbb{P} \in \mathcal{P}$ and Equation (2), we have $\mathbb{P}\left(\tilde{\xi}_1 = 1\right) = p$. Consider any $k \in \{1, \ldots, N\}$, the chance that the first subject is positive while the remaining subjects have exactly $(k-1)$ positives is

$$\sum_{\boldsymbol{\xi} \in \{0,1\}^N : \xi_1 = 1, \sum_{i=2}^{N} \xi_i = k-1} \mathbb{P}\left(\tilde{\boldsymbol{\xi}} = \boldsymbol{\xi}\right) = \sum_{\boldsymbol{\xi} \in \{0,1\}^N : \xi_1 = 1, \sum_{i=2}^{N} \xi_i = k-1} d_{\mathbb{P},k} = \binom{N-1}{k-1} d_{\mathbb{P},k},$$

where the equalities arise from the same logic as those in (EC.23). Therefore, we have

$$p = \mathbb{P}\left(\tilde{\xi}_1 = 1\right) = \sum_{k=1}^{N} \sum_{\boldsymbol{\xi} \in \{0,1\}^N : \xi_1 = 1, \sum_{i=2}^{N} \xi_i = k-1} \mathbb{P}\left(\tilde{\boldsymbol{\xi}} = \boldsymbol{\xi}\right) = \sum_{k=1}^{N} \binom{N-1}{k-1} d_{\mathbb{P},k}. \quad \text{(EC.25)}$$

Similarly, consider any $k \in \{2, \ldots, N\}$, the chance that the first two subjects are positive and the remaining subjects have exactly $(k-2)$ positives is

$$\sum_{\boldsymbol{\xi} \in \{0,1\}^N : \xi_1 = \xi_2 = 1, \sum_{i=3}^{N} \xi_i = k-2} \mathbb{P}\left(\tilde{\boldsymbol{\xi}} = \boldsymbol{\xi}\right) = \sum_{\boldsymbol{\xi} \in \{0,1\}^N : \xi_1 = \xi_2 = 1, \sum_{i=3}^{N} \xi_i = k-2} d_{\mathbb{P},k} = \binom{N-2}{k-2} d_{\mathbb{P},k}.$$

Therefore, by the second constraint in Equation (2), we have

$$p^2 + \rho p(1-p) = \mathbb{P}\left(\tilde{\xi}_1 = \tilde{\xi}_2 = 1\right) = \sum_{k=2}^{N} \sum_{\boldsymbol{\xi} \in \{0,1\}^N : \xi_1 = \xi_2 = 1, \sum_{i=3}^{N} \xi_i = k-2} \mathbb{P}\left(\tilde{\boldsymbol{\xi}} = \boldsymbol{\xi}\right) = \sum_{k=2}^{N} \binom{N-2}{k-2} d_{\mathbb{P},k}. \quad \text{(EC.26)}$$

Finally, $\boldsymbol{d}_{\mathbb{P}} \geq \mathbf{0}$ is by definition. Hence, by Equations (EC.24), (EC.25), (EC.26), we conclude $\boldsymbol{d}_{\mathbb{P}} \in \mathcal{D}$.

We now show $\mathbb{Q}_{\boldsymbol{d}_{\mathbb{P}}} = \mathbb{P}$. Consider any $\boldsymbol{\xi} \in \{0,1\}^N$, denote $k = \mathbf{1}^\top \boldsymbol{\xi}$, we then observe

$$\mathbb{Q}_{\boldsymbol{d}_{\mathbb{P}}}(\tilde{\boldsymbol{\xi}} = \boldsymbol{\xi}) = d_{\mathbb{P},k} = \mathbb{P}\left(\tilde{\boldsymbol{\xi}} = \sum_{i=1}^{k} \boldsymbol{e}_i\right) = \mathbb{P}\left(\tilde{\boldsymbol{\xi}} = \boldsymbol{\xi}\right),$$

where the first two qualities arise from the definitions, and the last equality follows from the exchangeability. Hence, we conclude $\mathbb{Q}_{\boldsymbol{d}_{\mathbb{P}}} = \mathbb{P}$.

To prove $\mathcal{P} = \{\mathbb{Q}_{\boldsymbol{d}} : \boldsymbol{d} \in \mathcal{D}\}$, it is easy to observe that $\mathcal{P} \subseteq \{\mathbb{Q}_{\boldsymbol{d}} : \boldsymbol{d} \in \mathcal{D}\}$ since for any $\mathbb{P} \in \mathcal{P}$ we have already shown $\mathbb{P} = \mathbb{Q}_{\boldsymbol{d}_{\mathbb{P}}} \in \{\mathbb{Q}_{\boldsymbol{d}} : \boldsymbol{d} \in \mathcal{D}\}$ (which is due to $\boldsymbol{d}_{\mathbb{P}} \in \mathcal{D}$). Now it remains to show that $\{\mathbb{Q}_{\boldsymbol{d}} : \boldsymbol{d} \in \mathcal{D}\} \subseteq \mathcal{P}$. That is, considering any $\boldsymbol{d} \in \mathcal{D}$, we need to show that $\mathbb{Q}_{\boldsymbol{d}} \in \mathcal{P}$. To this end, we first observe that

$$\sum_{\boldsymbol{\xi} \in \{0,1\}^N} \mathbb{Q}_{\boldsymbol{d}}(\tilde{\boldsymbol{\xi}} = \boldsymbol{\xi}) = \sum_{k=0}^{N} \sum_{\boldsymbol{\xi} \in \{0,1\}^N : \mathbf{1}^\top \boldsymbol{\xi} = k} \mathbb{Q}_{\boldsymbol{d}}(\tilde{\boldsymbol{\xi}} = \boldsymbol{\xi}) = \sum_{k=0}^{N} \sum_{\boldsymbol{\xi} \in \{0,1\}^N : \mathbf{1}^\top \boldsymbol{\xi} = k} d_k = \sum_{k=0}^{N} \binom{N}{k} d_k = 1, \tag{EC.27}$$

where the second and last equalities arise from the definition of $\mathbb{Q}_{\boldsymbol{d}}$ and the first constraint in the definition of $\mathcal{D}$, respectively.

We then confirm the marginal distribution for each subject $i \in [N]$ as follows,

$$\mathbb{Q}_{\boldsymbol{d}}\left(\tilde{\xi}_i = 1\right) = \sum_{\boldsymbol{\xi} \in \{0,1\}^N : \xi_i = 1} \mathbb{Q}_{\boldsymbol{d}}(\tilde{\boldsymbol{\xi}} = \boldsymbol{\xi}) = \sum_{k=1}^{N} \sum_{\boldsymbol{\xi} \in \{0,1\}^N : \xi_i = 1, \mathbf{1}^\top \boldsymbol{\xi} = k} \mathbb{Q}_{\boldsymbol{d}}(\tilde{\boldsymbol{\xi}} = \boldsymbol{\xi}) = \sum_{k=1}^{N} \binom{N-1}{k-1} d_k = p, \tag{EC.28}$$

where the equalities follow from similar logic as in (EC.27).

With the same logic, for the correlation between each pair of distinct subjects i, j in $[N]$,

$$\mathbb{Q}_{\boldsymbol{d}}\left(\tilde{\xi}_i = \tilde{\xi}_j = 1\right) = \sum_{\boldsymbol{\xi} \in \{0,1\}^N : \xi_i = \xi_j = 1} \mathbb{Q}_{\boldsymbol{d}}(\tilde{\boldsymbol{\xi}} = \boldsymbol{\xi}) = \sum_{k=2}^{N} \binom{N-2}{k-2} d_k = p^2 + \rho p(1-p). \tag{EC.29}$$

Finally, for exchangeability, consider any $\boldsymbol{\xi} \in \{0,1\}^N$ and permutation matrix $\boldsymbol{P}_\pi$, we have $\mathbf{1}^\top \boldsymbol{\xi} = \mathbf{1}^\top (\boldsymbol{P}_\pi \boldsymbol{\xi})$ since $\boldsymbol{P}_\pi$ is a permutation matrix. Hence, by the definition of $\mathbb{Q}_{\boldsymbol{d}}$,

$$\mathbb{Q}_{\boldsymbol{d}}(\tilde{\boldsymbol{\xi}} = \boldsymbol{\xi}) = d_{\mathbf{1}^\top \boldsymbol{\xi}} = d_{\mathbf{1}^\top (\boldsymbol{P}_\pi \boldsymbol{\xi})} = \mathbb{Q}_{\boldsymbol{d}}(\tilde{\boldsymbol{\xi}} = \boldsymbol{P}_\pi \boldsymbol{\xi}). \tag{EC.30}$$

Therefore, we conclude that $\mathbb{Q}_{\boldsymbol{d}} \in \mathcal{P}$ by the definition of $\mathcal{P}$ in (2) and (EC.27), (EC.28), (EC.29), (EC.30). Q.E.D.

EC.C.15. Proof of Lemma EC.2

Consider any given partition $\mathcal{X} \in \mathcal{B}$ and distribution $\mathbb{P} \in \mathcal{P}$, we define $\boldsymbol{d}_{\mathbb{P}}$ by definition of $\boldsymbol{d}$ and vector $\boldsymbol{z}(\mathcal{X}) \in \mathcal{Z}$ such that $z_i(\mathcal{X}) = |\{\Omega \in \mathcal{X} : |\Omega| = i\}|$, i.e., $z_i(\mathcal{X})$ is the number of groups in partition $\mathcal{X}$ with exactly i subjects. In this proof, we first demonstrate how to calculate the objectives $\mathbb{E}_{\mathbb{P}}[FN(\mathcal{X}, \tilde{\boldsymbol{\xi}})]$, $\mathbb{E}_{\mathbb{P}}[FP(\mathcal{X}, \tilde{\boldsymbol{\xi}})]$ and $\mathbb{E}_{\mathbb{P}}[T(\mathcal{X}, \tilde{\boldsymbol{\xi}})]$, and then show that it depends only on $\boldsymbol{z}(\mathcal{X})$ and $\boldsymbol{d}_{\mathbb{P}}$, and is independent of the identities of the subjects in the partition.

- Expected number of false negative classifications $\mathbb{E}_{\mathbb{P}}\left[FN(\mathcal{X}, \tilde{\boldsymbol{\xi}})\right]$

We first calculate $\mathbb{E}_{\mathbb{P}}[FN_i(\mathcal{X},\tilde{\boldsymbol{\xi}})]$, which denotes the expected number of false negative classifications for subject $i,\ i\in[N]$, then the summation over all subjects is exactly $\mathbb{E}_{\mathbb{P}}\left[FN(\mathcal{X},\tilde{\boldsymbol{\xi}})\right]$. Note that false negative classification is non-zero only when subject i is positive, i.e., $\tilde{\xi}_i=1$. Then, we have,

$$\mathbb{E}_{\mathbb{P}}[FN_i(\mathcal{X},\tilde{\boldsymbol{\xi}})]=\mathbb{E}_{\mathbb{P}}\left[FN_i(\mathcal{X},\tilde{\boldsymbol{\xi}})\ \middle|\ \tilde{\xi}_i=1\right]\mathbb{P}\left(\tilde{\xi}_i=1\right)=\mathbb{E}_{\mathbb{P}}\left[FN_i(\mathcal{X},\tilde{\boldsymbol{\xi}})\ \middle|\ \tilde{\xi}_i=1\right]p.$$

Now we consider the case where the subject i is positive.

If subject i is directly tested individually, then, the conditional expectation above is exactly the false negative probability, β_n.

If the subject i is first tested in a group with size $n\geq 2$, then, the pooled sample is positive, and the subject i would be classified as false negative in two cases. The first one is that, the test result for the pooled sample is correct, i.e., positive, but the subject itself is then tested as false negative in the second stage. The second one is the test result for the pooled sample is wrong, i.e., false negative and hence this subject is classified as negative without going to the second stage. Hence, $\mathbb{E}_{\mathbb{P}}\left[FN_i(\mathcal{X},\tilde{\boldsymbol{\xi}})\ \middle|\ \tilde{\xi}_i=1\right]=(1-\beta_n)\cdot\beta_n+\beta_n=\beta_n(2-\beta_n)$.

To sum up,

$$\mathbb{E}_{\mathbb{P}}[FN_i(\mathcal{X},\tilde{\boldsymbol{\xi}})]=\begin{cases}\beta_n p, & \text{if subject } i \text{ is individually tested;}\\ \beta_n(2-\beta_n)p, & \text{if subject } i \text{ is first tested in a group with other subjects.}\end{cases}$$

Therefore,

$$\begin{aligned}\mathbb{E}_{\mathbb{P}}[FN(\mathcal{X},\tilde{\boldsymbol{\xi}})]&=\sum_{i=1}^{N}\mathbb{E}_{\mathbb{P}}[FN_i(\mathcal{X},\tilde{\boldsymbol{\xi}})]=z_1(\mathcal{X})\beta_n p+(N-z_1(\mathcal{X}))\beta_n(2-\beta_n)p\\&=N\beta_n(2-\beta_n)p-\beta_n(1-\beta_n)pz_1(\mathcal{X})\\&\triangleq FN_{\boldsymbol{d}_{\mathbb{P}}}(\boldsymbol{z}(\mathcal{X})).\end{aligned}$$

- Expected number of false positive classifications $\mathbb{E}_{\mathbb{P}}\left[FP(\mathcal{X},\tilde{\boldsymbol{\xi}})\right]$

We first calculate $\mathbb{E}_{\mathbb{P}}[FP_i(\mathcal{X},\tilde{\boldsymbol{\xi}})]$, which denotes the expected number of false positive classifications for subject $i,\ i\in[N]$, then the summation over all subjects is exactly $\mathbb{E}_{\mathbb{P}}\left[FP(\mathcal{X},\tilde{\boldsymbol{\xi}})\right]$. Note that false positive classification is non-zero only when subject i is negative, i.e., $\tilde{\xi}_i=0$. Then, we have,

$$\mathbb{E}_{\mathbb{P}}[FP_i(\mathcal{X},\tilde{\boldsymbol{\xi}})]=\mathbb{E}_{\mathbb{P}}\left[FP_i(\mathcal{X},\tilde{\boldsymbol{\xi}})\ \middle|\ \tilde{\xi}_i=0\right]\mathbb{P}\left(\tilde{\xi}_i=0\right)=\mathbb{E}_{\mathbb{P}}\left[FP_i(\mathcal{X},\tilde{\boldsymbol{\xi}})\ \middle|\ \tilde{\xi}_i=0\right](1-p).$$

Now we consider the case where the subject i is negative.

If subject i is directly tested individually, then, the conditional expectation is exactly the false positive probability, β_p.

If subject i is first tested in group $\Omega \in \mathcal{X}$ whose group size is denoted by $n \geq 2$, then the above conditional expectation depends on whether the pooled sample of Ω is negative or not (i.e., all subjects in Ω are negative or at least one of them is positive). We define I as the number positive subjects in Ω. Then, we have,

$$\begin{aligned}
&\mathbb{E}_{\mathbb{P}}\left[FP_i(\mathcal{X},\tilde{\boldsymbol{\xi}}) \;\middle|\; \tilde{\xi}_i = 0\right]\mathbb{P}\left(\tilde{\xi}_i = 0\right)\\
=&\mathbb{E}_{\mathbb{P}}\left[FP_i(\mathcal{X},\tilde{\boldsymbol{\xi}}) \;\middle|\; \tilde{\xi}_i = 0, I = 0\right]\mathbb{P}\left(\tilde{\xi}_i = 0, I = 0\right) + \mathbb{E}_{\mathbb{P}}\left[FP_i(\mathcal{X},\tilde{\boldsymbol{\xi}}) \;\middle|\; \tilde{\xi}_i = 0, I \neq 0\right]\mathbb{P}\left(\tilde{\xi}_i = 0, I \neq 0\right)\\
=&\mathbb{E}_{\mathbb{P}}\left[FP_i(\mathcal{X},\tilde{\boldsymbol{\xi}}) \;\middle|\; I = 0\right]\mathbb{P}\left(I = 0\right) + \mathbb{E}_{\mathbb{P}}\left[FP_i(\mathcal{X},\tilde{\boldsymbol{\xi}}) \;\middle|\; \tilde{\xi}_i = 0, I \neq 0\right]\mathbb{P}\left(\tilde{\xi}_i = 0, I \neq 0\right).
\end{aligned}$$

Now we focus on the first term, $\mathbb{E}_{\mathbb{P}}\left[FP_i(\mathcal{X},\tilde{\boldsymbol{\xi}}) \;\middle|\; I = 0\right]$. Conditional on there being no positive in the group, the false positive classification of subject i is nonzero only if both the pooled sample of Ω (in the first stage) and the individual sample i itself (in the second stage) are tested as false positive, i.e.,

$$\mathbb{E}_{\mathbb{P}}\left[FP_i(\mathcal{X},\tilde{\boldsymbol{\xi}}) \;\middle|\; I = 0\right] = \beta_p^2.$$

Then we calculate the second term, $\mathbb{E}_{\mathbb{P}}\left[FP_i(\mathcal{X},\tilde{\boldsymbol{\xi}}) \;\middle|\; \tilde{\xi}_i = 0, I \neq 0\right]\mathbb{P}\left(\tilde{\xi}_i = 0, I \neq 0\right)$. Here the negative subject i is first tested with some positive subjects in the group and then tested individually if the first stage result is true positive. Hence,

$$\mathbb{E}_{\mathbb{P}}\left[FP_i(\mathcal{X},\tilde{\boldsymbol{\xi}}) \;\middle|\; \tilde{\xi}_i = 0, I \neq 0\right] = (1-\beta_n)\beta_p.$$

For $\mathbb{P}\left(\tilde{\xi}_i = 0, I \neq 0\right)$, we note that,

$$\mathbb{P}\left(\tilde{\xi}_i = 0, I \neq 0\right) = \mathbb{P}\left(\tilde{\xi}_i = 0\right) - \mathbb{P}\left(\tilde{\xi}_i = 0, I = 0\right) = \mathbb{P}\left(\tilde{\xi}_i = 0\right) - \mathbb{P}\left(I = 0\right) = 1 - p - \mathbb{P}\left(I = 0\right).$$

Therefore, in this case where subject i is tested in the group Ω,

$$\begin{aligned}
\mathbb{E}_{\mathbb{P}}\left[FP_i(\mathcal{X},\tilde{\boldsymbol{\xi}})\right] &= \beta_p^2\mathbb{P}\left(I = 0\right) + (1-\beta_n)\beta_p\left(1 - p - \mathbb{P}\left(I = 0\right)\right)\\
&= (1-\beta_n)\beta_p(1-p) - \beta_p(1-\beta_n-\beta_p)\mathbb{P}\left(I = 0\right)\\
&= (1-\beta_n)\beta_p(1-p) - \beta_p(1-\beta_n-\beta_p)\sum_{k=0}^{N-n}\binom{N-n}{k}d_{\mathbb{P},k}
\end{aligned}$$

where the last equality follows from the following equation (EC.31). Under any probability distribution $\mathbb{P} \in \mathcal{P}$ and a group Ω of subjects with $|\Omega| = n$, we have

$$\mathbb{P}\left(\tilde{\xi}_i = 0, \forall i \in \Omega\right) = \mathbb{P}\left(\sum_{i\in\Omega}\tilde{\xi}_i = 0\right) = \sum_{k=0}^{N-n}\mathbb{P}\left(\sum_{i\in\Omega}\tilde{\xi}_i = 0, \sum_{j\notin\Omega}\tilde{\xi}_j = k\right) = \sum_{k=0}^{N-n}\binom{N-n}{k}d_{\mathbb{P},k}. \qquad \text{(EC.31)}$$

To sum up,

$$\mathbb{E}_{\mathbb{P}}[FP_i(\mathcal{X},\tilde{\boldsymbol{\xi}})] = \begin{cases} \beta_p(1-p), & \text{if } i \text{ is individually tested;} \\ (1-\beta_n)\beta_p(1-p) - \beta_p(1-\beta_n-\beta_p)\sum_{k=0}^{N-n}\binom{N-n}{k}d_{\mathbb{P},k}, & \text{if } i \text{ is first tested in } \Omega. \end{cases}$$

Therefore,

$$\begin{aligned}
&\mathbb{E}_{\mathbb{P}}[FP(\mathcal{X},\tilde{\boldsymbol{\xi}})] = \sum_{i=1}^{N}\mathbb{E}_{\mathbb{P}}[FP_i(\mathcal{X},\tilde{\boldsymbol{\xi}})] \\
=&\beta_p(1-p)\cdot z_1(\mathcal{X}) + (1-\beta_n)\beta_p(1-p)\cdot(N-z_1(\mathcal{X})) - \beta_p(1-\beta_n-\beta_p)\sum_{n=2}^{\Gamma}\left(z_n(\mathcal{X})n\left(\sum_{k=0}^{N-n}\binom{N-n}{k}d_{\mathbb{P},k}\right)\right) \\
=&N(1-\beta_n)\beta_p(1-p) + \beta_n\beta_p(1-p)z_1(\mathcal{X}) - \beta_p(1-\beta_n-\beta_p)\sum_{n=2}^{\Gamma}\left(z_n n\left(\sum_{k=0}^{N-n}\binom{N-n}{k}d_{\mathbb{P},k}\right)\right) \\
=&FP_{\boldsymbol{d}_{\mathbb{P}}}(\boldsymbol{z}(\mathcal{X})).
\end{aligned}$$

- Expected number of tests $\mathbb{E}_{\mathbb{P}}\left[T(\mathcal{X},\tilde{\boldsymbol{\xi}})\right]$.

For any given group $\Omega_w \in \mathcal{X}$ whose size is denoted by $n_w = |\Omega_w|$, we first calculate $\mathbb{E}_{\mathbb{P}}[T_w(\mathcal{X},\tilde{\boldsymbol{\xi}})]$, which represents the expected number of tests for this group. Then we sum over w to obtain the expected number of tests for the whole population.

If $n_w = 1$, then this group includes only one subject and the number of tests $\mathbb{E}_{\mathbb{P}}[T_w(\mathcal{X},\tilde{\boldsymbol{\xi}})] = 1$.

If $n_w \geq 2$, then the expected number of tests depends on whether the group includes positive subjects. Here we use I_w to denote the number of positive subjects in group Ω_w. Then,

$$\begin{aligned}
\mathbb{E}_{\mathbb{P}}[T_w(\mathcal{X},\tilde{\boldsymbol{\xi}})] &= \mathbb{E}_{\mathbb{P}}[T_w(\mathcal{X},\tilde{\boldsymbol{\xi}}) \mid I_w = 0]\mathbb{P}(I_w = 0) + \mathbb{E}_{\mathbb{P}}[T_w(\mathcal{X},\tilde{\boldsymbol{\xi}}) \mid I_w \neq 0]\mathbb{P}(I_w \neq 0) \\
&= \mathbb{E}_{\mathbb{P}}[T_w(\mathcal{X},\tilde{\boldsymbol{\xi}}) \mid I_w = 0]\mathbb{P}(I_w = 0) + \mathbb{E}_{\mathbb{P}}[T_w(\mathcal{X},\tilde{\boldsymbol{\xi}}) \mid I_w \neq 0]\left[1 - \mathbb{P}(I_w = 0)\right].
\end{aligned}$$

We now focus on $\mathbb{E}_{\mathbb{P}}[T_w(\mathcal{X},\tilde{\boldsymbol{\xi}}) \mid I_w = 0]$. In this case, if the group is tested as true negative in the first stage, then the number of tests is 1; if the group is tested as false positive in the first stage, then the number of tests is $1 + n_w$. Hence, we have,

$$\mathbb{E}_{\mathbb{P}}[T_w(\mathcal{X},\tilde{\boldsymbol{\xi}}) \mid I_w = 0] = (1-\beta_p)\cdot 1 + \beta_p\cdot(1+n_w) = 1 + \beta_p n_w.$$

Then we move on to $\mathbb{E}_{\mathbb{P}}[T_w(\mathcal{X},\tilde{\boldsymbol{\xi}}) \mid I_w \neq 0]$. If the group is tested as true positive in the first stage, then the number of tests is $1 + n_w$; if the group is tested as false negative in the first stage, then the number of tests is 1. Hence, we have

$$\mathbb{E}_{\mathbb{P}}[T_w(\mathcal{X},\tilde{\boldsymbol{\xi}}) \mid I_w \neq 0] = (1-\beta_n)(1+n_w) + \beta_n\cdot 1 = 1 + (1-\beta_n)n_w.$$

To sum up, we can have

$$\begin{aligned}\mathbb{E}_{\mathbb{P}}[T_w(\mathcal{X},\tilde{\boldsymbol{\xi}})] &= (1+\beta_p n_w)\mathbb{P}(I_w=0) + (1+(1-\beta_n)n_w)\,(1-\mathbb{P}(I_w=0)) \\ &= 1+(1-\beta_n)n_w - (1-\beta_n-\beta_p)n_w\mathbb{P}(I_w=0) \\ &= 1+(1-\beta_n)n_w - (1-\beta_n-\beta_p)n_w\left(\sum_{k=0}^{N-n_w}\tbinom{N-n_w}{k}d_{\mathbb{P},k}\right)\end{aligned}$$

where the last equality follows from Equation (EC.31).

Hence,

$$\mathbb{E}_{\mathbb{P}}[T_w(\mathcal{X},\tilde{\boldsymbol{\xi}})] = \begin{cases} 1 & \text{if } n_w = 1, \\ 1+(1-\beta_n)n_w - (1-\beta_n-\beta_p)n_w\left(\sum_{k=0}^{N-n_w}\tbinom{N-n_w}{k}d_{\mathbb{P},k}\right) & \text{if } n_w \geq 2. \end{cases}$$

Therefore, we can simplify $\mathbb{E}_{\mathbb{P}}[T(\mathcal{X},\tilde{\boldsymbol{\xi}})]$ as follows,

$$\begin{aligned}\mathbb{E}_{\mathbb{P}}[T(\mathcal{X},\tilde{\boldsymbol{\xi}})] =& 1\cdot z_1(\mathcal{X}) + \sum_{n=2}^{\Gamma}\left(z_n(\mathcal{X})\cdot\left(1+(1-\beta_n)n - (1-\beta_n-\beta_p)n\left(\sum_{k=0}^{N-n}\tbinom{N-n}{k}d_{\mathbb{P},k}\right)\right)\right) \\ =& N(1-\beta_n) + \beta_n z_1(\mathcal{X}) + \sum_{n=2}^{\Gamma}\left(z_n(\mathcal{X})\left(1-(1-\beta_n-\beta_p)n\left(\sum_{k=0}^{N-n}\tbinom{N-n}{k}d_{\mathbb{P},k}\right)\right)\right) \\ =& N(1-\beta_n) + \beta_n z_1(\mathcal{X}) + \sum_{n=2}^{\Gamma}\left(z_n - (1-\beta_n-\beta_p)z_n n\left(\sum_{k=0}^{N-n}\tbinom{N-n}{k}d_{\mathbb{P},k}\right)\right) \\ =& T_{\boldsymbol{d}_{\mathbb{P}}}(\boldsymbol{z}(\mathcal{X})).\end{aligned}$$

For notationaly simplicity, we let $\boldsymbol{d} = \boldsymbol{d}_{\mathbb{P}}$ and $\boldsymbol{z} = \boldsymbol{z}(\mathcal{X})$. Hence, the equivalence between $(\mathbb{E}_{\mathbb{P}}[FN(\mathcal{X},\tilde{\boldsymbol{\xi}})], \mathbb{E}_{\mathbb{P}}[FP(\mathcal{X},\tilde{\boldsymbol{\xi}})], \mathbb{E}_{\mathbb{P}}[T(\mathcal{X},\tilde{\boldsymbol{\xi}})])$ and $(FN_{\boldsymbol{d}}(\boldsymbol{z}), FP_{\boldsymbol{d}}(\boldsymbol{z}), T_{\boldsymbol{d}}(\boldsymbol{z}))$ can be established easily.

Q.E.D.